

Integer Dimension in Convex Mixed-Integer Approximation of Nonlinear Graphs

Abstract

We study the minimum number of integer variables in a convex lift that contains a nonlinear graph and admits only a prescribed vertical error. The competing lifts may have unrestricted continuous dimension, real coefficients and unbounded integer variables. For a fixed quadratic system on a full-dimensional box, we prove that both this minimum and the minimum binary linear dimension equal one half of the noncommutative rank of the Hessian space times $\log_2(1/\varepsilon)$, up to an additive constant. At finite accuracy, a covariance determinant characterizes the count within $O(n \log(n+1))$, with rational polynomial-time constructions, certificates and sharper bounds for positive structure. For continuous convex scalar graphs, binary linear dimension is within two of the general convex integer minimum. For dense rational convex polynomials, certified curvature integration and implicit knot compilation give a polynomial-size rational MILP within eleven binaries of that minimum, constructed in polynomial time. Further results treat positive sparse polynomials and binary-encoded powers under their stated representations. For coupled convex outputs, shared curvature bases give additive overheads logarithmic in curvature rank; for separable inputs the overhead also scales with input dimension. Strong-oracle unconditional error bodies admit explicit linear bands in the constructed formulations. Explicit examples show limits of scalarization, exact binary/general-integer separations, and a fixed-error exponential separation between rational MILP and second-order-cone encoding lengths. Throughout, integer count, continuous size, coefficient encoding and construction time are distinguished.

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Formal verification scope. The repository’s topic 20 coverage map and verification record identify the Lean proofs and targeted checks for the exact square count, scalar quadratic rank and one-sided inertia laws, their finite linear upper constructions, the exact convex one-sided product count, and the continuous square-epigraph inequality lower bound. The vector quadratic results and the rest of this manuscript are outside that scope. Earlier review records remain records of their dated sources.

1 Introduction

How many integer variables are intrinsically necessary to approximate a nonlinear graph to a prescribed accuracy? A formulation can use many interval selectors, a few integer variables with large ranges, continuous lifting variables, or long coefficients. An efficient encoding of one chosen partition gives an upper bound, but it does not establish what is possible with all convex mixed-integer formulations. We study this minimum and identify geometric and algebraic quantities that govern it.

For a continuous map $F : D \rightarrow \mathbb{R}^m$ and a prescribed error body K , an admissible projected relaxation R must satisfy

$$\{(x, F(x)) : x \in D\} \subset R \subset \{(x, w) : x \in D, w - F(x) \in K\}.$$

Thus the approximation retains every input-output pair and controls the output error at the same input. We minimize the number of general integer coordinates in an arbitrary convex lift of any such R , denoting the result by p_{conv} . Integer ranges, finite continuous dimension, and real coefficients are unrestricted; the lift need not be closed. The comparison quantity p_{bin} restricts the lift to linear constraints and binary coordinates, so $p_{\text{conv}} \leq p_{\text{bin}}$. When we give a polynomial-time algorithm producing a formulation of polynomial rational encoding length, we report its binary count separately as p_{out} .

The minimum dimension of a convex integer representation is already the *MICP rank* of Lubin et al. [25, Definition 4.3], who also prove a parity-based lower bound. Our accuracy-dependent question minimizes over all intermediate graph relaxations R , and allows the broader class of

nonclosed convex lifts. On the constructive side, logarithmic encodings of bounded polyhedral unions and compact quadratic approximations are established [5, 39]. The issue addressed here is whether such constructions use close to the fewest integer coordinates allowed by the geometry, even when compared with general integers and arbitrary convex constraints.

This comparison separates three questions that matter in formulation design: the integer count required by accuracy, the benefit of using nonlinear convex constraints or unbounded integer ranges, and the cost of writing down a formulation. In particular, an asymptotically optimal count can coexist with an impractically long description. Our results provide benchmarks for these resources and identify when one discretization can serve several nonlinear expressions. They concern approximation of the entire graph on the stated domain; restricting that domain by additional nonlinear equations can change the lower bounds. Small integer dimension also does not imply fast optimization or an ideal linear relaxation.

1.1 Main contributions

The principal developments are an exact quadratic precision law, a finite quadratic comparison uniform over coefficients and outputs, and transfers from scalar approximation to minimum integer dimension.

An algebraic precision law for quadratic systems. Let F be a fixed real quadratic system on a full-dimensional bounded box $B \subset \mathbb{R}^n$, and let $\mathcal{H}$ be the span of its Hessians. With componentwise absolute error $\varepsilon > 0$, theorem 4.4 proves

$$p_{\text{conv}}(\varepsilon), p_{\text{bin}}(\varepsilon) = \frac{1}{2} \text{ncrank}(\mathcal{H}) \log_2(1/\varepsilon) + O_{F,B}(1) \quad (\varepsilon \downarrow 0). \quad (1)$$

Both minima have this expansion; they need not agree at a fixed tolerance. Here noncommutative rank is an established matrix-space invariant [19, 23]. One useful characterization is

$$\text{ncrank}(\mathcal{H}) = n - \max_{U \subseteq \mathbb{R}^n} (\dim U - \dim \mathcal{H}(U)), \quad \mathcal{H}(U) = \text{span}\{Hu : H \in \mathcal{H}, u \in U\},$$

where the maximum is over linear subspaces. The maximum records the largest loss of dimension forced by all Hessians on a common subspace; noncommutative rank subtracts that loss from the input dimension. For symmetric Hessian spaces the real form of this characterization follows from the usual complex one, as shown in lemma 4.2.

The joint invariant is essential. For the map $(u, v) \mapsto u \times v$ on six inputs, every scalar combination of the three outputs has Hessian rank at most four, whereas the Hessian space has noncommutative rank six. The coefficient in (1) is therefore three, which no single scalar combination detects. To the best of our knowledge, the identification of half the noncommutative rank with the minimum integer-dimension precision coefficient for quadratic graph approximation is new. The algebraic invariant and its algorithms are not new; the contribution is the lower bound for all admissible convex lifts together with a matching binary linear construction.

The proof makes the connection explicit. Contacts with the exact graph whose integer witnesses have the same parity have admissible midpoints. Bounds on the covariance and volume of these contacts limit how much of the input domain one parity class can cover. Symmetric shrinking supplies a coordinate decomposition with discretization depths proportional to $\log_2(1/\varepsilon)$, with coefficients in $\{0, 1/2, 1\}$ summing to $\text{ncrank}(\mathcal{H})/2$. Shared product relaxations then attain the lower coefficient. Scalar quadratic rank, one-sided inertia, and fractional vertex-cover formulas for bilinear interaction graphs are related exact cases. Local/global Hessian-rank bounds, constant scalar Hessian rank, and positive perspectives extend parts of the analysis beyond quadratics (theorems 5.3 and 5.4 and proposition 5.5).

A finite-accuracy benchmark and a rational construction. An asymptotic coefficient does not describe accuracy thresholds caused by small or highly unequal coefficients. For quadratic outputs with Hessians H_j on $[0, 1]^n$ and positive component tolerances ε_j , define

$$D_* = \max\{\det P : 0 \preceq P \preceq I, \operatorname{tr}(H_j P H_j P) \leq \varepsilon_j^2 \forall j\}, \quad \Phi = -\frac{1}{2} \log_2 D_*.$$

Theorem 6.1 places both minima within $O(n \log(n+1))$ of Φ , with constants independent of the coefficients, tolerances, and number of outputs. Theorem 7.3 constructs, for rational data, a rational MILP with $p_{\text{out}} \leq p_{\text{conv}} + O(n \log(n+1))$ in polynomial time and encoding length. The determinant constraints need not be Euclidean convex; the algorithm uses their geodesic convexity, rational approximations, and explicit feasibility repairs.

The contribution is the common finite benchmark for arbitrary convex integer lifts and binary linear constructions, including the rational algorithm. Anisotropic interpolation, determinant optimization, and geodesic optimization have their own established theories, discussed below. Further results give finite certificates, output-body and nonlinear-input reductions, and sharper positive semidefinite block and diagonal bounds. A thin-domain example shows why a containing box can misrepresent a correlated input domain. A complexity reduction also limits polynomial-time additive approximation of the minimum count: achieving error $O(n^{1-\delta})$ for any fixed $\delta \in (0, 1)$ on all rational quadratic instances would imply $P = NP$.

Scalar and vector comparisons with controlled descriptions. For every continuous convex scalar function on a compact interval and every positive absolute tolerance, lemma 11.1 proves

$$p_{\text{conv}} \leq p_{\text{bin}} \leq p_{\text{conv}} + 2.$$

Thus general integer ranges and nonlinear convex constraints save at most two integer coordinates in this setting. The bound uses real coefficients and does not control continuous formulation size. A separate result, theorem 12.4, constructs a rational MILP within eleven binaries of p_{conv} for every rational convex polynomial on $[0, 1]$ given in dense encoding and every positive rational tolerance. Its running time and total encoding length are polynomial in the input length, including the tolerance. This requires compact access to the selected intervals even when their number is exponential in the input length. An accuracy-dependent truncated curvature measure and a hybrid of greedy segmentation and curvature integration provide that access.

For several convex outputs of one input, shared breakpoints are already available [26]. Let r be the dimension of the outputs' span modulo affine functions, called their *curvature rank*. For $r \geq 1$, our comparison bounds the additional binary count by $\lceil \log_2 r \rceil$ plus an absolute constant; affine outputs need no binaries. It also gives a polynomial rational construction for dense polynomial data (theorem 16.6). Selecting a basis from original convex outputs preserves the nonnegative chord gaps needed for this bound. Positive facet scalarizations and positive polar spanners extend the comparison to the stated unconditional error bodies, including a strong-separation-oracle model (theorem 17.2), where the compiled overhead is $17 + 2\lceil \log_2 r \rceil$. The constructed optimization model has an explicit rational inner band. Positive separable polynomial systems admit allocation comparisons and constructions polynomial in sparse binary-exponent encoding (theorem 13.2). Separable vector maps admit a common curvature-basis comparison, with the rational construction stated for dense polynomial data (theorem 18.2).

The limits of these comparisons are part of the results. Fixed-degree convex product graphs have exact counts n and $\lceil n \log_2 3 \rceil$; a nonconvex scalar polynomial family has two general integers and growing binary dimension (theorems 20.1 and 20.2). A tilted error body transfers this gap to componentwise convex outputs while its condition number grows. A root graph has a four-bit

formulation but requires long rational MILP descriptions, whereas a short rational mixed-integer second-order cone description exists (theorems 15.2 and 15.4). These examples show why count, rational description, input dimension, and error geometry must be specified separately.

1.2 Relationship to prior work

Table 1 summarizes the closest comparisons. Its last column describes results proved here; it does not attribute novelty to all of their ingredients.

Prior work	Established result	Development in this paper
Lubin et al. [25]	Minimum MICP rank and midpoint lower bound	Accuracy-dependent contact-volume bounds for all intermediate graph relaxations
Beach et al. [5], Garg et al. [19], Ivanyos et al. [23]	Compact quadratic approximations; noncommutative rank and capacity theory	Exact joint quadratic coefficient and uniform finite covariance comparison
Codsi et al. [13], Rebennack and Kallrath [32], Simchowicz et al. [35]	Minimum-breakpoint models, chord estimates, and adaptive or greedy approximation	Two-bit scalar comparison and eleven-bit polynomial rational construction
Awerbuch and Kleinberg [4], Lyu et al. [26]	Common SOS2 encodings and barycentric spanners	Curvature-rank count comparisons with box, facet, and oracle error models

Table 1: Established methods and the complete graph-approximation comparisons developed here.

Representability and formulation size. Lubin et al. [25, Definition 4.3 and Lemma 4.1] define minimum MICP rank and show that w pairwise midpoint-incompatible points require at least $\lceil \log_2 w \rceil$ integer variables. We use the same parity mechanism, then control whole contact classes by covariance or chord geometry. Taking closures of contacts, rather than of the original lift, makes the volume arguments valid without measurable witness selections or closed lifts. On the upper-bound side, Vielma [39, Proposition 1] embed finite unions of bounded rational polyhedra using distinct binary codes. Our elementary version, lemma 2.3, also permits real coefficients. This provides an encoding once the local graph bands have been chosen; it does not by itself bound their number or the size of an implicit family.

Mixed-integer extension complexity likewise distinguishes integer count from linear formulation size. Cevallos et al. [12] derive lower bounds on integer count when the number of linear inequalities is restricted, using formulations whose projected mixed-integer hull represents a combinatorial polytope; Schade et al. [34] strengthen these bounds for stable set and knapsack. Those size measures count inequalities, rather than rational encoding bits. Our accuracy-dependent graph model instead requires the projected set itself to stay in the prescribed error band, and its minimum integer count allows arbitrary convex lifts without a size restriction.

Quadratic discretization and matrix-space geometry. Beach et al. [5] give quadratic approximations whose binary, row, and continuous-variable counts grow logarithmically in reciprocal error. Product envelopes [27], multiparametric disaggregation [37], and its refinements [6, 7] supply related shared discretization tools. Accordingly, logarithmic growth and binary product linearization are antecedents of our constructions. Garg et al. [19] develop operator-scaling and capacity theory, and

Ivanyos et al. [23] give constructive polynomial-time noncommutative-rank certificates. We import their algebraic results; the symmetric coordinate allocation and contact-volume arguments connect them to graph accuracy.

Anisotropic interpolation uses Hessian information to select local element shapes [11]. Classical determinant maximization treats linear matrix inequality constraints [38]. Our covariance constraints are generally quadratic in P , and the asserted approximation is to minimum integer dimension over arbitrary convex lifts. Geodesic optimization on positive definite matrices [36, 44] and convex-body separation/optimization [20] are established tools. We supply the rational error bounds, feasibility repairs, and residual certificates needed to use them in the stated bit-complexity model.

Adaptive approximation and coupled outputs. Rebennack and Kallrath [32] formulate minimum-breakpoint continuous piecewise-linear approximation, including shifted under- and overestimators. Simchowicz et al. [35, Lemma 2.1] give the factor-two chord/midpoint estimate and use a local approximation modulus with endpoint corrections in convex regression. Codsí et al. [13, Sections 3–4] reduce corridor fitting to successive maximal segments. For continuously differentiable convex corridor boundaries, their maximal-segment routine uses logarithmically many function and derivative oracle calls in the reciprocal numerical precision, per segment. Although their general model permits discontinuous piecewise-linear functions, the convex-corridor algorithm produces a continuous approximant [13, Remark 3]. These precedents establish adaptive knots, chord estimates, and greedy segmentation. The additional claims here compare to arbitrary convex integer lifts and, for the polynomial classes, control total rational encoding without enumerating all segments. The compiler uses established continuous encodings of Boolean circuits [3] and products involving discrete functions [1].

Lyu et al. [26, Proposition 1] explicitly share one SOS2 constraint among several piecewise-linear functions of the same input. We build on this possibility, comparing the required common partition with p_{conv} through curvature rank and error-body geometry. The determinant selection is a barycentric-spanner method [4]; approximate spanner construction [31] and separation/optimization equivalence [20] are also inherited. For the separation examples, the cap-set estimate is due to Ellenberg and Gijswijt [17], and difference-of-convex decomposition is classical [21]. Our claims concern the stated graph-contact reductions and formulation separations.

The main novelty assertion is therefore the quadratic characterization (1); the finite and nonlinear contributions are the specific comparison and encoding theorems above. We do not claim optimality of every additive constant or a characterization of all smooth and vector maps. In particular, a universal additive binary/general-integer gap for one-input convex vectors with growing output dimension and box error remains open here.

The paper proceeds from contact geometry and quadratic precision laws to finite quadratic accuracy and rational constructions, then scalar convex geometry and coupled vector comparisons. Relative error and the root-power encoding barrier clarify the dependence on the accuracy model and coefficient representation. Each section states its own hypotheses and defines the rank or allocation benchmark it uses; these quantities are not interchangeable.

2 Formulation dimension and convex contact geometry

We study the number of auxiliary integer coordinates needed to retain a nonlinear graph with a prescribed vertical error. This quantity measures one resource of a formulation. It does not measure the time needed to optimize over that formulation. In particular, an upper bound on integer dimension need not give a small linear description, and a small linear description need not

have short rational coefficients. We state these additional properties separately whenever they are available.

2.1 The representation model

Let $D \subset \mathbb{R}^n$ be compact and convex, and let $F : D \rightarrow \mathbb{R}^m$ be continuous. Write $\Gamma_D(F) = \{(x, F(x)) : x \in D\}$. A compact, convex set $K \subset \mathbb{R}^m$ with $K = -K$ and nonempty interior is called an *error body*. The basic example is $K = \prod_{j=1}^m [-\varepsilon_j, \varepsilon_j]$, with $\varepsilon_j > 0$.

Definition 2.1. A graph relaxation with error body K is a set $R \subset D \times \mathbb{R}^m$ satisfying

$$\Gamma_D(F) \subset R \subset \{(x, w) : x \in D, w - F(x) \in K\}.$$

A convex integer lift of integer dimension p represents R as

$$R = \{(x, w) : \exists a \in \mathbb{R}^q, z \in \mathbb{Z}^p, (x, w, a, z) \in \mathcal{C}\}, \quad \mathcal{C} \text{ convex}, \quad (2)$$

where q is finite but unrestricted. The convex set need not be closed, its coefficients need not be rational, and the integer coordinates need not be bounded. Let $p_{\text{conv}}(F, D; K)$ denote the least such p , or $+\infty$ if none exists. Let $p_{\text{bin}}(F, D; K)$ denote the least number of binary coordinates when $\mathcal{C}$ is a polyhedron and $z \in \{0, 1\}^p$. For the common component tolerance ε , abbreviate these quantities as $p_{\text{conv}}(\varepsilon)$ and $p_{\text{bin}}(\varepsilon)$ when F, D are fixed.

One can equivalently allow inputs outside D and measure error only on D : intersection with $D \times \mathbb{R}^{m+q+p}$ preserves convexity. In the binary linear model this equivalence applies when D is a polytope, which is the case for all linear constructions below. Always $p_{\text{conv}} \leq p_{\text{bin}}$. This notation deliberately reserves p_{bin} for *linear* binary lifts; a lower bound valid for arbitrary convex binary lifts will be stated explicitly when useful.

Adding an affine function to F changes neither minimum: subtract it from w in the lift. Invertible affine changes of input coordinates also preserve the minima when the domain is transformed along with the map. For an invertible linear output map A , the pairs (F, K) and (AF, AK) are equivalent. Restricting a formulation to an affine input slice preserves its integer dimension and gives the same error on that slice. These observations justify all normalizations and restrictions below.

Lemma 2.2 (Integer sections and parity contacts). *For a lift (2), define its projected section at a real index z by*

$$R_z = \{(x, w) : \exists a, (x, w, a, z) \in \mathcal{C}\}.$$

Every R_z is convex, and $(1-t)R_z + tR_{z'} \subset R_{(1-t)z+tz'}$ for $0 \leq t \leq 1$. If the lift has error body K , there are at most 2^p compact sets S_α covering D such that, for every $x, y \in S_\alpha$,

$$\frac{F(x) + F(y)}{2} - F\left(\frac{x+y}{2}\right) \in K. \quad (3)$$

More generally, if exact graph witnesses have integer indices z_i and nonnegative weights λ_i summing to one satisfy $\sum_i \lambda_i z_i \in \mathbb{Z}^p$, then $\sum_i \lambda_i F(x_i) - F(\sum_i \lambda_i x_i) \in K$. For binary lifts, fixing the binary vector gives at most 2^p convex sections whose union is the relaxation.

Proof. Take witnesses for two projected points and average their continuous and index coordinates. Convexity proves the section statements. For $\alpha \in \{0, 1\}^p$, let T_α contain those exact graph inputs with some integer witness congruent to α modulo two. These sets cover D ; they may overlap and need not be measurable. The midpoint of two witnesses in one class has an integer index, so its visible error belongs to K . Set $S_\alpha = \overline{T_\alpha}$, taking closure in D . Continuity of F and closedness of K preserve the inequality for pairs of limits. These compact sets still cover D . The finite-mixture assertion follows by taking the same convex combination of all witnesses. $\square$

The parity mechanism is the midpoint obstruction of Lubin et al. [25, Lemma 4.1]; the proof above specifies the approximation and closure conventions used here. A parity class is generally not a convex section. Convexity of the common lift relates different integer sections, whereas the parity argument extracts only the midpoint condition (3). Keeping this distinction is essential when comparing unrestricted integers with binaries.

Lemma 2.3 (Finite disjunction). *A union of $N \geq 1$ nonempty bounded polyhedra has a linear formulation with $\lceil \log_2 N \rceil$ binaries and unrestricted continuous size. The same construction applies to finitely many polyhedra having a common recession cone.*

Proof. Write $Q_i = \{v : A_i v \leq b_i\}$ and choose distinct codes $c_i \in \{0, 1\}^{\lceil \log_2 N \rceil}$. Use continuous variables $\lambda_i \geq 0, v_i$, and impose

$$\sum_i \lambda_i = 1, \quad A_i v_i \leq \lambda_i b_i, \quad v = \sum_i v_i, \quad z = \sum_i \lambda_i c_i, \quad z \text{ binary}.$$

An integral convex combination of binary vectors has every positive weight on the same code. Thus there is one active member. If Q_i is bounded, $\lambda_i = 0$ forces $v_i = 0$. With a common recession cone, the inactive v_i instead belong to that cone and their sum can be absorbed into the active member. The converse follows by placing weight one on any selected member. $\square$

This is the classical binary-code embedding construction; see Vielma [39, Proposition 1 and Corollary 1] for rational polyhedra with a common recession cone. The proof here also permits real coefficients and does not require extreme points.

2.2 Squares, products, and shared input bits

For a scalar function, write $J_f(x, y) = (f(x) + f(y))/2 - f((x + y)/2)$. If f or $-f$ is μ -strongly convex, then

$$|J_f(x, y)| \geq \frac{\mu}{8} \|x - y\|_2^2. \quad (4)$$

Consequently a parity contact has diameter at most $\sqrt{8\varepsilon/\mu}$. Let ω_d denote the volume of the unit Euclidean ball in $\mathbb{R}^d$. The isodiametric inequality and lemma 2.2 yield, on a d -dimensional domain of volume V ,

$$\varepsilon \geq \frac{\mu}{2} \left(\frac{V}{\omega_d} \right)^{2/d} 2^{-2p/d}. \quad (5)$$

The same estimate holds with 2^p replaced by the number of available integer assignments when that number is finite. Strong convexity follows, for example, from $\nabla^2 f \succeq \mu I$ throughout the domain.

Corollary 2.4 (C^2 strong-curvature law). *Let $B \subset \mathbb{R}^d$, $d \geq 1$, be a compact full-dimensional box, and let f be real C^2 on a neighborhood of B . If f or $-f$ is μ -strongly convex on B for some $\mu > 0$, then each of $p_{\text{conv}}(\varepsilon)$ and $p_{\text{bin}}(\varepsilon)$ has the expansion*

$$\frac{d}{2} \log_2(1/\varepsilon) + O_{f,B}(1), \quad \varepsilon \downarrow 0.$$

The binary upper bound permits real coefficients and unrestricted continuous size.

Proof. The lower bound is (5). Choose $M > 0$ bounding $\|\nabla^2 f\|_{\text{op}}$ on B and set $h = \sqrt{\varepsilon/(Md)}$. Divide each coordinate interval into intervals of length at most h . There are $N = O_{f,B}(h^{-d})$ rectangular cells as $h \downarrow 0$. On a cell choose c in the cell and put $\ell_c(x) = f(c) + \nabla f(c)^T(x - c)$. Its

diameter is at most $\sqrt{d}h$, and the segment from c to x stays in B , so the integral Taylor formula gives

$$|f(x) - \ell_c(x)| \leq \frac{1}{2}M\|x - c\|_2^2 \leq \frac{1}{2}Mdh^2 = \varepsilon/2.$$

The bounded polyhedron given by that cell and $|w - \ell_c(x)| \leq \varepsilon/2$ contains the cell's exact graph and has vertical error at most ε . Their union therefore is an admissible relaxation. By lemma 2.3 it uses $\lceil \log_2 N \rceil \leq (d/2) \log_2(1/\varepsilon) + O_{f,B}(1)$ binaries. $\square$

We will repeatedly use an elementary linear construction. For $0 \leq u \leq U$ and a binary β , the equality $v = \beta u$ is enforced by

$$0 \leq v \leq U\beta, \quad v \leq u, \quad v \geq u - U(1 - \beta). \quad (6)$$

For $0 \leq s \leq h$ and $0 \leq t \leq k$, use the four inequalities

$$q \geq 0, \quad q \geq ks + ht - hk, \quad q \leq ks, \quad q \leq ht \quad (7)$$

These contain $q = st$ and give $|q - st| \leq hk/4$. Indeed after $s = ha, t = kb$, the upper error is $hk \min\{a(1-b), b(1-a)\} \leq hk/4$; the lower error follows by replacing one of a, b by its complement. This is the classical McCormick relaxation [27]. For a square on $[0, h]$, its specialization

$$q \geq 0, \quad q \geq 2hs - h^2, \quad q \leq hs \quad (8)$$

contains the graph and has absolute error at most $h^2/4$.

To share precision across expressions, write

$$y_i = A_i + \rho_i, \quad A_i = \sum_{\ell=1}^{L_i} 2^{-\ell} \beta_{i\ell}, \quad 0 \leq \rho_i \leq h_i = 2^{-L_i}. \quad (9)$$

Every $y_i \in [0, 1]$ has a representation, including 1, by using the all-one prefix and residual h_i . Empty prefixes are allowed. The identity

$$y_i y_k = A_i y_k + \rho_i A_k + \rho_i \rho_k \quad (10)$$

uses exact binary products in the first two terms and only one relaxed residual product. It therefore has error at most $h_i h_k/4$, including $i = k$ when (8) is used. The same input bits and approximate monomials can serve all outputs. A required monomial uses $O(L_i + L_k + 1)$ continuous variables and inequalities.

Theorem 2.5 (Exact square law). *For $f(x) = x^2$ on $[0, 1]$, every convex integer lift with p integer coordinates has error at least 2^{-2p-2} . A binary linear lift attains this error with p binaries and $O(p+1)$ rows and variables. Thus, for $\varepsilon > 0$,*

$$p_{\text{conv}}(\varepsilon) = p_{\text{bin}}(\varepsilon) = \max \left\{ 0, \left\lceil \frac{\log_2(1/\varepsilon) - 2}{2} \right\rceil \right\}.$$

Proof. Here $J_f(x, y) = (x - y)^2/4$. A compact set covering length at least 2^{-p} has diameter at least 2^{-p} , so the parity cover gives the lower bound. For the upper bound, use (9) in one variable at depth p and (10) for its square. The only error is the residual square error $h^2/4 = 2^{-2p-2}$. At a residual midpoint its upper error equals this quantity, so the bound is attained. Exact graph points lift by selecting their exact residual square values. $\square$

This construction proves attainment directly. Shared binary discretization and residual McCormick relaxations also underlie the constructions of Beach et al. [7]. The same square upper bound follows from an established sawtooth formulation: Beach et al. [5, Section 2.1 and Proposition 1] formulate the dyadic interpolant F_p with p binaries and linear size, where $0 \leq F_p(x) - x^2 \leq e_p = 2^{-2p-2}$. Adding the band $F_p(x) - e_p \leq w \leq F_p(x)$ retains the exact graph with error e_p and the same binary count. See also Beach et al. [6] for enhanced sawtooth relaxations. The lower bound in theorem 2.5 proves that this established count is optimal against arbitrary convex lifts with unrestricted integer ranges.

Lemma 2.6 (Product contact geometry). *Suppose a compact $S \subset \mathbb{R}^2$ satisfies $|(x_1 - y_1)(x_2 - y_2)| \leq \delta$ for every $x, y \in S$. If its coordinate widths are d_1, d_2 , then*

$$d_1 d_2 \leq 5\delta, \quad \text{vol}_2(S) \leq 4\delta \ln 2.$$

Proof. If $d_1 = 0$, both conclusions follow. Otherwise choose points a, b attaining the extreme first coordinates. Their second coordinates differ by at most δ/d_1 . Every other point is at first-coordinate distance at least $d_1/2$ from one of them, hence its second coordinate is within $2\delta/d_1$ of a_2 or b_2 . This gives the width bound. At first coordinate $a_1 + t$, $0 < t < d_1$, the vertical section lies in two intervals, of lengths $2\delta/t$ and $2\delta/(d_1 - t)$. Integration gives

$$\text{vol}_2(S) \leq \int_0^{d_1} \min \left\{ \frac{2\delta}{t}, \frac{2\delta}{d_1 - t} \right\} dt = 4\delta \ln 2.$$

□

For the product xy on $[0, 1]^2$, $J_f(a, b) = (a_1 - b_1)(a_2 - b_2)/4$. The lemma and parity therefore give

$$\frac{1}{16 \ln 2 \cdot 2^p} \leq E_{\min, \text{conv}}(p) \leq E_{\min, \text{bin}}(p) \leq 2^{-p-2}, \quad (11)$$

where each $E_{\min}$ is the infimum achievable error with at most p integer coordinates in its stated class. The upper bound uses depth p in the first input and depth zero in the second in (10). Hence both minimum dimensions are $\log_2(1/\varepsilon) + O(1)$. For $0 < \varepsilon \leq 1/4$ the explicit bounds are

$$\lceil \log_2(1/\varepsilon) - \log_2(16 \ln 2) \rceil \leq p_{\text{conv}}(\varepsilon) \leq p_{\text{bin}}(\varepsilon) \leq \lceil \log_2(1/\varepsilon) \rceil - 2.$$

Their difference is at most two. No exact optimum constant for the whole product graph is claimed.

The width constant cannot be reduced from five to four. Put $s = \sqrt{5} - 2$ and consider

$$(0, (1 - s)/2), \quad (1, (1 + s)/2), \quad ((1 + s)/2, 0), \quad ((1 - s)/2, 1).$$

Both widths are one, while all six pairwise difference products have absolute value s , since $1 - s^2 = 4s$. Any universal width constant must therefore be at least $1/s = 2 + \sqrt{5}$. This obstruction concerns width; it does not identify the best area constant or a realizable optimal contact partition.

Theorem 2.7 (Bilinear interaction graphs). *Let $G = (V, E)$ be a simple graph with n vertices, and approximate all outputs $x_i x_j$, $ij \in E$, on $[0, 1]^V$ with positive tolerances ε_{ij} . For $C > 0$ define*

$$L_C = \min \left\{ \sum_i s_i : s_i \geq 0, \quad s_i + s_j \geq \lceil \log_2(1/(C\varepsilon_{ij})) \rceil_+ \quad (ij \in E) \right\},$$

where $\lceil t \rceil_+ = \max\{t, 0\}$. If s minimizes L_4 , then

$$\lceil L_{20} \rceil \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \sum_i \lceil s_i \rceil \leq L_4 + n. \quad (12)$$

Let $\tau^*(G) = \min\{\sum_i a_i : a_i \geq 0, a_i + a_j \geq 1 \text{ } (ij \in E)\}$ be the fractional vertex-cover number. Then $L_4 - L_{20} \leq \tau^*(G) \log_2 5$. For common $\varepsilon \downarrow 0$, each minimum has asymptotic expansion

$$\tau^*(G) \log_2(1/\varepsilon) + O_G(1).$$

Proof. For any nonempty closed parity contact, let $d_i \leq 1$ be its coordinate widths. Its two-coordinate projections satisfy $d_i d_j \leq 20\varepsilon_{ij}$ by lemma 2.6. A zero width gives zero n -volume. Otherwise $s_i = -\log_2 d_i$ is feasible for L_{20} , so the contact volume is at most $\prod_i d_i \leq 2^{-L_{20}}$. Summing over at most 2^p contacts proves $p \geq L_{20}$.

For the upper bound give coordinate i depth $L_i = \lceil s_i \rceil$ in (9), shared across all incident edges. Equation (10) has error at most $2^{-L_i - L_j}/4 \leq \varepsilon_{ij}$. The formulation has $O(n + |E| + \sum_i \deg(i)L_i)$ continuous variables and rows. Adding $\log_2(5)$ times an optimal fractional cover to an optimizer for L_{20} makes it feasible for L_4 , because each edge demand increases by at most $\log_2 5$. Finally, for $\varepsilon < 1/20$, common demands give $L_C = \tau^*(G) \log_2(1/(C\varepsilon))$, for $C = 4, 20$. $\square$

For rational tolerances this is also an effective rational construction. Replace each L_4 demand by $d_{ij} = \max\{0, \lceil \log_2(1/(4\varepsilon_{ij})) \rceil\}$, computed by integer comparisons with powers of two, and solve the resulting rational linear program. Rounding the demands increases its optimum by at most $\tau^*(G)$. Thus a polynomial-time construction uses at most $L_{20} + n + \tau^*(G)(1 + \log_2 5)$ binaries and has polynomial encoding length in the graph and tolerance data. The real logarithmic linear program in theorem 2.7 is an exact benchmark; computing it exactly is unnecessary for this guarantee.

Stars have coefficient one; $K_{r,s}$ has coefficient $\min\{r, s\}$; a matching of k edges has coefficient k ; an odd cycle on $2k + 1$ vertices has coefficient $k + 1/2$; and K_n , $n \geq 2$, has coefficient $n/2$. Thus shared bits and fractional allocation are both consequential. On a general nondegenerate box replace ε_{ij} by $\varepsilon_{ij}/((u_i - l_i)(u_j - l_j))$. Affine terms introduced by the normalization have no effect.

3 Scalar quadratic rank and one-sided inertia

An indefinite quadratic is not strongly convex on its full domain, but its graph still has the precision exponent of its rank. Cancellation along some directions does not allow large contact volume in all independent directions.

Lemma 3.1 (Indefinite contact volume). *Let $d \geq 1$, let M be a nonsingular real symmetric $d \times d$ matrix, and let $\delta \geq 0$. If a compact set $S \subset \mathbb{R}^d$ satisfies $|\frac{1}{2}(x - y)^T M(x - y)| \leq \delta$ for all $x, y \in S$, then*

$$\text{vol}_d(S) \leq \frac{2^d (3\sqrt{d}\delta)^{d/2}}{\sqrt{|\det M|}}.$$

Proof. If the affine span has dimension below d , its volume is zero. Otherwise choose $s_0, \dots, s_d \in S$ maximizing $|\det A|$, where $A = [s_1 - s_0, \dots, s_d - s_0]$. The maximum is positive and attained. Replacing column i by $s - s_0$ shows that each coordinate of $A^{-1}(s - s_0)$ has absolute value at most one. Thus $S \subset s_0 + A[-1, 1]^d$ and $\text{vol}_d(S) \leq 2^d |\det A|$. Writing $q(v) = v^T M v/2$, polarization gives for $G = A^T M A$

$$|G_{ii}| \leq 2\delta, \quad |G_{ij}| = |q(s_i - s_0) + q(s_j - s_0) - q(s_i - s_j)| \leq 3\delta.$$

Hadamard's inequality gives $|\det G| \leq (3\sqrt{d}\delta)^d$. Since $\det G = (\det A)^2 \det M$, the conclusion follows. This also excludes a full-dimensional span when $\delta = 0$. $\square$

Theorem 3.2 (Scalar quadratic rank). *Let $B = \prod_i [l_i, u_i]$ be a bounded box with $l_i < u_i$, and $f(x) = x^T H x / 2 + a^T x + b$, where $H = H^T$ has rank $r > 0$. Then each of $p_{\text{conv}}(\varepsilon), p_{\text{bin}}(\varepsilon)$ equals*

$$\frac{r}{2} \log_2(1/\varepsilon) + O_{H,B}(1), \quad \varepsilon \downarrow 0. \quad (13)$$

A binary upper formulation uses $O_{H,B,n}(1 + \log(1/\varepsilon))$ rows and variables. For any nonsingular principal submatrix $H[I, I]$ of order r , every admissible p -integer lift satisfies

$$\varepsilon \geq \frac{|\det H[I, I]|^{1/r} (\prod_{i \in I} (u_i - l_i))^{2/r}}{48\sqrt{r}} 2^{-2p/r}.$$

If $r = 0$, the exact graph is affine and both minima are zero.

Proof. A symmetric rank- r matrix has a nonsingular principal submatrix of order r : the sum of its principal minors of that order is the product of its nonzero eigenvalues. Fix complementary coordinates in their intervals. The remaining Hessian is $H[I, I]$ and the slice has volume $V_I = \prod_{i \in I} (u_i - l_i)$. On any parity contact,

$$J_f(x, y) = \frac{1}{8} (x - y)^T H[I, I] (x - y),$$

so lemma 3.1 applies with $\delta = 4\varepsilon$. The sum of at most 2^p contact volumes covers the slice. Rearranging $V_I \leq 2^p 2^r (12\sqrt{r}\varepsilon)^{r/2} / \sqrt{|\det H[I, I]|}$ gives the explicit bound.

For the upper bound, diagonalize the quadratic and normalize its r nonconstant linear coordinates to $y_j \in [0, 1]$ on B . Then $f = \ell + \sum_{j=1}^r c_j y_j^2$, where ℓ is affine and $c_j \neq 0$. Let $A = \sum_j |c_j|$. Apply theorem 2.5 independently at common depth $L = \max\{0, \lceil \frac{1}{2} \log_2(A/(4\varepsilon)) \rceil\}$. The output equation $w = \ell + \sum_j c_j t_j$ contains the graph and has error at most $A 2^{-2L-2} \leq \varepsilon$. It uses rL binaries and $O(r(L+1))$ auxiliary variables and rows, in addition to the affine coordinate equations and original box. This proves the upper bound. $\square$

The formal upper construction uses a positive enclosing radius rather than requiring exact extrema of each spectral coordinate. For an eigenvector v_j , put

$$R_j = 1 + \sum_i |v_{ji}| \max\{|l_i|, |u_i|\}, \quad y_j = \frac{v_j^T x + R_j}{2R_j}, \quad c_j = \frac{\lambda_j}{2} (2R_j)^2.$$

The box inequalities imply $0 \leq y_j \leq 1$. After omitting zero eigenvalues, the actual depth- L graph system has rL binary coordinates, $r(3 + 2L)$ real auxiliary coordinates, and $2n + r(11 + 10L) + 2$ inequality rows. Affine substitution adds no rows; these counts include the original box and the two rows for the output equality. Inputs and the output are not counted as auxiliary coordinates. The enclosing radii change the constant A , not the rank coefficient.

For example $(x_1 + \dots + x_n)^2$ has rank one, independently of its number of monomials. Conversely the inertia of a nonsingular indefinite form does not change its whole-graph coefficient. Approximating a scalar sum and approximating every monomial in that sum are different tasks.

3.1 Epigraphs and hypographs

An epigraph relaxation contains every (x, w) with $x \in B, w \geq f(x)$ and admits only $w \geq f(x) - \varepsilon$. Define $p_{\text{epi,conv}}(\varepsilon)$ and $p_{\text{epi,bin}}(\varepsilon)$ with the same lifting conventions. Hypograph relaxations contain $w \leq f(x)$ and require $w \leq f(x) + \varepsilon$. The allowed output is unbounded in its respective epigraph or hypograph direction.

We first record a continuous linear square underestimator of small size. The underlying dyadic folding identity is the square approximation of Yarotsky [43, Proposition 2], used in the MIP formulations of Beach et al. [5]. Put $G(t) = \min\{2t, 2(1-t)\}$ on $[0, 1]$, and

$$F_L(t) = t - \sum_{j=1}^L 4^{-j} G^{\circ j}(t).$$

Induction on the dyadic subdivision shows that F_L is the linear interpolant of t^2 at $k2^{-L}$: subdivision subtracts a triangular function of height 4^{-L} at the new midpoints and zero at the old knots. Therefore

$$0 \leq F_L(t) - t^2 \leq e_L := 4^{-L}/4. \quad (14)$$

The epigraph $w \geq F_L(t) - e_L$ has a continuous linear lift of size $O(L+1)$. Indeed replace the equalities $g_j = G(g_{j-1})$ by

$$g_0 = t, \quad 0 \leq g_j \leq 2g_{j-1}, \quad g_j \leq 2(1 - g_{j-1}), \quad w \geq t - \sum_{j=1}^L 4^{-j} g_j - e_L.$$

For fixed t , the weighted sum of g_j is maximized by the exact folding sequence. To check this, optimize successively backwards. If g_k decreases by $\Delta \geq 0$ from its allowed maximum, the loss $4^{-k}\Delta$ exceeds the largest possible gain in the optimized future sum, whose Lipschitz constant is at most $\sum_{j>k} 4^{-j}2^{j-k} < 4^{-k}$. This proves the claim by backward induction; at the final level there is no future sum. Equation (14) then gives a zero-binary epigraph relaxation of error e_L . Using depth $L+1$ gives error 2^{-2L-4} . This is the folding principle underlying the sawtooth epigraph construction [6].

Theorem 3.3 (One-sided inertia law). *For the scalar quadratic in theorem 3.2, let $k_- = n_-(H)$ and $k_+ = n_+(H)$ be its negative and positive inertia. If $k_- > 0$, both epigraph minima have expansion*

$$\frac{k_-}{2} \log_2(1/\varepsilon) + O_{H,B}(1).$$

If $k_- = 0$, both epigraph minima are zero for every $\varepsilon > 0$, and the convex-lift minimum is zero also at exact accuracy. For hypographs, replace k_- by k_+ . Binary linear upper formulations have $O_{H,B,n}(1 + \log(1/\varepsilon))$ rows and variables.

Proof. If $H = 0$, the function is affine and its exact epigraph and hypograph are polyhedra, so all stated minima are zero. For the lower bound when $k_- > 0$, choose an orthonormal basis T of the negative eigenspace and an interior point x_0 of B . For a sufficiently small fixed $\rho > 0$, $x_0 + T[-\rho, \rho]^{k_-} \subset B$. The restricted quadratic is μ -strongly concave for some $\mu > 0$. Two graph witnesses in the same parity class give an admitted midpoint below the true graph by at least $\mu\|s - t\|^2/8$. The one-sided error bound thus yields (5) in dimension k_- and proves the lower rate. For $k_- = 0$, the lower bound is simply $p \geq 0$.

For the upper bound normalize a signed-square decomposition as $f = \ell + \sum_{j \in P} c_j y_j^2 - \sum_{j \in N} d_j y_j^2$, where $c_j, d_j > 0$ and $|N| = k_-$. Approximate each positive square from below by the continuous linear epigraph just constructed, and each negative square from above by the upper half of the graph construction in theorem 2.5. The latter also contains the whole square hypograph if the lower output bound is dropped. Impose

$$w \geq \ell + \sum_{j \in P} c_j t_j - \sum_{j \in N} d_j t_j.$$

Put $A = \sum_{j \in P} c_j + \sum_{j \in N} d_j > 0$ and choose

$$L = \max\{0, \lceil \frac{1}{2} \log_2(A/(4\varepsilon)) \rceil\}.$$

Then $A2^{-2L-2} \leq \varepsilon$, and the construction uses $k_-L = (k_-/2) \log_2(1/\varepsilon) + O_{H,B}(1)$ binaries as $\varepsilon \downarrow 0$. Choosing $t_j = y_j^2$ lifts every exact epigraph point, including arbitrarily large w . If $k_- = 0$, no binaries are used; the exact epigraph is itself convex. Apply the argument to $-f$ for the hypograph statement. $\square$

The formal lower proof also establishes the sharp Euclidean isodiametric inequality for compact sets, without a convexity assumption on those sets. It maximizes a compactly supported radial weighted measure over a compact family of sets with bounded diameter. Reflection polarization would strictly increase this measure if a support point lay outside the half-diameter ball; maximality excludes that case. A sufficiently large finite cutoff transfers a putative volume excess to a weighted-measure excess. Applied to the actual negative-curvature slice, this proves the unit-ball-volume constant in (5), not only its precision exponent.

In the formal epigraph construction, depth L is used for both kinds of square component. The positive-square error $e_L = 4^{-L}/4$ already suffices for the aggregate bound Ae_L ; the sharper depth- $(L+1)$ estimate remains available independently. Positive components use L real auxiliaries and $3L+3$ rows each, and negative components use $2+2L$ auxiliaries and $11+10L$ rows each. Storing one output per component gives at most $r(3+2L)$ real auxiliaries. Including the original box and output inequality, the implemented system has at most $2n + r(11+10L) + 2$ rows; one of the final two rows is the harmless inequality $0 \leq 0$. Output reflection preserves these bounds for the hypograph.

For one product there is an exact one-sided convex-lift formula:

$$E_{\text{epi,conv}}(p) = E_{\text{hyp,conv}}(p) = 2^{-2p-2}. \quad (15)$$

To prove the lower bound restrict to $(x, y) = (t, 1-t)$, where $t(1-t)$ has concavity midpoint gap $(s-t)^2/4$; the interval parity argument of theorem 2.5 applies. For the upper bound use $u = (x+y)/2$, $v = (x-y+1)/2$, so $xy = u^2 - v^2 + v - 1/4$. Represent the positive square epigraph exactly by the convex inequality $t_+ \geq u^2$, and use a depth- p square hypograph for t_- . The inequality $w \geq t_+ - t_- + v - 1/4$ attains error 2^{-2p-2} ; take $u = 1/2$ and v at a dyadic residual midpoint. Hypographs follow from the affine transformation $(x, y, w) \mapsto (x, 1-y, x-w)$. In a purely linear lift, the exact positive square epigraph can be approached with arbitrarily small extra error and no binaries. Exact finite-LP attainment at the threshold in (15) is not asserted.

The formal interval lower bounds use a finite-grid pigeonhole argument: $2^p + 1$ equally spaced graph inputs have only 2^p parity codes. Two witnesses of the same parity give the required midpoint error directly. This proves the exact square and one-sided product thresholds without assuming that a nonclosed lift attains a largest contact diameter. The higher-dimensional scalar proofs instead use compact closures of parity contacts. For each fixed p and every $\delta > 0$, the product construction also gives an actual finite binary linear lift with error at most $2^{-2p-2} + \delta$ and the same p binaries.

Zero binaries do not mean bounded continuous description size. Every linear extended formulation with M inequalities approximating the square epigraph within ε satisfies

$$M \geq \frac{1}{2} \log_2(1/\varepsilon) - 1. \quad (16)$$

To see this, write the projected lower boundary as the maximum of its q affine lower facets. On an active interval of length h , linearity at its endpoints and the bounds $x^2 - \varepsilon \leq g(x) \leq x^2$ imply $h^2/4 \leq \varepsilon$. Hence $q \geq 1/(2\sqrt{\varepsilon})$. Every projected facet pulls back to a distinct exposed face of the

extended polyhedron, which has at most 2^M faces, since each face is determined by a subset of tight inequalities. Equalities and lineality do not affect this count. This proves (16); the folding construction gives the matching logarithmic order.

4 The exact precision exponent for quadratic systems

Let $F = (f_1, \dots, f_m)$ with

$$f_j(x) = \frac{1}{2}x^T H_j x + a_j^T x + b_j, \quad H_j = H_j^T,$$

on a bounded full-dimensional box $B \subset \mathbb{R}^n$. The simultaneous problem depends on the Hessian *matrix space*, rather than on the largest rank of a scalar combination of its members.

The proof has two geometric parts. A rank witness limits the volume of every parity contact, yielding a lower bound even for lifts with unbounded integer ranges. A shrinking subspace then supplies coordinates in which shared input bits attain the same coefficient. The algebraic characterizations used to obtain these witnesses are established results; the contact-volume bound and its matching precision allocation connect them to the formulation minimum studied here.

4.1 Algebraic structure of a symmetric matrix space

Let $\mathcal{H} = \text{span}_{\mathbb{C}}\{H_1, \dots, H_m\}$. Its noncommutative rank r is the rank of the pencil $\sum_j t_j H_j$ over the complex free skew field, whose indeterminates t_j do not commute. This is a finite-dimensional invariant of the coefficient matrices; the actual optimization inputs remain ordinary real variables. We use the established rank and evaluation characterizations [19, Theorems 1.4 and 1.17]:

$$n - r = \max_{U \subset \mathbb{C}^n} (\dim U - \dim \mathcal{H}(U)), \quad \mathcal{H}(U) = \sum_j H_j U, \quad (17)$$

$$r = n \iff \exists d, B_j \in \mathbb{C}^{d \times d} : \sum_j B_j \otimes H_j \text{ is invertible.} \quad (18)$$

In particular, (17) is an equivalent definition that requires only ordinary subspaces and matrix multiplication. A subspace U is *shrunk* when $\dim \mathcal{H}(U) < \dim U$; the maximum dimension deficit, allowing $U = \{0\}$, is exactly $n - r$. No manipulation of noncommutative rational expressions is needed in the geometric proofs below. The positive-capacity characterization, due to Gurvits, is discussed in Garg et al. [19, Section 1.5, concluding discussion on p. 237, and Section 2]:

$$r = n \iff \kappa > 0, \quad \kappa = \inf_{P \succ 0} \frac{\det(\sum_j H_j P H_j^*)}{\det P}. \quad (19)$$

The infimum in (19) is over positive definite Hermitian matrices. The involution on the free skew field conjugates scalars and fixes the free variables, reversing the order of multiplication [40, Section 2.1]. Thus a pencil with real symmetric coefficients is Hermitian over that division ring. These algebraic characterizations are imported results. The geometric and formulation consequences are proved below.

Lemma 4.1 (Principal compression). *A Hermitian matrix of rank r over a division ring with involution has an invertible principal submatrix of order r . Consequently there is a set I of r original coordinates such that $\{H_j[I, I]\}_j$ has full noncommutative rank r .*

Proof. Induct on matrix order. A nonzero diagonal entry is an invertible one-by-one Hermitian pivot. If all diagonal entries vanish but the matrix is nonzero, a nonzero off-diagonal entry a supplies the principal pivot

$$D = \begin{pmatrix} 0 & a \\ a^* & 0 \end{pmatrix}, \quad D^{-1} = \begin{pmatrix} 0 & (a^*)^{-1} \\ a^{-1} & 0 \end{pmatrix}.$$

After placing a pivot first write the matrix as $\begin{pmatrix} D & E \\ E^* & G \end{pmatrix}$. Its Schur complement $S = G - E^* D^{-1} E$ is Hermitian and block elimination gives $\text{rank } M = \text{order}(D) + \text{rank } S$. Induction selects a principal invertible submatrix $S[J, J]$ of order $\text{rank } S$. The original principal submatrix on the pivot indices and J has this Schur complement, so it is invertible. The zero matrix uses the empty submatrix. Apply the lemma to the free-field pencil to obtain I . $\square$

Lemma 4.2 (Real symmetric shrinking). *There are mutually orthogonal real subspaces Z, W, Q with $\mathbb{R}^n = Z \oplus W \oplus Q$ such that*

$$H_j Z \subset W \quad \text{for every } j, \quad 2 \dim W + \dim Q = r. \quad (20)$$

In orthonormal coordinates ordered as Z, W, Q , each Hessian has form

$$\begin{pmatrix} 0 & B_j & 0 \\ B_j^T & C_j & D_j \\ 0 & D_j^T & E_j \end{pmatrix}. \quad (21)$$

Proof. Put $d = n - r$ and $g(U) = \dim U - \dim \mathcal{H}(U)$. The identities $\mathcal{H}(U + V) = \mathcal{H}(U) + \mathcal{H}(V)$ and $\mathcal{H}(U \cap V) \subset \mathcal{H}(U) \cap \mathcal{H}(V)$ show that $g(U + V) + g(U \cap V) \geq g(U) + g(V)$. A maximizing complex subspace exists because the deficiencies take finitely many integer values. Since the matrices are real, its conjugate also has deficiency d . Supermodularity forces both the sum and intersection with its conjugate to have deficiency d . The sum is conjugation invariant and hence is the complexification of a real space U_0 : for any member, its real and imaginary parts belong to the sum and span it over $\mathbb{C}$. The image likewise complexifies its real image. Consequently a real pair $U, V = \sum_j H_j U$ has $\dim U - \dim V = d$.

Set $Z = U \cap V^\perp$ and $W = V \cap U^\perp$. The restricted orthogonal projections $P_V : U \rightarrow V$ and $P_U : V \rightarrow U$ are adjoints, and therefore have the same rank. Their kernels are Z, W , so $\dim Z - \dim W = d$. For $z \in Z$ we have $H_j z \in V$, and for $u \in U$

$$u^T H_j z = (H_j u)^T z = 0.$$

Thus $H_j z \in V \cap U^\perp = W$. Moreover $Z \perp W$. Let $Q = (Z + W)^\perp$. The dimension identity gives $2 \dim W + \dim Q = n - d = r$, and symmetry gives the displayed zero blocks. $\square$

Assign an exponent α_i equal to zero on Z , one on W , and one half on Q . Every nonzero quadratic coefficient has $\alpha_i + \alpha_k \geq 1$, including square coefficients with $i = k$, while

$$\sum_i \alpha_i = r/2. \quad (22)$$

This converts an algebraic rank witness into a precision allocation.

4.2 Covariance bounds for contact volume

Lemma 4.3 (Contact covariance). *Let $S \subset \mathbb{R}^s$ be compact with positive volume, and let Σ be the covariance matrix of its uniform distribution. For real symmetric G_j , suppose $|\frac{1}{2}(x-y)^T G_j (x-y)| \leq \delta_j$ for all $x, y \in S$. Then $\Sigma \succ 0$ and*

$$\sum_j \text{tr}(G_j \Sigma G_j \Sigma) \leq \sum_j \delta_j^2, \quad (23)$$

$$\text{vol}_s(S) \leq \omega_s(s+2)^{s/2} \sqrt{\det \Sigma}. \quad (24)$$

Proof. A singular covariance would place almost every point in an affine hyperplane, contradicting positive volume. Center independent uniform points X, Y at zero. Expanding the square and using independence gives

$$\mathbb{E} \left[\left(\frac{1}{2}(X-Y)^T G_j (X-Y) \right)^2 \right] = \frac{1}{2} \mathbb{E}[(X^T G_j X)^2] + \frac{1}{2} (\mathbb{E}[X^T G_j X])^2 + \text{tr}(G_j \Sigma G_j \Sigma).$$

Mixed terms containing $X^T G_j Y$ vanish after conditioning on one centered factor. This proves (23). For (24), whiten the centered set by $\Sigma^{-1/2}$. Its mean squared norm is s . Among sets of a fixed volume, a centered ball minimizes the integral of squared norm: subtracting the ball's integral pairs exterior points with missing interior points, and the former have no smaller squared norm. A ball of radius R has mean squared norm $sR^2/(s+2)$. Thus the whitened set has volume at most $\omega_s(s+2)^{s/2}$, proving the result. $\square$

Suppose the tuple G_j has full noncommutative rank s . Write $T(P) = \sum_j G_j P G_j$ and let $\kappa > 0$ be its capacity. The ordinary eigenvalue arithmetic–geometric mean inequality implies

$$\text{tr}(\Sigma T(\Sigma)) \geq s[\det \Sigma \det T(\Sigma)]^{1/s} \geq s\kappa^{1/s}(\det \Sigma)^{2/s}. \quad (25)$$

Here $T(\Sigma)$ is positive definite by positive capacity, and the first inequality is applied to $\Sigma^{1/2} T(\Sigma) \Sigma^{1/2}$. Combining lemma 4.3 with $\delta_j = 4\varepsilon$ yields

$$\text{vol}_s(S) \leq C_s \varepsilon^{s/2}, \quad C_s = \omega_s(s+2)^{s/2} \left(\frac{16m}{s\kappa^{1/s}} \right)^{s/4}. \quad (26)$$

Zero-volume sets satisfy this inequality as well.

There is also an elementary way to obtain a positive constant in (25), without using capacity. For an orthogonal O define $M_{ab}(O) = \sum_j ((O^T G_j O)_{ab})^2$. If the support bipartite graph of $M(O)$ failed Hall's condition, a coordinate column subspace would be mapped by every $O^T G_j O$ into a smaller row subspace. This would be a real shrunk subspace, contradicting (17). Hence its permanent is positive. Continuity and compactness of the orthogonal group give $\mu = \min_O \text{per } M(O) > 0$. If $\Sigma = O \text{diag}(\lambda) O^T$, some permutation π has $\prod_a M_{a, \pi(a)} \geq \mu/s!$. Therefore

$$\text{tr}(\Sigma T(\Sigma)) = \sum_{a,b} M_{ab} \lambda_a \lambda_b \geq s(\mu/s!)^{1/s} (\det \Sigma)^{2/s}.$$

Thus the qualitative rate uses only the shrinking characterization. The constant $\mu/s!$ need not equal capacity; capacity remains useful for explicit rational lower estimates.

Theorem 4.4 (Quadratic-system precision law). *For a fixed real quadratic system F on a full-dimensional bounded box B , let r be the noncommutative rank of its Hessian space. If $r > 0$, each minimum has expansion*

$$p_{\text{conv}}(\varepsilon), p_{\text{bin}}(\varepsilon) = \frac{r}{2} \log_2(1/\varepsilon) + O_{F,B}(1). \quad (27)$$

The binary upper formulation has $O_{F,B,n,m}(1 + \log(1/\varepsilon))$ rows and variables, with real coefficients. If $r = 0$, all outputs are affine and both minima are exactly zero. For $r > 0$ and a principal compression I from lemma 4.1, write $V_I = \prod_{i \in I} (u_i - l_i)$ and let $\kappa_I > 0$ be the capacity of the compressed tuple. Every admissible p -integer lift satisfies

$$\varepsilon \geq \frac{\sqrt{r\kappa_I^{1/r}/m} V_I^{2/r}}{4(r+2)\omega_r^{2/r}} 2^{-2p/r}. \quad (28)$$

Proof. If $r = 0$, all Hessians vanish, so the exact graph has a zero-binary linear formulation. Hence assume $r > 0$. Fix the coordinates outside I in their intervals. The restricted quadratic system has Hessians $G_j = H_j[I, I]$ of full noncommutative rank r . On a parity contact its midpoint errors give $|\frac{1}{2}(x - y)^T G_j (x - y)| \leq 4\varepsilon$. Equations (26) and lemma 2.2 imply $V_I \leq 2^p C_r \varepsilon^{r/2}$. Rearranging proves (28) and the lower rate.

For the upper bound take the coordinates of lemma 4.2, enclose the transformed box in a box, and normalize its coordinates to $[0, 1]^n$. Retain the original box inequalities and coordinate identities. Diagonal rescaling and translation do not create quadratic coefficients in zero blocks. Write each output as an affine function plus $\sum_{i \leq k} c_{j,ik} y_i y_k$, and put $C = \max_j \sum_{i \leq k} |c_{j,ik}| > 0$. Take $T = \max\{0, \log_2(C/(4\varepsilon))\}$ and depths $L_i = \lceil \alpha_i T \rceil$ in (9). For every nonzero monomial $2^{-L_i - L_k} \leq 2^{-T}$, by (22) and its preceding coefficient condition. Shared residual-product relaxations therefore give every output error at most $C 2^{-T}/4 \leq \varepsilon$. Exact graph points lift with exact residual values. The binary count is

$$\sum_i L_i \leq \frac{r}{2} T + n,$$

and at most $n(n+1)/2$ monomials are needed. The size statement follows from (10). $\square$

Equations such as (27) assert the same asymptotic expansion for two minima; they do not assert equality at every accuracy. Also, a graph on a lower-dimensional feasible set need not obey the full-box lower bound. In particular the theorem cannot be transferred to the zero set of the output equations without a separate argument.

4.3 A certificate and a strict scalar-rank gap

If $\sum_j H_j^2 = cI_n$ with $c > 0$, one can replace capacity by an explicit constant. In a covariance eigenbasis set $M_{ab} = \sum_j ((O^T H_j O)_{ab})^2$. Its row and column sums are c . Weighted arithmetic-geometric mean with weights $M_{ab}/(cn)$ gives

$$\sum_j \text{tr}(H_j \Sigma H_j \Sigma) \geq cn(\det \Sigma)^{2/n}.$$

Thus lemma 4.3 gives the finite bound

$$\varepsilon \geq \frac{\sqrt{cn/m} \text{vol}_n(B)^{2/n}}{4(n+2)\omega_n^{2/n}} 2^{-2p/n}. \quad (29)$$

Together with uniform half-depths in all coordinates, this proves coefficient $n/2$ directly. It also certifies full noncommutative rank: for a real or complex subspace U with image contained in V , writing P_U, P_V for orthogonal projections gives

$$c \dim U = \sum_j \|H_j P_U\|_F^2 \leq \sum_j \|P_V H_j\|_F^2 = c \dim V,$$

where $P_V H_j P_U = H_j P_U$ and symmetry is used in the second total. There is no shrunk subspace.

For $F(x, y) = x \times y$, $x, y \in [0, 1]^3$, let

$$A(a) = \begin{pmatrix} 0 & a_3 & -a_2 \\ -a_3 & 0 & a_1 \\ a_2 & -a_1 & 0 \end{pmatrix}, \quad H(a) = \begin{pmatrix} 0 & A(a) \\ -A(a) & 0 \end{pmatrix}.$$

Then $a^T F = x^T A(a) y$ has Hessian $H(a)$. Since $A(a)^T A(a) = \|a\|^2 I - aa^T$, every nonzero scalar combination has rank four. Yet $\sum_{j=1}^3 H(e_j)^2 = 2I_6$, so both simultaneous minimum dimensions have expansion $3 \log_2(1/\varepsilon) + O(1)$. The explicit lower bound is $\varepsilon \geq (16\omega_6^{1/3})^{-1} 2^{-p/3}$, where $\omega_6 = \pi^3/6$. For k independent copies the coefficient is $3k$, while half the maximum scalar rank is $2k$. Thus maximum scalar rank misses an arbitrarily large additive amount in the precision coefficient. The alternating matrix space itself is classical; the distinction here is between its simultaneous graph and any single scalar output.

The two independently proved laws also identify the algebraic invariant for an interaction graph: comparing theorems 2.7 and 4.4 as $\varepsilon \downarrow 0$ gives $\text{ncrank span}_{ij \in E} \{e_i e_j^T + e_j e_i^T\} = 2\tau^*(G)$. This identity follows by equality of the precision coefficients and needs no additional graph-rank theorem.

4.4 Rational construction and its scope

For rational data there is an effective coordinate transformation. The algorithm for noncommutative rank due to Ivanyos et al. [23, Theorem 1.5 and Lemma 5.3] returns a maximal rational shrunk subspace with polynomial bit complexity and preserves its rank under field extension. Rational bases of Z, W, Q in lemma 4.2 suffice: they need not be orthonormal, because their spaces are orthogonal. Rational nullspace computations therefore preserve (21) without introducing square roots.

More precisely, for a rational quadratic system and rational full-dimensional box with total binary encoding length s , there are universal polynomials P, Q such that for every integer $k \geq 0$

$$\frac{r}{2}k - Q(s) \leq p_{\text{conv}}(2^{-k}) \leq p_{\text{bin}}(2^{-k}) \leq \frac{r}{2}k + P(s). \quad (30)$$

A deterministic algorithm constructs the upper relaxation in time and total encoding length polynomial in $s + k$. Its individual coefficient lengths are $\text{poly}(s) + O(k)$, and its number of rows and variables is $\text{poly}(s)(k + \text{poly}(s) + 1)$. The detailed bit bounds, including a uniform lower estimate from the integral capacity bound [19, Theorem 2.18], are developed with the finite-accuracy quadratic constructions later in the paper. This statement concerns construction, not polynomial-time optimization of the resulting MILP. For arbitrary real data theorem 4.4 makes no claim about finding or encoding the coordinate change in a bit model.

5 Smooth maps, varying null spaces, and perspectives

For a smooth nonlinear map, Hessians at different inputs need not have a common null space or a common optimal precision allocation. We therefore distinguish two ranks. For $F = (f_1, \dots, f_m)$ smooth on a neighborhood of a full-dimensional compact box B , set

$$r_{\text{loc}} = \max_{x \in \text{int } B} \text{ncrank span}_j \{\nabla^2 f_j(x)\},$$

$$r_{\text{all}} = \text{ncrank span}_{x \in B, j} \{\nabla^2 f_j(x)\}.$$

The global span is finite-dimensional, being a space of symmetric $n \times n$ matrices. These are matrix-space ranks, not dimensions of the spans, and $0 \leq r_{\text{loc}} \leq r_{\text{all}} \leq n$.

5.1 A local contact-volume estimate

We give the analytic ingredient with its local proof, since it is what allows contacts of arbitrary aspect ratio. The estimate is the classical nondegenerate mixed-Hessian oscillatory-integral bound [22, Theorem 1.1]; see also Wolff [42, Theorem A, pp. 50–51].

Lemma 5.1 (Local oscillatory estimate). *Suppose a real C^∞ phase $\Phi(X, Y)$, with $X, Y \in \mathbb{R}^N$, has invertible mixed Hessian at (X_0, Y_0) . On a sufficiently small product neighborhood, for every smooth compactly supported amplitude a ,*

$$T_\lambda g(X) = \int e^{i\lambda\Phi(X, Y)} a(X, Y) g(Y) dY \quad \text{satisfies} \quad \|T_\lambda\|_{L^2 \rightarrow L^2} \leq C\lambda^{-N/2}, \quad \lambda \geq 1.$$

The constant depends on the fixed phase, amplitude, and neighborhood.

Proof. Let $A_0 = \Phi_{XY}(X_0, Y_0)$ and let $\sigma > 0$ be its least singular value. Shrink a convex product neighborhood until $\|\Phi_{XY} - A_0\| \leq \sigma/2$. The gradient in Y of $\psi(Y) = \Phi(X, Y) - \Phi(Z, Y)$ is the integral of $\Phi_{XY}(Z + t(X - Z), Y)^T (X - Z)$, hence has norm at least $(\sigma/2)|X - Z|$. Every fixed-order Y derivative of ψ is $O(|X - Z|)$, by smoothness and integration in X . The kernel of $T_\lambda T_\lambda^*$ is the integral of $e^{i\lambda\psi(Y)} a(X, Y) \overline{a(Z, Y)}$. For $X \neq Z$ integrate by parts with

$$\mathcal{L} = \frac{\nabla\psi \cdot \nabla_Y}{i\lambda|\nabla\psi|^2}, \quad \mathcal{L}e^{i\lambda\psi} = e^{i\lambda\psi}.$$

The derivative bounds show that each integration gains a factor $O((\lambda|X - Z|)^{-1})$; the amplitude has compact support, so there are no boundary terms. Combining any fixed number M of integrations with the trivial kernel bound gives $|K_\lambda(X, Z)| \leq C_M(1 + \lambda|X - Z|)^{-M}$. Choose $M > N$. Both Schur integrals are $O(\lambda^{-N})$, so Schur's test bounds $\|T_\lambda T_\lambda^*\|$ by that quantity. Taking the square root proves the result. Indicator functions are permitted by L^2 continuity; no boundary regularity of their support is needed. $\square$

Lemma 5.2 (Smooth contact volume). *If the Hessian tuple of F has full noncommutative rank n at an interior point x_0 , then there are a compact box Q about x_0 and constants $C, \varepsilon_0 > 0$ such that every compact $S \subset Q$ with*

$$|f_j(x) + f_j(y) - 2f_j((x + y)/2)| \leq 2\varepsilon \quad (x, y \in S, \ 1 \leq j \leq m)$$

has volume at most $C\varepsilon^{n/2}$ for $0 < \varepsilon \leq \varepsilon_0$.

Proof. By (18) there exist d and matrices B_j for which $\sum_j B_j \otimes \nabla^2 f_j(x_0)$ is invertible. They can be chosen real: its determinant is a real-coefficient polynomial in the matrix entries, and a nonzero real polynomial cannot vanish at every real tuple. Put $D_j(x, y) = f_j(x) + f_j(y) - 2f_j((x + y)/2)$, and for $X = (x_1, \dots, x_d)$, $Y = (y_1, \dots, y_d)$ define

$$\Phi(X, Y) = \sum_{a, b, j} (B_j)_{ab} D_j(x_a, y_b).$$

At the repeated base point its mixed Hessian is $-\frac{1}{2} \sum_j B_j \otimes \nabla^2 f_j(x_0)$. Choose Q small enough that a smooth cutoff equal to one on $Q^d \times Q^d$ has support in the neighborhood of lemma 5.1. Set $K_0 = \sum_{a, b, j} |(B_j)_{ab}| > 0$. On $S^d \times S^d$, $|\Phi| \leq 2K_0\varepsilon$, so with $\lambda = (4K_0\varepsilon)^{-1} \geq 1$ the real part of the exponential is at least $1/2$. Writing $v = \text{vol}_n(S)$ gives

$$\frac{1}{2}v^{2d} \leq |\langle 1_{S^d}, T_\lambda 1_{S^d} \rangle| \leq C\lambda^{-nd/2}v^d = C(4K_0\varepsilon)^{nd/2}v^d.$$

If $v > 0$, divide by v^d and take the d th root. If $v = 0$ the conclusion is immediate. The evaluation dimension cancels from the exponent, although it affects the fixed constant. $\square$

A direct small perturbation of the quadratic determinant argument would not supply this lemma: a remainder controlled by $\eta\|x - y\|^2$ can overwhelm a determinant bound on highly anisotropic sets. The oscillatory estimate controls arbitrary compact contacts without an aspect-ratio assumption.

Theorem 5.3 (Local and global smooth ranks). *Suppose F is real C^∞ on an open neighborhood of B . Then*

$$\frac{r_{\text{loc}}}{2} \log_2(1/\varepsilon) - O_{F,B}(1) \leq p_{\text{conv}}(\varepsilon) \leq p_{\text{bin}}(\varepsilon) \leq \frac{r_{\text{all}}}{2} \log_2(1/\varepsilon) + O_{F,B}(1). \quad (31)$$

If both ranks are r , both minima have leading coefficient $r/2$. In particular, full pointwise rank at one interior point gives coefficient $n/2$. The general upper construction may have size polynomial in $1/\varepsilon$ for fixed data. If every output is a polynomial of degree at most a fixed $D \geq 2$, its upper formulation instead has $O_{F,B,n,m,D}((1 + \log(1/\varepsilon))^D)$ rows and variables, with the same binary count. Affine maps have exact zero-binary linear formulations.

Proof. Choose an interior point of maximum local rank, which exists because occurring ranks are integers in a finite range. For positive rank, lemma 4.1 supplies a principal slice of dimension r_{loc} whose Hessian tuple has full rank at that point. Apply lemma 5.2 on a smaller box in the slice and use the compact parity cover. Its positive volume is at most $2^p C \varepsilon^{r_{\text{loc}}/2}$, proving the lower bound. Rank zero requires only $p \geq 0$.

For the upper bound apply lemma 4.2 to a finite basis of the global Hessian span. Its coordinates give fixed exponents $\alpha_i \in \{0, 1, 1/2\}$ summing to $r_{\text{all}}/2$, and every possibly nonzero Hessian entry has endpoint exponent sum at least one. Enclose the transformed domain P in a box of side lengths $b_i > 0$. Divide side i into $2^{\lceil \alpha_i T \rceil}$ intervals and intersect the cells with P . For each nonempty cell choose a center c in that intersection. Along the segment from c to any point x of the cell, every Hessian is bounded and has the stated zero blocks. The integral Taylor remainder therefore satisfies, uniformly over cells and outputs,

$$|f_j(x) - f_j(c) - \nabla f_j(c)^T(x - c)| \leq C_0 2^{-T}.$$

Indeed each contributing product $(x - c)_i(x - c)_k$ has absolute value at most $b_i b_k 2^{-T}$. Enlarge C_0 to be at least one and take

$$T = \max\{0, \lceil \log_2(2C_0/\varepsilon) \rceil\}, \quad 2C_0 2^{-T} \leq \varepsilon.$$

Use an affine output band of radius $\varepsilon/2$ on each cell. Each band contains its exact graph and has error at most ε . There are at most $2^{\sum_i \lceil \alpha_i T \rceil}$ bounded polyhedral members, so lemma 2.3 proves the upper binary count. If all Hessians vanish, the map is affine on the convex box.

For polynomial outputs, each forbidden Hessian entry vanishes on an open set and thus identically. The same structure is valid on the whole enclosing box, which may now be normalized to $[0, 1]^n$. Retain the original affine coordinate identities and domain inequalities in the formulation. Enlarge $C_0 \geq 1$ to bound the Taylor remainder on this entire normalized box, using bounds on all second derivatives there, and choose T by the same formula with this constant. Use the input prefixes (9), with $L_i = \lceil \alpha_i T \rceil$, and the first-order expression

$$P_j(A, \rho) = f_j(A) + \sum_i \partial_i f_j(A) \rho_i. \quad (32)$$

The Taylor error is again at most $C_0 2^{-T}$, since the segment from A to $A + \rho$ stays in the enclosing box, so impose $|w_j - P_j(A, \rho)| \leq \varepsilon/2$. Expand (32). Each term is a product of at most D bits, or

at most $D - 1$ bits and one residual. Repeated bits can be removed. For a set J of distinct bits, continuous variables satisfying

$$0 \leq v_J \leq 1, \quad v_J \leq \beta_b \ (b \in J), \quad v_J \geq \sum_{b \in J} \beta_b - |J| + 1$$

are forced to equal the bit product whenever the input bits are integral. A product $v_J \rho_i$ is then represented by (6); v_J needs no separate integrality declaration. Empty products equal one. With fixed D, n, m , the expansion contains $O((1 + \sum_i L_i)^D)$ terms and $O(D)$ rows per term. All graph points are retained because the affine band covers the Taylor remainder. No additional integer coordinates are introduced. $\square$

A product $f(x) = \prod_{i=1}^k x_i$, $k \geq 2$, has coefficient $k/2$ on every full-dimensional compact box in the positive orthant. At any point of that box its Hessian is

$$\nabla^2 f(x) = f(x) \operatorname{diag}(x_i^{-1})(\mathbf{1}\mathbf{1}^T - I_k) \operatorname{diag}(x_i^{-1}).$$

The middle matrix has eigenvalues $k - 1$ once and -1 with multiplicity $k - 1$, so the Hessian is nonsingular at every positive point. Thus the full local rank condition applies even when the box does not meet the line on which all coordinates are equal. A rank-deficient example is

$$f(z_1, z_2, w, t) = z_1 w^3 + z_2(w + w^2) + wt^2 + t^4.$$

Its Hessians have $Z = (z_1, z_2)$, $W = w$, $Q = t$ in (21), so their global rank is at most three by the shrinking characterization. At $(1, 1, 1, 1)$ the Hessian is

$$\begin{pmatrix} 0 & 0 & 3 & 0 \\ 0 & 0 & 3 & 0 \\ 3 & 3 & 8 & 2 \\ 0 & 0 & 2 & 14 \end{pmatrix},$$

which has rank three. Thus its coefficient is $3/2$ on a box containing that point in its interior, with an upper formulation of size $O((1 + \log(1/\varepsilon))^4)$.

There is a representation-dependent computational boundary. Given a rational arithmetic circuit for a polynomial $g(z)$ in d variables, form

$$F(x, y, z) = (xyg(z), z_1^2, \dots, z_d^2) \quad \text{on } [-1, 1]^{d+2}.$$

If $g \equiv 0$, its precision coefficient is $d/2$. Otherwise $g(z_0) \neq 0$ at some interior point, and at $(0, 0, z_0)$ the Hessian of the scalar combination $xyg(z) + \sum_i z_i^2$ is

$$\begin{pmatrix} 0 & g(z_0) & 0 \\ g(z_0) & 0 & 0 \\ 0 & 0 & 2I_d \end{pmatrix},$$

which is invertible. The coefficient is then $(d+2)/2$ by theorem 5.3. Computing the exact coefficient, or an additive approximation with error strictly below $1/2$, for such circuits would therefore decide polynomial identity testing. This is a reduction, not an unconditional running-time lower bound. In a dense coefficient representation identity testing itself is immediate.

5.2 Constant scalar Hessian rank

The global-span upper bound in theorem 5.3 need not be sharp. For scalar maps with constant Hessian rank one can follow the local null spaces by polyhedral tubes. The affine-gradient-fiber geometry is classical [29, Definition 1.1 and the following paragraph, pp. 208–209]; we derive it locally to show why these tubes are linear objects. The resulting formulation law uses this geometry to count a cover by polyhedral tubes and compares that count with the unrestricted-integer lower bound.

Theorem 5.4 (Constant-rank scalar law). *Let f be C^∞ on a neighborhood U of the compact full-dimensional box B , and assume $\text{rank } \nabla^2 f(x) = r$ for every $x \in U$. Both minimum dimensions have expansion*

$$\frac{r}{2} \log_2(1/\varepsilon) + O_{f,B}(1).$$

The upper bound is a union of $O_{f,B}(\varepsilon^{-r/2})$ bounded polyhedra, encoded with the logarithm of that many binary codes. It does not assert a formulation of size polynomial in $\log(1/\varepsilon)$.

Proof. Rank zero gives an affine function on B . For $r > 0$ the lower bound is the scalar case of theorem 5.3. At any point of B choose coordinates $(u, v) \in \mathbb{R}^r \times \mathbb{R}^{n-r}$ with f_{uu} invertible. The inverse function theorem makes $(u, v) \mapsto (p, v) = (f_u(u, v), v)$ a local smooth coordinate map. Restrict its inverse $(u(p, v), v)$ to an open product of boxes and put $g(p, v) = f(u(p, v), v) - p^T u(p, v)$. The chain rule gives

$$g_p = -u, \quad g_v = f_v, \quad g_{vv} = f_{vv} - f_{vu} f_{uu}^{-1} f_{uv} = 0.$$

The last equality follows from constant rank and the Schur complement. Thus, on the product chart,

$$g(p, v) = b(p)^T v + c(p), \quad u(p, v) = -Db(p)^T v - \nabla c(p). \quad (33)$$

For fixed p the fiber is affine in v , its gradient is $(p, b(p))$, and the affine function $\ell_p(u, v) = p^T u + b(p)^T v + c(p)$ agrees with both f and its gradient along that fiber. Convexity of f is unnecessary for this differential use of Legendre coordinates.

Choose smaller closed parameter boxes P_0, V_0 within the chart and P_1 with $P_0 \subset \text{int } P_1$, still within it. There are constants $L, \eta, M > 0$ such that $u(p, v)$ is L -Lipschitz in p in the maximum norm on $P_1 \times V_0$, and a transverse η -neighborhood of this compact image lies in U with Hessian operator norm at most M . For small h , cover P_0 by $O(h^{-r})$ cubes of radius h and centers $p_\nu \in P_1$. Form

$$C_\nu(h) = \{(u, v) : v \in V_0, \quad \|u - u(p_\nu, v)\|_\infty \leq Lh\}.$$

This is a bounded polytope by (33), and the Lipschitz bound shows that the polytopes cover the image of $P_0 \times V_0$. For any point in a tube compare it with the central fiber point of the same v . Function and gradient errors vanish there, and the displacement has only r coordinates, each at most Lh . Taylor's theorem gives, throughout the *whole* tube,

$$|f - \ell_{p_\nu}| \leq \frac{1}{2} M r L^2 h^2,$$

provided $Lh \leq \eta$. This estimate does not assume that the tube point's own gradient parameter lies in the selected parameter cube.

The interiors of images of smaller parameter products cover B . Compactness selects finitely many charts, including boundary points because U is a neighborhood. Intersect all tubes with B and use uniform constants across this finite collection. There are $O(h^{-r})$ bounded polytopes covering B . Choose h proportional to $\sqrt{\varepsilon}$ so the Taylor error is at most $\varepsilon/2$, and impose $|w - \ell_{p_\nu}(x)| \leq \varepsilon/2$ on each. Their union contains the graph and has error at most ε . Apply lemma 2.3. $\square$

For $f(x) = \|x\|_2$ on $[1, 2]^n$, $n \geq 2$, the Hessian is $\|x\|^{-1}(I - xx^T/\|x\|^2)$ and has constant rank $n - 1$. Thus its coefficient is $(n - 1)/2$ although its null line varies. In dimension two the global Hessian span is the whole symmetric matrix space: three distinct ratios x/y give independent vectors of coefficients of $(y^2, -xy, x^2)$ by the Vandermonde determinant. Thus $r_{\text{all}} = 2$ while the exact coefficient is $1/2$. An elementary upper construction uses angular sectors of width h . On a sector with central unit direction u and bounded radius R , $0 \leq \|x\| - u^T x \leq R(1 - \cos(h/2)) \leq Rh^2/8$. Intersecting the sectors with the positive box gives polyhedral cells, and their affine bands use $O(\varepsilon^{-1/2})$ members.

The theorem assumes constant scalar rank on a neighborhood of the whole box. Rank-changing singularities and general vector maps with $r_{\text{loc}} < r_{\text{all}}$ are not characterized by these results. A nonlinear coordinate substitution is not a free operation on convex lifts; the proof uses its charts only to select actual polyhedra in the original coordinates.

5.3 Positive perspectives

Proposition 5.5 (Perspective transfer). *Let $F : D \rightarrow \mathbb{R}^m$ have a binary linear graph relaxation of component error δ . For $0 < \ell \leq u < \infty$ define*

$$G(x, t) = tF(x/t), \quad D_{\text{pers}} = \{(x, t) : \ell \leq t \leq u, x/t \in D\}.$$

There is a binary linear graph relaxation of G with the same binaries and error at most $u\delta$. For each $t_0 \in [\ell, u]$,

$$\begin{aligned} p_{\text{conv}}(F, D; \varepsilon/t_0) &\leq p_{\text{conv}}(G, D_{\text{pers}}; \varepsilon), \\ p_{\text{bin}}(F, D; \varepsilon/t_0) &\leq p_{\text{bin}}(G, D_{\text{pers}}; \varepsilon) \leq p_{\text{bin}}(F, D; \varepsilon/u), \end{aligned}$$

where scalar tolerances denote componentwise boxes. Hence a matching logarithmic coefficient for F is preserved by G .

Proof. Write the original rows, including the domain restrictions, as $Ay + Bv + Ca + Ez \leq h$, with $z \in \{0, 1\}^p$. Introduce a' and $v' = tz$, imposing the exact binary-product rows

$$\ell z_i \leq v'_i \leq u z_i, \quad t - u(1 - z_i) \leq v'_i \leq t - \ell(1 - z_i).$$

Replace the original rows by $Ax + Bw + Ca' + Ev' \leq ht$ and impose $\ell \leq t \leq u$. Division by positive t recovers the original formulation with $y = x/t, v = w/t, a = a'/t$, and conversely every original feasible point scales to one of these points. Thus graph containment is exact and $|w_j - tF_j(x/t)| \leq t\delta \leq u\delta$. The transformation adds p continuous product variables and $4p + 2$ rows, besides the new input t ; original continuous auxiliaries need no finite bounds. Rational data retain polynomial encoding size. Finally, fixing $t = t_0$ and dividing visible variables by t_0 gives the two lower transfers for arbitrary convex or binary linear lifts. $\square$

Perspective geometry is classical [9, Sections 2.3.3 and 3.2.6]. Here the relevant fact is that homogenizing rows adds no binary coordinates. The product encoding is not valid for an unrestricted integer variable in place of a binary, and no such upper transfer is asserted. The positive lower bound on t excludes the cone apex.

For quadratic F , the perspective outputs are

$$G_j(x, t) = \frac{x^T H_j x}{2t} + a_j^T x + b_j t.$$

By theorem 4.4 and proposition 5.5, their precision coefficient is $r/2$ and their binary upper formulation has size $O(1 + \log(1/\varepsilon))$ for fixed data. The domain is the bounded polytope $\ell \leq t \leq u$, $l_i t \leq x_i \leq u_i t$ when $D = \prod_i [l_i, u_i]$. The same coefficient holds on an ordinary full-dimensional (x, t) box with positive t : enclose all ratios in a larger fixed box for the upper bound, retain the requested input box, and fix $t = t_0$ for the lower bound. In particular x^2/t on $[0, 1] \times [1, 2]$ has coefficient $1/2$. Its Hessian is $(2/t^3)(t, -x)(t, -x)^T$, of rank one; two nonparallel such vectors give a positive definite sum, so the global Hessian span has rank two. This provides a rational, compactly formulable example in which the global smooth-span upper invariant is strictly larger than the exact coefficient.

6 Finite accuracy and a covariance determinant

The rank law describes a fixed system as its tolerance tends to zero. A different question keeps the tolerances finite and asks for constants uniform over the coefficients and the number of outputs. Covariance geometry answers this question, including when the output tolerances are unequal. Throughout this section

$$f_j(x) = \frac{1}{2}x^T H_j x + a_j^T x + b_j, \quad x \in [0, 1]^n, \quad H_j = H_j^T.$$

Let ω_n be the volume of the Euclidean unit ball, and set

$$c_n = \max\{4, n/4\}, \quad A_n = \log_2[\omega_n(n+2)^{n/2}] + \frac{n}{2} \log_2 c_n, \quad B_n = n \log_2 n + n. \quad (34)$$

The Gaussian integral gives $\omega_n \leq (2\pi e/n)^{n/2}$, so $A_n + B_n = O(n \log(n+1))$.

The matrix variable P below plays two roles. After a fixed rescaling, the covariance of any positive-volume parity contact is feasible for its constraints. Conversely, a positive definite feasible P has eigenvectors specifying grid directions, and its eigenvalues specify squared residual widths. The quantity $-\frac{1}{2} \log_2 \det P$ measures the total directional precision cost, up to an overhead depending only on dimension. Relating these two uses of the same matrix gives a comparison with arbitrary convex lifts, uniformly in output count and coefficient size.

Theorem 6.1 (Finite covariance law). *For positive tolerances ε_j , define*

$$D_* = \max \left\{ \det P : 0 \preceq P \preceq I, \ E_j(P) := \text{tr}(H_j P H_j P) \leq \varepsilon_j^2 \ (1 \leq j \leq m) \right\}, \quad \Phi = -\frac{1}{2} \log_2 D_*. \quad (35)$$

Then $0 < D_ \leq 1$ and*

$$\max\{0, \Phi - A_n\} \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \Phi + B_n. \quad (36)$$

The upper construction has a binary count $p_{\text{grid}} \leq \Phi + B_n$ and $O(np_{\text{grid}} + n^2 + mn^2)$ rows and continuous variables. The statement permits real coefficients; a rational polynomial-time construction is proved in theorem 7.3.

Proof. The feasible set in (35) is compact and contains a positive scalar multiple of I , so a determinant maximizer is positive definite. Let S be a positive-volume parity contact from lemma 2.2, with covariance Σ . The midpoint identity and lemma 4.3 give $E_j(\Sigma) \leq 16\varepsilon_j^2$. A unit linear functional on the cube has range at most $\sqrt{n}$, hence variance at most $n/4$; thus $\Sigma \preceq (n/4)I$. Consequently Σ/c_n is feasible and $\det \Sigma \leq c_n^n D_*$. The volume inequality (24) bounds every contact by $2^{A_n} \sqrt{D_*}$. The same bound holds for zero-volume contacts. The at most 2^p contacts cover a cube of volume one, proving the lower bound without any measurability assumption on the initial lift projection.

For the upper bound, take any feasible $P = U \text{diag}(\lambda_i)U^T \succ 0$ with U orthogonal. After rotation and translation, the cube lies in $0 \leq y_i \leq \Delta_i$, where $\Delta_i = \sum_k |U_{ki}| \in [1, \sqrt{n}]$. Keep the original cube inequalities and the affine coordinate identities. The transformed Hessians are $G_j = U^T H_j U$. Choose

$$L_i = \left\lceil \log_2 \left(\Delta_i \sqrt{n/\lambda_i} \right) \right\rceil, \quad h_i = \Delta_i 2^{-L_i} \leq \sqrt{\lambda_i/n}. \quad (37)$$

Use the prefix and residual construction (9), scaled by Δ_i . The exact binary products and residual envelopes (6)–(10) provide a shared symmetric monomial approximation with errors δ_{ik} satisfying $|\delta_{ik}| \leq h_i h_k / 4$. This includes squares, empty prefixes, and the upper endpoints of the bounding intervals. The same approximate monomial is used in every output. Thus

$$|w_j - f_j(x)| \leq \frac{1}{8} \sum_{i,k} |G_{j,ik}| h_i h_k \leq \frac{1}{8n} \sum_{i,k} |G_{j,ik}| \sqrt{\lambda_i \lambda_k} \leq \frac{1}{8} \sqrt{E_j(P)} \leq \varepsilon_j / 8.$$

The last inequality before the energy uses Cauchy–Schwarz on n^2 entries. Choosing every approximate monomial equal to its exact value proves graph containment. Finally,

$$\sum_i L_i \leq -\frac{1}{2} \log_2 \det P + \sum_i \log_2 \Delta_i + \frac{n}{2} \log_2 n + n \leq -\frac{1}{2} \log_2 \det P + B_n.$$

Use an optimal P . Each bit occurs in $O(n)$ binary products; there are $O(n^2)$ residual monomials and $O(mn^2)$ output coefficients. $\square$

On a full-dimensional compact convex $\Omega \subset [0, 1]^n$, exactly the same lower proof gives the useful variant

$$p_{\text{conv}}(F, \Omega; K) \geq \Phi - A_n + \log_2 \text{vol}(\Omega). \quad (38)$$

Here K is the tolerance box, and Φ uses the same Hessians on the containing cube. The upper grid can be restricted to Ω by any available exact linear lift of that domain. The volume term cannot in general be discarded.

A feasible covariance also describes a directional-error ellipsoid. For $d = P^{1/2}u$ with $\|u\|_2 \leq 1$, $|d^T H_j d| \leq \|P^{1/2} H_j P^{1/2}\|_F \leq \varepsilon_j$. Using the operator norm in place of the Frobenius norm changes the log-determinant benchmark by at most $(n/4) \log_2 n$: the norm inequalities $\|M\|_2 \leq \|M\|_F \leq \sqrt{n} \|M\|_2$ permit rescaling a covariance by $1/\sqrt{n}$. This connects the finite invariant to anisotropic quadratic interpolation and established derivative-based ellipsoid metrics [11], while the formulation comparison still requires the contact and shared-grid arguments above.

6.1 Geodesic convexity and finite certificates

The feasible covariances need not be convex in the ordinary matrix coordinates: a product Hessian on positive diagonal matrices has energy $2p_1 p_2$. They are convex along the positive definite geodesics

$$P(t) = P_0^{1/2} \exp(tA) P_0^{1/2}, \quad A = A^T.$$

If $A = V \text{diag}(a_i) V^T$ and $C_j = V^T P_0^{1/2} H_j P_0^{1/2} V$, then

$$E_j(P(t)) = \sum_{i,k} C_{j,ik}^2 e^{t(a_i + a_k)}. \quad (39)$$

Each energy is convex, and its logarithm is convex unless the Hessian vanishes. The log determinant is affine. Between two endpoints the geodesic obeys $P(t) \preceq (1-t)P(0) + tP(1)$: conjugate by

$P(0)^{-1/2}$ and use $d^t \leq 1 - t + td$ on each positive eigenvalue. Thus the cap and all energy constraints are geodesically convex. Every local determinant maximum in their positive definite part is global. These are applications of established positive definite geometry; see Sra and Hosseini [36].

A feasible covariance is already an upper-construction certificate. The following residual certificate also bounds its suboptimality and can be used without executing the conservative algorithm below.

Proposition 6.2 (A posteriori determinant certificate). *Let $0 \prec P \preceq I$ satisfy every energy budget. For $\mu_j \geq 0$ and $S \succeq 0$, put*

$$c = \sum_j \mu_j (\varepsilon_j^2 - E_j(P)) + \text{tr}[S(I - P)], \quad R = -P^{-1} + 2 \sum_j \mu_j H_j P H_j + S, \quad L = -\log \det P.$$

Then $c \geq 0$, and, with natural logarithms,

$$0 \leq \log D_* - \log \det P \leq c + \sqrt{n} L \sqrt{\text{tr}(PRPR)}. \quad (40)$$

In particular, $c = 0$ and $R = 0$ certify global optimality. For rational Hessians, tolerances, and certificate data P, μ, S , the slack c and residual square $\text{tr}(PRPR)$ are exactly computable nonnegative rational numbers; scalar logarithm and square-root intervals certify the right side.

Proof. The Lagrangian $J(X) = -\log \det X + \sum_j \mu_j (E_j(X) - \varepsilon_j^2) + \text{tr}[S(X - I)]$ is geodesically convex by (39) and the corresponding positive-coefficient expansion of $\text{tr}(SX)$. Its tangent gradient at P , in isometric coordinates, is $G = P^{1/2} R P^{1/2}$. An optimal feasible P_* has $\det P_* \geq \det P$. Since both matrices are capped by I , their minimum eigenvalues are at least $\det P$. Therefore the eigenvalues of $P^{-1/2} P_* P^{-1/2}$ lie between $\det P$ and $(\det P)^{-1}$. Writing $A = \log(P^{-1/2} P_* P^{-1/2})$, we obtain $\|A\|_F \leq \sqrt{n} L$. Convexity gives

$$-\log D_* \geq J(P_*) \geq J(P) + \text{tr}(GA) \geq -\log \det P - c - \sqrt{n} L \|G\|_F.$$

Now $\|G\|_F^2 = \text{tr}(PRPR)$, proving the claim. For fixed P , the right side is convex in the multipliers, since its slack is linear and its residual norm is a norm of an affine map. No necessity of stationarity or constraint qualification is asserted. $\square$

6.2 Exact reductions and a dimension gap

For an edge output $x_i x_k$, direct multiplication gives $E_{ik}(P) = 2(P_{ii} P_{kk} + P_{ik}^2)$. Replacing P by its diagonal preserves the cap, decreases these energies and increases its determinant by Hadamard's inequality. With $P_{ii} = 2^{-2s_i}$, the benchmark becomes

$$\Phi = \min \left\{ \sum_i s_i : s_i \geq 0, \quad s_i + s_k \geq \log_2(\sqrt{2}/\varepsilon_{ik}) \text{ for every edge} \right\}. \quad (41)$$

This recovers the fractional-cover allocation of theorem 2.7 with its finite tolerance constants.

Proposition 6.3 (Commuting Hessians). *If all Hessians commute, the benchmark has a maximizer diagonal in a common orthonormal eigenbasis. Writing its eigenvalues as p_i and the Hessian eigenvalues as h_{ji} , it reduces to*

$$\max \left\{ \sum_i \log t_i : 0 < t_i \leq 1, \quad \sum_i h_{ji}^2 t_i \leq \varepsilon_j^2 \right\}, \quad t_i = p_i^2,$$

whose optimal objective is $2 \log D_$. Hessians may be indefinite.*

Proof. In the common eigenbasis every coordinate sign flip S commutes with every Hessian. Thus P and SPS have the same determinant and energies. Their geometric mean $M = P\#(SPS)$ is feasible by geodesic convexity and has determinant $\det P$. The identity $SMS = M$ follows from symmetry and orthogonal congruence invariance of the geometric mean. These identities follow in turn from its unique positive definite solution of $MP^{-1}M = Q$: this equation also implies $MQ^{-1}M = P$. A previously imposed commuting sign symmetry is preserved. Apply the operation to the n coordinate flips in succession; the result is an optimal diagonal covariance. Substitution proves the scalar program. $\square$

For one Hessian put $a_i = h_i^2$. If $\varepsilon^2 \geq \sum_i a_i$, then $D_* = 1$. Otherwise the unique $s > 0$ solving $\sum_i \min(a_i, s) = \varepsilon^2$ gives $t_i = \min(1, s/a_i)$ for $a_i > 0$ and $t_i = 1$ for $a_i = 0$. Indeed, the budget multiplier $1/s$ and upper-cap multipliers $\max(0, 1 - a_i/s)$ satisfy the stationarity equations for the strictly concave scalar objective. Sorting the a_i determines s by a linear equation on the correct interval. A rational common diagonalization, when supplied, gives rational scalar allocation data; arbitrary rational commuting matrices need not have a rational common eigenbasis.

Example 6.4 (The covariance benchmark loses order $n \log n$). For $f(x) = \frac{1}{2} \sum_i x_i^2$ on $[0, 1]^n$ at unit tolerance, $E(P) = \text{tr}(P^2)$. The arithmetic–geometric mean inequality gives $D_* = n^{-n/2}$, attained by $n^{-1/2}I$, so $\Phi = (n/4) \log_2 n$. A parity contact has diameter at most $\sqrt{8}$, by its midpoint gap $\|x - y\|^2/8 \leq 1$. Its volume is at most $\omega_n 8^{n/2}$, which yields

$$p_{\text{conv}} \geq \frac{n}{2} \log_2 n - \frac{n}{2} \log_2(16\pi e).$$

Conversely, put $L = \max\{0, \lceil \frac{1}{2} \log_2(n/8) \rceil\}$. Shared coordinate squares with residual width 2^{-L} have total error at most $n2^{-2L}/8 \leq 1$ and use nL binaries. Hence both minima are $(n/2) \log_2 n + O(n)$ as dimension grows. The order $O(n \log(n+1))$ in (36) cannot be improved to $O(n)$ for this particular benchmark. This is an unconditional geometric limitation, separate from computational hardness.

7 Rational quadratic constructions

We first make the optimal rank coefficient effective, then give an algorithm with a dimension-only finite-accuracy loss. These are different guarantees: the former allows an additive polynomial in input height; the latter has an additive $O(n \log(n+1))$ independent of that height.

7.1 Uniform rational bounds at the noncommutative-rank rate

Theorem 7.1. *Equation (30), including its construction-time, coefficient-length and formulation-size assertions, holds for rational quadratic data on a rational full-dimensional box. Its parameter r is the noncommutative rank, not the common nonlinear input rank used later in theorem 8.4.*

Proof. Let s be the total binary input length. Exact elimination computes a basis for the Hessian span. The rational algorithm of Ivanyos et al. [23, Theorem 1.5 and Lemma 5.3] returns a maximal shrunk subspace U of polynomial bit length with the same deficiency as over $\mathbb{C}$. Set $V = \sum_j H_j U$ and construct rational bases for $Z = U \cap V^\perp$, $W = V \cap U^\perp$, and $Q = (Z + W)^\perp$. All operations are rational nullspaces and intersections. Their chosen bases need not be orthonormal: the spaces themselves are orthogonal. Thus $T = [Z_b \ W_b \ Q_b]$ is rational and invertible, and lemma 4.2 gives

$$T^T H_j T = \begin{pmatrix} 0 & B_j & 0 \\ B_j^T & C_j & D_j \\ 0 & D_j^T & E_j \end{pmatrix}, \quad 2 \dim W + \dim Q = r.$$

If $r = 0$, all Hessians vanish and the affine graph needs no integers. If $r = n$, choosing $U = V = 0$ is sufficient. Rational bases, their inverses, and the displayed coefficients have polynomial height: clear denominators and bound the determinants of the square subsystems used by elimination. Only a fixed number of such linear-algebra operations follows the imported polynomial-bit algorithm.

For $v = T^{-1}x$, compute each coordinate's exact minimum and maximum on the original box by summing the smaller or larger endpoint contribution of each entry of T^{-1} . Every width is positive. Normalize this enclosing box to $y \in [0, 1]^n$, obtaining $x = c + Ay$. Retain all original bounds on $c + Ay$ in the formulation. Translation and positive diagonal scaling preserve the zero blocks. Write the transformed quadratics as an exact affine part plus $\sum_{i \leq t} c_{j,it} y_i y_t$, and set

$$C = \max_j \sum_{i \leq t} |c_{j,it}| > 0, \quad b = \max\{0, \lceil \log_2(C/4) \rceil\}, \quad K = k + b.$$

The integer b is computed by rational comparisons with powers of two and satisfies $b \leq \text{poly}(s)$. Give Z -coordinates depth zero, W -coordinates depth K , and Q -coordinates depth $\lceil K/2 \rceil$. Every nonzero monomial has a W endpoint or two Q endpoints, so its residual-width product is at most 2^{-K} . The shared prefix construction has output error at most $C2^{-K}/4 \leq 2^{-k}$ and binary count

$$\dim(W)K + \dim(Q)\lceil K/2 \rceil \leq \frac{r}{2}k + \frac{r}{2}b + \frac{n}{2}.$$

All transformed fixed coefficients have polynomial bit length; the new dyadics have $O(k + b + 1)$ bits. There are $O(n^2)$ monomials, and a monomial needs $O(L_i + L_t + 1)$ rows and auxiliaries. Consequently individual coefficients have length $\text{poly}(s) + O(k)$, while the numbers of rows and variables are $\text{poly}(s)(k + \text{poly}(s) + 1)$. This proves the algorithmic upper bound, with no algebraic-number coefficient model.

For the lower bound let D_H be a common denominator of the Hessian entries and put $E_B = \prod_i \text{den}(l_i) \text{den}(u_i)$ for the rational box endpoints. Both logarithms are at most s after using the product of the input denominators. By lemma 4.1 there is a principal coordinate set I of size r whose restricted Hessians G_j have full noncommutative rank. The matrices $D_H G_j$ are integral. The integral capacity bound [19, Theorem 2.18] gives

$$\kappa = \text{cap}(T_G) = D_H^{-2r} \text{cap}(T_{D_H G}) \geq D_H^{-2r} r^{-2r}.$$

The power $2r$ is essential, since capacity is a determinant ratio. Fixing the other coordinates leaves an r -box of volume $V_I \geq E_B^{-1}$: each positive rational width is at least the reciprocal product of its endpoint denominators. The finite capacity bound (28), together with $\omega_r^{2/r} \leq 4$, now gives

$$\varepsilon \geq \frac{2^{-2p/r}}{16D_H(r+2)\sqrt{mr}E_B^{2/r}}.$$

At $\varepsilon = 2^{-k}$ this implies

$$p \geq \frac{r}{2}k - \frac{r}{2} \log_2 D_H - \log_2 E_B - \frac{r}{2} \log_2 [16(r+2)\sqrt{mr}]. \quad (42)$$

The subtracted terms are $O(rs + r \log(mr + 2) + s)$ and hence polynomial in s . This lower bound permits arbitrary real convex lifts. If desired, the principal set I can be found by repeated coordinate-deletion rank tests: while more than r coordinates remain, an invertible principal r -submatrix guarantees a deletion preserving rank r . $\square$

An arbitrary rational $\varepsilon \in (0, 1]$ can be handled by the integer $k = \lceil \log_2(1/\varepsilon) \rceil$, computed by integer comparisons. Running time includes the length of the rational tolerance input, not only its numerical logarithm. Polynomial encoding permits exponential numerical coefficient magnitudes and does not give polynomial-time MILP optimization.

7.2 Certified rational matrix functions

The finite construction requires matrix functions, but it does not require well-separated eigenvalues. We give the numerical details because a polynomial arithmetic-operation count alone would not establish a rational formulation theorem. Rational Jacobi near-diagonalization is an established spectral ingredient; see also Del Pia [16, Theorem 2].

Lemma 7.2 (Rational spectral residual). *Given rational symmetric A and rational $\sigma > 0$, one can compute rational matrices V, B in deterministic polynomial time such that*

$$V^T V = I, \quad A = V B V^T, \quad \|B - \text{diag}(B)\|_F \leq \sigma.$$

The bit bound is polynomial in the input lengths, including the precision length of σ . No spectral-gap hypothesis is needed.

Proof. For order one, take $V = I$. Otherwise let $N = n(n-1)$ and choose a rational $M \geq \max(1, \|A\|_F)$, for example the larger of one and its entrywise absolute sum. Maintain exact rational identities $A = V B V^T$ and $V^T V = I$, starting at $B = A, V = I$. If the off-diagonal Frobenius norm s exceeds σ , select a largest off-diagonal entry B_{ik} . An exact Jacobi rotation C_* annihilates it and changes the squared off-norm from s^2 to $s^2 - 2B_{ik}^2 \leq (1 - 2/N)s^2$.

Choose an exactly orthogonal rational plane rotation C with $\|C - C_*\|_2 \leq \sigma/(8NM)$. This is possible with polynomially many bits: take a rational approximation of the half-angle tangent t and use $c = (1 - t^2)/(1 + t^2)$ and $s_\theta = 2t/(1 + t^2)$. The annihilating angle may be chosen in $[-\pi/4, \pi/4]$; its tangent follows from the two-by-two quadratic formula, and the half-angle conversion has denominator bounded away from zero. While a rotation is needed, $|B_{ik}| \geq \sigma/\sqrt{N}$, so divisions used to find that angle require only polynomial precision depth. Rational bisection computes the square roots and the parameter approximation.

Set $B \leftarrow C^T B C$ and $V \leftarrow V C$. Because $\|B\|_F = \|A\|_F \leq M$, the perturbation from exact conjugation is at most $2M\|C - C_*\|_2$. Hence the new off-norm is at most

$$(1 - 1/N)s + \sigma/(4N) \leq (1 - 3/(4N))s \quad (s > \sigma).$$

Thus $O(N(1 + \log^+(M/\sigma)))$ rotations suffice; termination is tested by comparing rational squared norms. A common denominator for each rotation has polynomial bit length. Each exact update adds only twice that length to a common denominator for B , and once to one for V . There are polynomially many updates. Norm preservation bounds the numerators once these denominators are fixed. This proves polynomial Turing complexity, as well as exact reconstruction and orthogonality. $\square$

For positive definite A with spectrum in $[a, b]$, put $A_0 = V \text{diag}(B) V^T$. Its eigenvalues also lie in $[a, b]$, because the entries B_{ii} are Rayleigh quotients. With $\|A - A_0\|_2 \leq \sigma$,

$$\begin{aligned} \|\log A - \log A_0\|_2 &\leq \sigma/a, \\ \|A^{1/2} - A_0^{1/2}\|_2 &\leq \sigma/(2\sqrt{a}), \\ \|A^{-1/2} - A_0^{-1/2}\|_2 &\leq \sigma/(2a^{3/2}). \end{aligned} \tag{43}$$

The first bound follows from the resolvent formula for logarithms and $\int_0^\infty (a+t)^{-2} dt = 1/a$. For the second, the difference $X = A^{1/2} - A_0^{1/2}$ solves $A^{1/2}X + XA_0^{1/2} = A - A_0$; its integral solution gives the bound. Multiplication by the two inverse square roots gives the third. For symmetric arguments of norm at most M_0 , the Duhamel formula also gives $\|e^A - e^{A_0}\|_2 \leq e^{M_0}\|A - A_0\|_2$.

Thus matrix functions reduce to scalar functions of rational diagonal entries, without approximating any particular exact eigenbasis. Scalar square roots use rational bisection. For logarithms, dyadic range reduction to $[1, 2]$ makes the arctanh series parameter at most $1/3$; the same series computes $\log 2$. Exponentials use range reduction and a Taylor series. When $|\log a|, |\log b|, M_0$ are polynomially bounded, prescribed inverse-exponential-polynomial absolute errors require polynomial work and output length. The largest diagonal entry also gives a rational unit vector v , a column of V , with

$$0 \leq \lambda_{\max}(A) - v^T A v \leq \sigma. \quad (44)$$

This avoids selecting an eigenvector at a repeated largest eigenvalue.

7.3 A polynomial-time finite-accuracy construction

The remaining algorithmic task is to find a nearly optimal covariance with certified rational data. A constant additive error in its log determinant costs only a constant number of additional binaries in theorem 6.1. We obtain that accuracy by geodesic optimization, repair feasibility by scalar rescaling, and construct the grid through rational near-diagonalization. Each step below includes the precision bounds needed to pass from real matrix formulas to a rational MILP.

Theorem 7.3. *For rational quadratic coefficients and positive rational component tolerances, a deterministic polynomial-time algorithm constructs a rational MILP graph relaxation with*

$$p_{\text{out}} \leq p_{\text{conv}} + O(n \log(n+1)). \quad (45)$$

Its total encoding length is polynomial in the complete input length. The implicit constant is independent of coefficients, tolerances and output dimension. The running-time assertion is a worst-case complexity bound, without a practical iteration-count claim.

Proof. All logarithms in this proof are natural. On the positive definite cone use the affine-invariant metric and its distance

$$\langle A, B \rangle_P = \text{tr}(P^{-1} A P^{-1} B), \quad d(P, Q) = \|\log(P^{-1/2} Q P^{-1/2})\|_F.$$

This is a Hadamard manifold with sectional curvature in $[-1/2, 0]$; we use the weaker lower bound -1 [14, Proposition I.1]. The comparison inequality of Zhang and Sra [44, Corollary 8] is the only geometric convergence import. The following error accounting applies it to rational stored iterates.

An exact penalty. Omit zero Hessians from logarithmic expressions and define

$$h(P) = \max \left\{ 0, \log \lambda_{\max}(P), \max_{j: H_j \neq 0} \frac{1}{2} \log \frac{E_j(P)}{\varepsilon_j^2} \right\}, \quad F(P) = -\log \det P + n h(P). \quad (46)$$

The covariance $\hat{P} = e^{-h(P)} P$ is feasible and satisfies $F(P) = -\log \det \hat{P}$. Hence $\min F = -\log D_*$, attained by any optimal feasible covariance. Each component of h is geodesically convex: use (39), and express the largest-eigenvalue term as the supremum over fixed nonzero v of $\log[(v^T P v)/(v^T v)]$. Each such Rayleigh branch is log-sum-exp along a geodesic. In tangent coordinates $A \mapsto P^{-1/2} A P^{-1/2}$, the half-log energy gradient and Rayleigh gradient are respectively

$$\frac{M_j^2}{\text{tr}(M_j^2)}, \quad M_j = P^{1/2} H_j P^{1/2}, \quad \frac{P^{1/2} v v^T P^{1/2}}{v^T P v}. \quad (47)$$

They are positive semidefinite of trace one and norm at most one. Therefore h is globally one-Lipschitz and F is globally $(n + \sqrt{n})$ -Lipschitz.

A polynomial-radius search region. Choose a nonnegative integer b with $\delta = 2^{-b} \leq 1$ and $\delta \sum_{a,c} |H_{j,ac}| \leq \varepsilon_j$ for every j . Rational dyadic comparisons find b of polynomial magnitude. Then δI is feasible, so an optimal P_* has $\det P_* \geq \delta^n$ and eigenvalues at most one. It follows that

$$d(I, P_*) \leq \sum_i -\log \lambda_i(P_*) \leq nb \log 2 < R := 1 + nb.$$

The closed metric ball $\mathcal{B}_R$ is geodesically convex. Its metric projection is

$$\Pi_R(Q) = \exp(t \log Q), \quad t = \frac{R}{\max\{R, \|\log Q\|_F\}}. \quad (48)$$

The reverse triangle inequality gives the lower distance to the ball, and the radial geodesic attains it. Projection is nonexpansive on a Hadamard manifold.

A recurrence for the actual rational iterates. Start at $P_0 = I$ and store iterates in $\mathcal{B}_{R+1}$. Put $D = 2R + 2$ and $Z = 1 + D$. At the current rational P obtain a symmetric tangent-coordinate matrix G , with $\|G\|_F \leq 3n$, such that

$$F(P) - F(P_*) \leq -\operatorname{tr}\left[G \log(P^{-1/2} P_* P^{-1/2})\right] + e. \quad (49)$$

Form the conceptual exact update $Q = P^{1/2} e^{-\eta G} P^{1/2}$, project it to $\bar{Q} = \Pi_R(Q)$, and store a rational symmetric P^+ with $d(P^+, \bar{Q}) \leq \xi$. For $\xi \leq 1$, the next iterate again lies in $\mathcal{B}_{R+1}$. The comparison inequality, using its curvature factor $D/\tanh D \leq 1 + D$, and nonexpansive projection give

$$F(P) - F(P_*) \leq \frac{d(P, P_*)^2 - d(P^+, P_*)^2}{2\eta} + \frac{Z\eta(3n)^2}{2} + e + \frac{2D\xi + \xi^2}{2\eta}. \quad (50)$$

For the last term, both the projected point and P_* are in $\mathcal{B}_R$, so changing the former by distance ξ changes its squared distance to P_* by at most $2D\xi + \xi^2$. Choose

$$\eta = [16Z(3n)^2]^{-1}, \quad T = \lceil 16D^2/\eta \rceil, \quad e \leq 1/32, \quad \xi \leq \eta/[128(D+1)].$$

Summing (50) over T steps telescopes on the stored iterates. The initial-distance, curvature, oracle and rounding terms in the average gap are at most $1/32, 1/32, 1/32, 1/64$ respectively. Thus some stored objective has gap below $1/4$. Selecting the smallest reported objective, each evaluated within $1/64$, adds at most $1/32$ and gives gap below $1/2$. No global exact trajectory is being tracked; rounding enters only the local inequality and the telescoping distances.

Implementation of the inexact oracle. All stored spectra lie in $[e^{-(R+1)}, e^{R+1}]$. If a symmetric approximation to a positive definite matrix with minimum eigenvalue a has operator error at most au , $u \leq 1/2$, then relative eigenvalues lie in $[1 - u, 1 + u]$, proving

$$d(A, \tilde{A}) \leq 2\sqrt{n}u. \quad (51)$$

For a nonzero rational Hessian, $E_j(P) \geq e^{-2(R+1)} \|H_j\|_F^2$; this is the minimum singular-value bound for two-sided multiplication by $P^{1/2}$. A nonzero input entry has magnitude at least 2^{-s} , where s is its total input length. Every normalization denominator therefore has an inverse of at most exponential-polynomial magnitude.

Evaluate each branch of h within $\tau = 1/(512n)$. For the cap branch use the fixed rational Rayleigh vector from (44), with its logarithmic shortfall at most τ , rather than a possibly unstable exact leading eigenvector. Choose a largest reported branch. Its exact value is within 3τ of $h(P)$. Approximate that branch's tangent gradient within ν/n , where $\nu = 1/[128(D+1)]$, and form

$G = -I + n\tilde{\nabla}h$. The full gradient error is at most ν , so (49) holds with $e \leq 3n\tau + D\nu < 1/32$ and $\|G\|_F \leq 3n$. The conditioning estimates and lemma 7.2 give all these approximations in polynomial bit complexity.

The exact update Q has logarithmic spectral bounds at most $R + 1 + 3n\eta$. Matrix functions are therefore covered by (43). To implement projection, first replace Q by its nearby rationally diagonalizable matrix. Equation (51) and nonexpansiveness control this replacement. Then (48) is a scalar spectral formula with denominator at least $R \geq 1$. Round its output symmetrically to a fresh dyadic matrix so that (51) gives the prescribed ξ . Polynomially many precision bits suffice and positive definiteness is preserved. Fresh rounding prevents accumulated symbolic denominator growth in the outer iteration.

Exact repair and a rational grid. For the selected rational P , compute a rational upper approximation $\bar{\kappa}$ to

$$\kappa = \max\{1, \lambda_{\max}(P), \max_j \sqrt{E_j(P)/\varepsilon_j}\}$$

within factor $e^{1/(4n)}$, using certified spectral and scalar intervals. Then $\hat{P} = P/\bar{\kappa}$ is exactly feasible and

$$\log \det \hat{P} \geq \log D_* - 1. \quad (52)$$

The previous radius and dyadic bounds give $\kappa \leq e^{R+1}2^b$, hence an explicit inverse-exponential-polynomial lower bound on $\lambda_{\min}(\hat{P})$. Apply lemma 7.2 at residual at most half this lower bound, writing $B = U^T \hat{P} U$, and set $\tilde{\lambda}_i = B_{ii}/4$ and $\tilde{P} = U \operatorname{diag}(\tilde{\lambda}_i) U^T$. Then

$$\hat{P}/8 \preceq \tilde{P} \preceq \hat{P}/2, \quad \det \tilde{P} \geq 8^{-n} \det \hat{P}. \quad (53)$$

Indeed $\tilde{P}$ differs from $\hat{P}/4$ by at most $\lambda_{\min}(\hat{P})/8$ in operator norm. Each energy is monotone in positive semidefinite order: its derivative in direction $A \succeq 0$ is $2 \operatorname{tr}(H_j P H_j A) \geq 0$. Thus $\tilde{P}$ remains feasible.

Use the rational orthogonal matrix U and rational scales $\tilde{\lambda}_i$ in (37). Coordinate widths, translations, transformed coefficients and prefix products are rational. The depths are chosen by squared rational comparisons, without evaluating an exact square root. By (52) and (53), their sum is at most $\Phi + B_n + 3n/2 + 1/(2 \log 2)$. Combine this with (36) to prove (45). Finally $-\log D_* \leq nb \log 2$, so the depths, row counts and coefficient lengths are polynomial in the input length. $\square$

8 Output budgets and nonlinear input coordinates

8.1 Correlated and partially measured errors

A finite family of positive semidefinite matrices $W_\ell \in \mathbb{R}^{m \times m}$ and tolerances $\varepsilon_\ell > 0$ defines the requirement

$$e^T W_\ell e \leq \varepsilon_\ell^2 \quad (1 \leq \ell \leq g), \quad e = w - F(x). \quad (54)$$

When $\bigcap_\ell \ker W_\ell = \{0\}$ this is an error body in definition 2.1. Otherwise its intersection is an unbounded closed symmetric convex set: errors in its common kernel are unmeasured. In this subsection only, extend the same minima by using this closed set in the graph-containment definition. This convention is explicit because the compact-body hypothesis elsewhere remains in force.

Theorem 8.1 (Grouped covariance law). *Replace the energies in (35) by*

$$E_\ell(P) = \sum_{j,k} (W_\ell)_{jk} \operatorname{tr}(H_j P H_k P)$$

and their bounds by ε_ℓ^2 . Then (36) holds with the same A_n, B_n for (54). For rational input, including W_ℓ , the algorithm of theorem 7.3 has the same dimension-only additive guarantee and polynomial rational encoding. It does not require a rational factorization of W_ℓ .

Proof. For the proof, factor $W_\ell = C_\ell^T C_\ell$ and define $G_a = \sum_j (C_\ell)_{aj} H_j$. The energy is the nonnegative sum $E_\ell(P) = \sum_a \text{tr}(G_a P G_a P)$. Apply the fourth-moment identity in lemma 4.3 to these scalar combinations and sum. The midpoint budget bounds the sum by $16\varepsilon_\ell^2$, and every discarded fourth-moment term is a square or squared mean. Therefore $E_\ell(\Sigma) \leq 16\varepsilon_\ell^2$, proving the covariance lower bound as before, including for singular budgets.

For the shared grid let $Z_{ik} = \delta_{ik}/\sqrt{\lambda_i \lambda_k}$ and $M_j = \text{diag}(\sqrt{\lambda}) U^T H_j U \text{diag}(\sqrt{\lambda})$. Every entry of Z is at most $1/(4n)$ in absolute value, so $\|Z\|_F \leq 1/4$, while $e_j = \frac{1}{2} \text{tr}(M_j Z)$. Consequently

$$e^T W_\ell e = \frac{1}{4} \sum_a \left\langle \sum_j (C_\ell)_{aj} M_j, Z \right\rangle_F^2 \leq \frac{1}{4} E_\ell(P) \|Z\|_F^2 \leq \varepsilon_\ell^2/64. \quad (55)$$

Thus the grid controls an entire correlated error vector, with no output count in the constants.

For computation retain the rational double sums and define $\widehat{M}_j = P^{1/2} H_j P^{1/2}$ in the original-coordinate isometric tangent frame used in theorem 7.3. These matrices satisfy $\widehat{M}_j = U M_j U^T$. Each nonzero energy is a positive-coefficient exponential sum along a geodesic. Its half-log gradient in this frame is

$$\frac{N_\ell(P)}{E_\ell(P)}, \quad N_\ell(P) = \sum_{j,k} (W_\ell)_{jk} \widehat{M}_j \widehat{M}_k = \sum_a \left(\sum_j (C_\ell)_{aj} \widehat{M}_j \right)^2.$$

The last identity proves symmetry, positive semidefiniteness and trace $E_\ell(P)$, without using the factorization in the algorithm. An energy is identically zero precisely when its rational value $E_\ell(I)$ is zero; omit those logarithmic branches. Such measured output combinations are affine and are still represented exactly. For every remaining group, $E_\ell(P) \geq \lambda_{\min}(P)^2 E_\ell(I)$, and the positive rational $E_\ell(I)$ has polynomial encoding length. A feasible dyadic covariance is found by exact comparisons $\delta^2 E_\ell(I) \leq \varepsilon_\ell^2$. All radius, gradient, matrix-function and repair estimates in theorem 7.3 now apply. The final common monomial grid satisfies (55). $\square$

The certificate proposition 6.2 also applies, replacing $H_j P H_j$ by $\sum_{a,b} (W_j)_{ab} H_a P H_b$. Its exact rational evaluation additionally requires rational budget matrices W_j . For commuting Hessians the scalar program of proposition 6.3 uses the nonnegative coefficients $h_{ji}^T W_\ell h_{ji}$ in place of h_{ji}^2 .

8.2 A specialized total-absolute-error benchmark

For the error requirement $\|w - F(x)\|_1 \leq \varepsilon$, define the Gram matrix and correlation semidefinite bound

$$\Gamma(P)_{jk} = \text{tr}(H_j P H_k P), \quad S(P) = \max\{\text{tr}(\Gamma(P)X) : X \succeq 0, \text{diag } X = \mathbf{1}\}. \quad (56)$$

The matrix $\Gamma(P)$ is positive semidefinite. The classical positive semidefinite Grothendieck bound is

$$\max_{s \in \{-1,1\}^m} s^T \Gamma s \leq S \leq \frac{\pi}{2} \max_{s \in \{-1,1\}^m} s^T \Gamma s; \quad (57)$$

see Briët et al. [10] for its attribution to Rietz and Nesterov. One proof of the upper bound factors a correlation optimizer into unit vectors and applies Gaussian hyperplane signs. Their expected sign product is $\frac{2}{\pi} \arcsin X_{jk}$. The entrywise power series of $\arcsin X - X$ is a nonnegative sum of positive semidefinite odd Hadamard powers. Its inner product with $\Gamma \succeq 0$ is nonnegative, proving the ratio. The lower bound uses $X = s s^T$.

Theorem 8.2. Let $D_1 = \max\{\det P : 0 \preceq P \preceq I, S(P) \leq \varepsilon^2\}$ and $\Phi_1 = -\frac{1}{2} \log_2 D_1$. Then

$$\max\{0, \Phi_1 - A'_n\} \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \Phi_1 + B_n, \quad A'_n = \log_2[\omega_n(n+2)^{n/2}] + \frac{n}{2} \log_2 \max\{6, n/4\}.$$

For rational input a polynomial-time rational MILP construction again has count at most $p_{\text{conv}} + O(n \log(n+1))$.

Proof. Every sign combination of the quadratic midpoint difference on a parity contact has absolute value at most 4ε in the normalization $q_j(x-y) = \frac{1}{2}(x-y)^T H_j(x-y)$. Its covariance therefore satisfies $s^T \Gamma(\Sigma) s \leq 16\varepsilon^2$. Equation (57) gives $S(\Sigma) \leq 8\pi\varepsilon^2 < 36\varepsilon^2$. Scale by $\max\{6, n/4\}$ and apply the volume proof of theorem 6.1. For the upper grid, its shared Z satisfies $\|Z\|_F \leq 1/4$, whence

$$|s^T e| \leq \frac{1}{2} \left\| \sum_j s_j M_j \right\|_F \quad \|Z\|_F \leq \frac{1}{8} \sqrt{s^T \Gamma(P) s} \leq \varepsilon/8.$$

Maximization over signs is exactly the ℓ_1 norm.

For each fixed correlation X , the energy $E_X(P) = \text{tr}(\Gamma(P)X)$ is a grouped positive semidefinite energy. Its nonzero half-log branch is one-Lipschitz and geodesically convex. Their maximum $\frac{1}{2} \log S(P)$ has these same properties. A nearly optimal rational feasible X supplies a near-active gradient branch, and a certified upper value of $S(P)$ supplies the final feasibility repair in theorem 7.3. We construct both next.

For rational P , Γ is rational. If it is nonzero, then $\text{tr } \Gamma \leq S \leq m \text{tr } \Gamma$. Normalize $\text{tr } \Gamma = 1$. For $m \geq 3$, parameterize $X = I + \text{off}(v)$ by its $d = m(m-1)/2$ independent off-diagonal entries. The correlation body contains the radius-1/4 ball around zero, since $\|\text{off}(v)\|_F = \sqrt{2}\|v\|_2$, and lies in the radius- m ball. It has a rational strong separation oracle: exact symmetric elimination either proves $X \succeq 0$ or returns rational a with $a^T X a < 0$, giving a separating inequality. A negative pivot provides such a vector; a zero diagonal with a nonzero off-diagonal entry provides an indefinite two-dimensional block and a rational negative vector. Rational congruences lift it back to the original coordinates. Fraction-free elimination gives polynomial bit bounds. The cases $m = 1, 2$ are an exact scalar or interval optimization.

The weak separation-optimization theorem of Grötschel et al. [20, Definition (5) and Theorem (3.1)], with these inner and outer balls, returns rational v within distance ρ of the body whose linear objective is within ρ below its optimum. The running time is polynomial in $\log(1/\rho)$ and the other input lengths. Writing $s = S/\text{tr } \Gamma$ before normalization and $a = \text{tr}(\Gamma X(v))$ after normalization, we have $a \geq s - \rho$ and $\lambda_{\min}(X(v)) \geq -\sqrt{2}\rho \geq -2\rho$. Thus

$$X_f = \frac{X(v) + 2\rho I}{1 + 2\rho} \tag{58}$$

is rational, exactly positive semidefinite, and has unit diagonal. Its normalized objective $a_f = (a + 2\rho)/(1 + 2\rho)$ obeys $0 \leq s - a_f \leq (2m+1)\rho$. Given $0 < \nu \leq 1/2$, choose $\rho = \nu/[4(m+1)]$. Undoing normalization, the rational numbers

$$L = \text{tr}(\Gamma(P)X_f), \quad U = L + \nu \text{tr } \Gamma(P)$$

satisfy $L \leq S(P) \leq U$, $L \geq \text{tr } \Gamma(P)/2$, and $U/L \leq 1 + 2\nu$. Thus the half-log branch error is at most ν , while U is a certified upper bound. These certificates follow from weak optimization; an exact dual semidefinite solution is not assumed.

Finally, $\text{tr } \Gamma(P) \geq \lambda_{\min}(P)^2 \text{tr } \Gamma(I)$, where the last trace is a positive rational unless all Hessians vanish. A dyadic feasible covariance follows from $\delta^2 m \text{tr } \Gamma(I) \leq \varepsilon^2$. The conditioning, branch-error and rational-repair bounds of theorem 7.3 therefore apply with inverse-polynomial ν . When every Hessian vanishes, use the exact affine graph instead. $\square$

Rationally transformed budgets $\|Ae\|_1 \leq \varepsilon$, finitely many such budgets, and their intersections with the grouped quadratic budgets are handled by taking the maximum of their penalty branches. The additive count is independent of the number of groups. The hardness reduction in section 10 uses separate component tolerances; it does not by itself establish the same hardness for a single ℓ_1 budget.

8.3 General symmetric bodies via the nonlinear output image

For a body given by an oracle, we use the following explicit input model. The body $K \subset \mathbb{R}^m$ is compact, convex, symmetric about zero, and has known positive rational radii $r_0 B_2^m \subset K \subset R_0 B_2^m$. At a rational query a strong separation oracle returns membership or a rational valid linear inequality strictly separating the query. Its running time and output encoding length are polynomial in the query length and a fixed encoding of the oracle. All subsequent oracle running-time bounds include these data.

We use one classical rounding consequence under this model. In dimension d , a well-bounded symmetric body C admits a polynomial-time rational ellipsoid E_0 satisfying

$$E_0 \subset C \subset \beta_d E_0, \quad \beta_d = (d+1)\sqrt{d}. \quad (59)$$

Here $E_0 = \{z : z^T W z \leq 1\}$ with rational $W \succ 0$. For precision about the import, Dadush et al. [15, Theorem B.5], a GLS rounding theorem, returns rational $A \succ 0$ and a possibly real center t . Define $E(A) = \{z : z^T A z \leq 1\}$. The theorem gives $C \subset t + E(A)$ and either $\text{vol } E(A) \leq \eta$ or $t + E(A)/\beta_d \subset C$. For a known inner radius r , choose $\eta = \min\{1/2, (r/d)^d/2\} < \text{vol } C$. The small-volume alternative is impossible. Symmetry and averaging remove t from both inclusions. Take $W = d(d+1)^2 A$. This proves (59) with rational data, without computing the center or a matrix square root.

Theorem 8.3 (General symmetric output body). *Under the stated oracle assumptions and rational quadratic data on the cube, a deterministic polynomial-time algorithm produces a rational MILP with $p_{\text{out}} \leq p_{\text{conv}}(F, [0, 1]^n; K) + O(n \log(n+1))$. The additive constant is independent of output dimension; runtime still depends on the complete input and oracle encoding.*

Proof. Write $F(x) = a(x) + Cu(x)$, where a is affine and u lists the $n(n+1)/2$ quadratic monomials. Choose a rational full-column-rank basis T for the image of C and a rational factorization $C = TB$. Put $d = \text{rank } C$, $g = Bu$, and $K_{\text{eff}} = \{z : Tz \in K\}$. If $d = 0$, use the exact affine graph. Otherwise rational elimination has polynomial bit complexity, and for either formulation model

$$p(F, [0, 1]^n; K) = p(g, [0, 1]^n; K_{\text{eff}}). \quad (60)$$

One direction adjoins $w = a(x) + Tz$ to a reduced formulation. For the other direction intersect an original formulation with this equation and introduce z . Every exact graph point remains feasible, and $T(z - g(x)) \in K$ is precisely the reduced error condition. This does not assume that the original formulation's errors already lay in $\text{im } T$.

Choose rational upper bounds $c_T \geq \|T\|_2$ and $c_L \geq \|(T^T T)^{-1} T^T\|_2$ using entrywise sums plus one. Then

$$(r_0/c_T)B_2^d \subset K_{\text{eff}} \subset (R_0 c_L)B_2^d.$$

The ambient separator at Tz pulls back exactly to a separator at z . Its normal cannot vanish if the query violates it, because $0 \in K$. Thus the effective body satisfies the same oracle model with polynomial encoding. Apply (59) to obtain $E_0 \subset K_{\text{eff}} \subset \alpha E_0$, where $\alpha = \beta_d$.

Let $D(t)$ be the grouped covariance determinant for g with error tE_0 , and $\Phi(t) = -\frac{1}{2} \log_2 D(t)$. Scaling a covariance by $1/\alpha$ proves

$$\Phi(1) \leq \Phi(\alpha) + \frac{n}{2} \log_2 \alpha. \quad (61)$$

The grouped lower bound and monotonicity in the allowed error give $\Phi(\alpha) - A_n \leq p_{\text{conv}}(g; \alpha E_0) \leq p_{\text{conv}}(F; K)$. The rational construction for the inner ellipsoid uses at most $\Phi(1) + B_n + O(n)$ binaries and admits only errors in E_0 . Restoring the output through T proves the claim, since $d \leq n(n+1)/2$ and hence $n \log \alpha = O(n \log(n+1))$. The formulation uses explicit rational bands from the common monomial grid; it does not try to describe the oracle body itself by finitely many linear inequalities. $\square$

The proof also applies if the effective body has the specified bounded strong oracle directly, even when unrelated ambient output directions are unmeasured. It makes no claim for an effective body with no known inner and outer radii. A factor α in output-norm rounding costs at most $(n/2) \log_2 \alpha$ in the covariance comparison; reducing to the quadratic output image is what removes m from this cost.

8.4 Quotienting the common Hessian kernel

The next parameter measures active input directions:

$$r_{\text{in}} = \text{rank} \begin{bmatrix} H_1 \\ \vdots \\ H_m \end{bmatrix}.$$

It is distinct from the noncommutative rank of the Hessian span. In this subsection abbreviate it by r .

Theorem 8.4 (Nonlinear input quotient). *Under the rational and oracle hypotheses of theorem 8.3, there is a polynomial-time rational MILP construction with*

$$p_{\text{out}} \leq p_{\text{conv}}(F, [0, 1]^n; K) + O(r \log(r+1)). \quad (62)$$

For $r = 0$, the exact affine graph needs no integers. Continuous size, runtime and encoding depend on the full input dimensions.

Proof. Let $U \in \mathbb{Q}^{r \times n}$ have independent rows spanning all Hessian rows and let $F_U = U^T(UU^T)^{-1}$. Then $G_j = F_U^T H_j F_U$ satisfies $H_j = U^T G_j U$, because $F_U U$ is the orthogonal projection onto their common row space. With $c = \frac{1}{2} \mathbf{1}$, write

$$z = U(x - c), \quad F(x) = a(x) + g(z), \quad g_j(z) = \frac{1}{2} z^T G_j z, \quad Z = U[-1/2, 1/2]^n.$$

The minima on the original cube and on Z agree. To project a formulation, subtract $a(x)$ from the output and use the affine visible map $(x, w) \mapsto (U(x - c), w - a(x))$. Every $z \in Z$ has a preimage, so the exact reduced graph is covered. Conversely, keep original continuous x , its bounds, $z = U(x - c)$ and $w = a(x) + v$. These operations preserve both models and add no integers, even when a varies along a fiber.

The full-dimensional rational zonotope Z has a polynomial strong separation oracle. Membership is the rational LP $U t = z$, $-1/2 \leq t_i \leq 1/2$. For an infeasible query, a separating normal can be obtained from the rational LP

$$v_i \geq \pm (U^T h)_i, \quad h^T z - \frac{1}{2} \sum_i v_i \geq 1.$$

Strict separation and scaling guarantee feasibility of this second LP. It returns the valid, violated inequality $h^T y \leq \frac{1}{2} \sum_i v_i$. Polynomial rational LP algorithms and basic-solution determinant bounds give polynomial lengths. Explicit radii follow by selecting nonsingular columns U_J : with $c_{\text{inv}} = 1 + \sum |(U_J^{-1})_{ik}|$ and $R_Z = 1 + \sum |U_{ik}|$,

$$(2c_{\text{inv}})^{-1} B_2^r \subset U_J[-1/2, 1/2]^r \subset Z \subset R_Z B_2^r.$$

Apply oracle rounding to obtain $E(A)/\beta_r \subset Z \subset E(A)$, with rational $A \succ 0$. An exact factorization $A = LDL^T$ has rational unit lower-triangular L and positive diagonal D . Choose positive dyadics b_i by squared comparisons so that $\sqrt{D_{ii}} \leq b_i \leq 2\sqrt{D_{ii}}$, and put $R = \text{diag}(b_i)L^T$. Then

$$A \preceq R^T R \preceq 4A, \quad \beta_r^{-1} B_2^r \subset RZ \subset 2B_2^r.$$

All maps have polynomial rational encoding. The normalized domain $\Omega = \frac{1}{2}\mathbf{1} + \frac{1}{4}RZ \subset [0, 1]^r$ contains a ball of radius $1/(4\beta_r)$ and hence a cube of side $1/[2r(r+1)]$. Therefore

$$-\log_2 \text{vol}(\Omega) \leq r \log_2 [2r(r+1)]. \quad (63)$$

Its exact linear lift keeps the original x and the equation $y = \frac{1}{2}\mathbf{1} + \frac{1}{4}RU(x - c)$.

Transform the quadratics to Ω and reduce their nonlinear output image as in (60). Its dimension is at most $r(r+1)/2$. Round its effective output body to an inner ellipsoid with factor $\alpha = \beta_d$. Equation (38), whose proof also holds for grouped budgets, and (61) imply

$$\Phi(1) \leq p_{\text{conv}}(F; K) + A_r - \log_2 \text{vol}(\Omega) + \frac{r}{2} \log_2 \alpha.$$

Construct the rational grid on the containing r -cube, then impose the exact domain lift and restore the original outputs. Its count is at most $\Phi(1) + B_r + O(r)$. Equation (63) proves (62). If the original rank is positive, the reduced quadratic output image is nonzero, so the factor α is used only in positive dimension. $\square$

9 Positive quadratic structure and sharper finite bounds

Positive semidefinite Hessians permit a first-moment trace estimate in place of the squared Frobenius energy. When the coordinates split into small independent blocks, this improves the dimension overhead. The error bodies here are *unconditional*: changing any coordinate sign preserves the body. A convex unconditional body has the domination property

$$v \in K, \quad |u_i| \leq |v_i| \text{ (all } i) \implies u \in K. \quad (64)$$

Indeed, shrink one coordinate by a convex combination of its two sign choices and repeat. This property allows a rational output rectangle to certify membership in a curved body.

9.1 A rational log-determinant allocation oracle

We give a common optimization lemma for matrix blocks and scalar allocations. Its scalar case will also support the nonlinear constructions later in the paper. The objective and weak-optimization machinery are classical [20, 38]; the explicit interior ball and rational feasibility repair specify the oracle and bit model used here.

Lemma 9.1 (Rational log-determinant allocation). *Let $P = (P_1, \dots, P_B)$ be symmetric blocks of sizes d_b , and let $N = \sum_b d_b \geq 1$. Let $\mathcal{L}(P)$ be a rational linear map into $\mathbb{R}^m$, and let K be closed and convex with a polynomial rational strong separation oracle and known rational $\rho_0 > 0$ satisfying $\rho_0 B_2^m \subset K$. Define*

$$D = \max \left\{ \prod_b \det P_b : 0 \preceq P_b \preceq I, \mathcal{L}(P) \in K \right\}.$$

Given rational $0 < \nu \leq 1$, a deterministic polynomial-time algorithm returns exactly feasible rational positive definite blocks with product of determinants at least $e^{-\nu} D$. The assertion requires neither positivity of $\mathcal{L}$ nor symmetry or boundedness of K . For blocks of size one it gives a positive rational allocation p with $Cp \in K$, $p \leq \mathbf{1}$, and $\prod_i p_i \geq e^{-\nu} D$.

Proof. Use the independent upper-triangular entries as coordinates $u \in \mathbb{R}^q$, $q = \sum_b d_b(d_b + 1)/2$. Their block matrix embedding satisfies $\|P(u) - P(v)\|_F \leq 2\|u - v\|_2$. Write $\mathcal{L}(P(u)) = Cu$, including the factor two on off-diagonal trace coefficients, and put $c = 1 + \sum_{j,i} |C_{ji}|$. Choose $\delta = 2^{-b} \leq 1$ with $\delta N c \leq \rho_0/4$. The block identity scaled by δ is feasible, so $D \geq \delta^N$. Every eigenvalue of an optimizer is at least δ^N , because all its eigenvalues are at most one. Put

$$\begin{aligned} a &= \delta^N/4, \quad B_0 = N(Nb + 2) + 4, \quad P_0 = (\delta/2)I, \\ t_0 &= -N(b + 1) - 2, \quad \sigma = \min\{\delta/(32N), \rho_0/(8c), 1/4\}. \end{aligned}$$

Let u_0 encode P_0 . The compact convex hypograph

$$Q = \left\{ (u, t) : aI \preceq P_b(u) \preceq I, \quad Cu \in K, \quad -B_0 \leq t \leq g(P) := \sum_b \log \det P_b \right\} \quad (65)$$

has optimal last coordinate $\log D$. It contains the Euclidean ball of radius σ about (u_0, t_0) . To check this, coordinate perturbations of norm σ change block spectra by at most 2σ , leaving every eigenvalue between $\delta/4$ and one, and above a . The center's image has norm at most $\rho_0/8$, and the image perturbation is at most $c\sigma \leq \rho_0/8$. On this ball, the independent-coordinate gradient of g has diagonal entries from P_b^{-1} and twice its off-diagonal entries, with norm at most $8N/\delta$. Hence $|g(P) - g(P_0)| \leq (8N/\delta)\sigma \leq 1/4$. At the center $g(P_0) - t_0 \geq 2$, whereas the last-coordinate perturbation is at most $1/4$; also $t_0 + B_0 \geq 3$. These prove all the ball inclusions. An outer ball about the same center of radius $2(B_0 + q + 1)$ follows from the spectral and last-coordinate bounds. All constants have polynomial rational length.

A weak separation oracle for Q checks the linear last-coordinate bounds and the spectral caps exactly; rational symmetric elimination supplies the latter separators as in the proof of theorem 8.2. It pulls back a violated separator for $Cu \in K$. For a query passing these tests, approximate $g(P)$ within $\eta/2$ by rational $\hat{g}$. If $t > \hat{g} + \eta/2$, the rational tangent

$$t' \leq \hat{g} + \eta/2 + \sum_b \text{tr}[P_b^{-1}(P'_b - P_b)] \quad (66)$$

is valid and strictly separates the query. Otherwise lowering t to $\min(t, g(P))$ proves distance at most η to Q . This correction respects the lower last-coordinate bound because $g(P) \geq N \log a = -N(Nb + 2) \log 2 > -B_0$. Only logarithms of exact positive rational determinants are approximated; the inverse matrices in the tangent are exact rational matrices. The spectral lower cap ensures polynomial precision depth. The tangent normal has norm at least one through its t coefficient; other normals can be divided by their nonzero infinity norm. Thus this is the weak separation convention of Grötschel et al. [20, Theorem (3.1)].

Weakly maximize the last coordinate to tolerance $\rho = \nu\sigma/[4(B_0 + \sigma + 1)]$. The result is rational $y = (u_y, t_y)$ with $\text{dist}(y, Q) \leq \rho$ and $t_y \geq \log D - \rho$. Define the rational repaired point

$$y_f = \frac{\sigma y + \rho(u_0, t_0)}{\sigma + \rho}. \quad (67)$$

For some $z \in Q$, $\|y - z\| \leq \rho$. Equation (67) is the convex combination of z and $(u_0, t_0) + (\sigma/\rho)(y - z)$ with weights $\sigma/(\sigma + \rho)$ and $\rho/(\sigma + \rho)$. The second point lies in the known interior ball. Thus $y_f \in Q$ exactly. Moreover $0 \leq \log D - t_0 \leq B_0$, so its objective loss is at most $\rho + (\rho/\sigma)B_0 \leq \nu$. The first coordinates give the claimed rational blocks. The same proof with $d_b = 1$ proves the scalar case. $\square$

9.2 Independent positive semidefinite blocks

Suppose the original coordinates have a common partition of sizes $d_1, \dots, d_B$, with $N = \sum_b d_b$, and

$$H_j = \text{diag}(H_{j1}, \dots, H_{jB}), \quad H_{jb} \succeq 0.$$

Any block absent from all quadratic parts can first be removed after affine output subtraction. If all blocks disappear, the graph is affine and needs no integers; henceforth $N \geq 1$. The Hessians need not commute within a block. For positive definite block variables set

$$E(P)_j = \sum_b \text{tr}(H_{jb}P_b), \quad \Phi_K = -\frac{1}{2} \log_2 D_K,$$

$$D_K = \max \left\{ \prod_b \det P_b : 0 \preceq P_b \preceq I, E(P) \in K \right\}.$$

Write $c_b = \max\{4, d_b/4\}$ and

$$A_{\mathcal{B}} = \log_2[\omega_N(N+2)^{N/2}] + \frac{1}{2} \sum_b d_b \log_2 c_b, \quad B_{\mathcal{B}} = N + \sum_b d_b \log_2 d_b. \quad (68)$$

Theorem 9.2 (Positive block allocation). *For an unconditional error body K ,*

$$\max\{0, \Phi_K - A_{\mathcal{B}}\} \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \Phi_K + B_{\mathcal{B}}. \quad (69)$$

For rational data and the strong-oracle hypotheses on K , a polynomial-time rational MILP construction satisfies

$$p_{\text{out}} \leq p_{\text{conv}} + O\left(\sum_b d_b \log(d_b + 1)\right).$$

If $r_b = \text{rank}[H_{1b}; \dots; H_{mb}]$, the sharper intrinsic bound is

$$p_{\text{out}} \leq p_{\text{conv}} + O\left(\sum_{b:r_b > 0} r_b \log(r_b + 1)\right). \quad (70)$$

For a tolerance box the feasibility condition is simply $\sum_b \text{tr}(H_{jb}P_b) \leq \varepsilon_j$ for every output.

Proof. The midpoint error of a parity contact is the nonnegative vector $J_j(x, y) = \frac{1}{8} \sum_b (x_b - y_b)^T H_{jb}(x_b - y_b) \in K$. For independent uniform contact points with full covariance Σ , convexity gives $\mathbb{E}J \in K$ and $(\mathbb{E}J)_j = \frac{1}{4} \sum_b \text{tr}(H_{jb}\Sigma_{bb})$. Set $P_b = \Sigma_{bb}/c_b$. Each block covariance is capped by $(d_b/4)I$ using its original cube coordinate range. Positivity gives $0 \leq E(P) \leq \mathbb{E}J$, so (64) makes P

feasible. The block determinant inequality $\det \Sigma \leq \prod_b \det \Sigma_{bb}$ follows by iterated Schur complements. Combining it with the covariance-volume inequality bounds every contact by $2^{A_B} \sqrt{D_K}$, proving the lower bound.

Diagonalize each P_b within its own block. The corresponding rotated coordinate has width at most $\sqrt{d_b}$. Choose residual widths $h_i \leq \sqrt{\lambda_i/d_b}$, giving at most $-\frac{1}{2} \log_2 \prod_b \det P_b + B_B$ binaries. The normalized shared residual error matrix Z_b has entries at most $1/(4d_b)$, hence operator norm at most $1/4$. The scaled transformed Hessian M_{jb} is positive semidefinite with trace $\text{tr}(H_{jb}P_b)$. Therefore

$$|e_j| \leq \frac{1}{2} \sum_b |\text{tr}(M_{jb}Z_b)| \leq \frac{1}{8} \sum_b \text{tr}(H_{jb}P_b) = E(P)_j/8.$$

For $M \succeq 0$, the inequality $|\text{tr}(MZ)| \leq \text{tr}(M)\|Z\|_2$ follows from $-\|Z\|_2 I \preceq Z \preceq \|Z\|_2 I$. Unconditional domination proves the whole-body error guarantee, and exact monomial values prove graph containment. An optimal allocation proves the finite upper bound.

For computation apply lemma 9.1 with $\nu = 1$. It gives exactly feasible rational blocks of determinant product at least $e^{-1}D_K$. Apply lemma 7.2 separately within each block and use (53); the resulting rational grid covariance loses at most a factor 8^N in determinant. Its trace vector is dominated by the original trace vector, so it remains feasible in K . The extra count is $O(N)$. Since the global volume factor in (68) has logarithm $O(N)$, this proves the first rational guarantee with polynomial encoding.

Finally quotient the common kernel separately inside each original block by the construction of theorem 8.4. Congruence preserves positive semidefiniteness. Because the original input blocks are disjoint, the projected domain is a product of zonotopes. Normalize each nonzero block quotient separately to $\Omega_b \subset [0, 1]^{r_b}$ with $\text{vol}(\Omega_b) \geq [2r_b(r_b + 1)]^{-r_b}$. The lower contact proof now covers the product of these volumes; its additional loss is at most $\sum_b r_b \log_2 [2r_b(r_b + 1)]$. The upper grid is built on the containing product cube and restricted by the original continuous input variables. Applying the preceding finite bounds with dimensions r_b proves (70). Zero-rank blocks remain only in affine output terms and cost no integers. $\square$

For separate component tolerances there is also a direct product-cone version of the geodesic algorithm. With $E_j(P) = \sum_b \text{tr}(H_{jb}P_b)$ use

$$h(P) = \max\{0, \max_b \log \lambda_{\max}(P_b), \max_{j: E_j(I) > 0} \log(E_j(P)/\varepsilon_j)\}, \quad F(P) = - \sum_b \log \det P_b + Nh(P).$$

The repair is $P_b \mapsto e^{-h(P)} P_b$. Along a block geodesic the nonzero trace energies are positive-coefficient exponential sums, and their normalized gradient blocks are $P_b^{1/2} H_{jb} P_b^{1/2} / E_j(P)$ with total trace one. Thus the same $2N$ Lipschitz bound applies. A feasible dyadic allocation has $\delta E_j(I) \leq \varepsilon_j$, so the optimizer lies in a polynomial-radius product-cone ball, and $E_j(P) \geq \min_b \lambda_{\min}(P_b) E_j(I)$ controls denominators. Block diagonal matrices form a totally geodesic subspace; logarithms, exponentials and identity-centered radial projection preserve it. Consequently the full inexact proof of theorem 7.3 applies with N replacing n , performing all numerical operations within blocks. This supplies the same rational allocation guarantee. The Euclidean hypograph oracle above additionally handles arbitrary unconditional strong-oracle budgets directly.

9.3 Diagonal Hessians and an explicit linear overhead

For original-axis diagonal positive semidefinite Hessians, the case with no active coordinate is the exact affine graph. Otherwise retain only $r \geq 1$ active coordinates and write $f_j(x) = \frac{1}{2} \sum_i h_{ji} x_i^2 +$

affine, with $h_{ji} \geq 0$. Define

$$D_{\text{tr}} = \max \left\{ \prod_i p_i : 0 \leq p_i \leq 1, H p \in K \right\}, \quad \Phi_{\text{tr}} = -\frac{1}{2} \log_2 D_{\text{tr}}, \quad A_r^{\text{tr}} = r + \log_2 [\omega_r (r+2)^{r/2}].$$

Here $H = (h_{ji})$ is the coefficient array rather than a Hessian matrix. For a tolerance box, the budget is $\sum_i h_{ji} p_i \leq \varepsilon_j$.

Corollary 9.3 (Diagonal trace law). *For any unconditional error body,*

$$\max\{0, \Phi_{\text{tr}} - A_r^{\text{tr}}\} \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \Phi_{\text{tr}} + r, \quad A_r^{\text{tr}} < 4r. \quad (71)$$

Under rational strong-oracle input, a polynomial-time rational MILP has

$$p_{\text{out}} \leq p_{\text{conv}} + 5r + 1. \quad (72)$$

Proof. Use blocks of size one in theorem 9.2: the contact allocation is $p_i = \Sigma_{ii}/4$, and Hadamard's inequality gives the lower constant displayed. Alternatively, this is the direct first-moment identity $\mathbb{E}J = H \text{diag}(\Sigma)/4 \in K$. A feasible allocation has coordinate depths $L_i = \lceil \frac{1}{2} \log_2(1/p_i) \rceil$. Its shared square error is componentwise bounded by $H p/8$, proving the upper bound. The Gaussian volume estimate gives

$$A_r^{\text{tr}} \leq r + \frac{r}{2} \log_2 [2\pi e(1 + 2/r)] < 4r.$$

The scalar case of lemma 9.1 provides rational p with product at least $e^{-1} D_{\text{tr}}$, adding at most $1/(2 \log 2) < 1$ to the real-valued count bound. No Jacobi approximation is needed in the original axes. This proves (72). $\square$

For tolerance boxes an elementary log-coordinate algorithm avoids a matrix or body oracle. Choose $\delta = 2^{-b}$ with $\delta \sum_i h_{ji} \leq \varepsilon_j$, and put $R = rb + 1$. Every optimal allocation has $p_i \geq \delta^r$, so its logarithms lie in $[-R, 0]^r$. Define

$$h(u) = \max\{0, \max_i u_i, \max_{j: \sum_i h_{ji} > 0} \log(\sum_i h_{ji} e^{u_i} / \varepsilon_j)\}, \quad F(u) = -\sum_i u_i + r h(u).$$

The repair $p_i = e^{u_i - h(u)}$ is feasible and has negative log product $F(u)$. Its minimum on the box equals $-\log D_{\text{tr}}$; its subgradients have norm at most $2r$ because each row-branch gradient is a probability vector. Set $D_0 = rR$, $G_0 = 3r$, $\eta = (16G_0^2)^{-1}$ and $T = \lceil 16D_0^2/\eta \rceil$. Starting at zero, approximate branch values within $1/(128r)$ and the full selected subgradient within $1/(64D_0)$, then take a rational step and clip exactly to $[-R, 0]^r$. The approximate subgradient inequality has error at most $1/32$. The usual squared-distance expansion for projected Euclidean subgradients telescopes to mean objective gap at most $1/32 + 1/32 + 1/32 = 3/32$. Choosing the least objective estimate at accuracy $1/64$ gives gap at most $1/8$. All nonzero row sums on the box are at least $e^{-R} \sum_i h_{ji}$, so scalar evaluation has polynomial precision depth. Store gradients on a fixed sufficiently fine dyadic grid; the polynomial number of rational steps has polynomial encoding length. Choose rational $q_i > 0$ with $|\log q_i - u_i| \leq 1/(32r^2)$ and set

$$\kappa = \max\{1, \max_i q_i, \max_j (\sum_i h_{ji} q_i) / \varepsilon_j\}, \quad p_i = q_i / \kappa.$$

This is exactly feasible, and the $2r$ Lipschitz bound gives $-\sum_i \log p_i = F(\log q) \leq -\log D_{\text{tr}} + 1$. Thus this scalar procedure proves the same rational count independently of general weak optimization.

The trace benchmark differs from the Frobenius benchmark. In example 6.4, $D_{\text{tr}} = r^{-r}$ and $\Phi_{\text{tr}} = (r/2) \log_2 r$, which agrees with the true count up to $O(r)$. Also, diagonal structure here means the original box axes, or a signed permutation of them. A simultaneous diagonalization in an arbitrary rotated basis can produce a correlated domain and does not by itself prove (72).

9.4 Independent integer features and forest differences

An explicit domain-volume estimate can preserve a linear overhead even when the square features overlap in the original coordinates. Let $T \in \mathbb{Z}^{r \times n}$ have full row rank, with every row active in

$$f_j(x) = \frac{1}{2} \sum_{i=1}^r a_{ji} (T_i x)^2 + \ell_j^T x + b_j, \quad a_{ji} \geq 0, \quad x \in [0, 1]^n.$$

Put $w_i = \sum_k |T_{ik}| \geq 1$, $l_i = \sum_{k: T_{ik} < 0} T_{ik}$, $C_{ji} = a_{ji} w_i^2$, and

$$\begin{aligned} \Omega &= \{\text{diag}(w)^{-1}(Tx - l) : x \in [0, 1]^n\} \subset [0, 1]^r, & V &= \text{vol}_r(\Omega), \\ D &= \max\{\prod_i p_i : 0 \leq p_i \leq 1, Cp \in K\}, & \Phi &= -\frac{1}{2} \log_2 D. \end{aligned}$$

Theorem 9.4 (Integer-feature precision). *For an unconditional error body K and $r \geq 1$,*

$$\max\{0, \Phi - A_r^{\text{tr}} + \log_2 V\} \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \Phi + r. \quad (73)$$

For rational coefficients and the strong-oracle assumptions on K , a polynomial-time rational MILP satisfies

$$\begin{aligned} p_{\text{out}} &\leq p_{\text{conv}} + A_r^{\text{tr}} + r - \log_2 V + 1/(2 \log 2) \\ &\leq p_{\text{conv}} + 5r + \sum_i \log_2 w_i + 1. \end{aligned} \quad (74)$$

The common nonlinear input rank is r . The feature representation is part of the input; discovering such a positive decomposition is not part of the theorem.

Proof. Choose a nonsingular r -column minor T_I . Fixing the other input coordinates at zero gives a parallelotope contained in Ω , so

$$V \geq \frac{|\det T_I|}{\prod_i w_i} \geq \frac{1}{\prod_i w_i}. \quad (75)$$

The last step uses integrality of its nonzero determinant. A minor can be found by exact elimination, and its actual determinant gives a sharper bound if useful; no volume computation for the whole zonotope is needed. In coordinates $u_i = (T_i x - l_i)/w_i$, the quadratic parts are $\frac{1}{2} \sum_i a_{ji} (w_i u_i + l_i)^2$ with diagonal Hessian entries C_{ji} . Subtracting the original affine output before projecting and restoring it afterward gives exact equality of formulation minima, by the fiber argument in theorem 8.4.

The proof of corollary 9.3 bounds each parity contact inside Ω by $2^{A_r^{\text{tr}}} \sqrt{D}$. Their cover has volume V , which proves the lower bound. The square-prefix construction on the containing cube uses at most $\Phi + r$ binaries, with error dominated by $Cp/8$. Keep original continuous x , its box bounds and the equation for u to restrict to the actual domain. This proves the upper bound and preserves exact graph points. A rational allocation with product at least $e^{-1}D$ from lemma 9.1 adds less than one to the count. Equation (75) proves the second inequality in (74). Finally, the common kernel of the Hessians is $\ker T$: positivity and activity force each $T_i x = 0$ when all quadratic forms vanish. Thus its codimension is r . $\square$

Rational feature rows may be cleared of denominators, with their row scales absorbed into a_{ji} , then divided by their greatest common divisors to make them primitive. These are polynomial-bit operations. The resulting w_i may be large, so the theorem does not imply a uniform linear overhead for arbitrary rational changes of coordinates. With a uniform row-length bound $w_i \leq s$, it gives overhead at most $(5 + \log_2 s)r + 1$, allowing overlapping features and noncommuting Hessians.

For a forest, the row-length estimate has a sharper exact-volume form. Orient a forest on n vertices and let B be its r -row edge incidence matrix, after removing edges absent from every output. If no edge remains, the graph is affine and has an exact LP; assume $r \geq 1$ below. For $f_j(x) = \frac{1}{2} \sum_e a_{je}(x_u - x_v)^2 + \text{affine}$, every incidence row has width two. Let n_c be the component vertex counts, including isolated vertices. Then the normalized incidence domain satisfies

$$V_F = \text{vol}_r((B[0, 1]^n + \mathbf{1})/2) = 2^{-r} \prod_c n_c \geq 2^{-r}. \quad (76)$$

For a tree on s vertices, subtracting the minimum coordinate gives the unique cube representative of its incidence image with minimum zero. Thus the image is tiled by the images of its s faces $x_v = 0$. Each has volume $|\det B_{-v}| = 1$, by repeated leaf-column expansion. Distinct face images overlap only where two coordinates vanish, a set of dimension at most $s - 2$. The image volume is therefore s . Different components give a product, and scaling all r coordinates by one half proves (76). This also proves incidence row independence; positivity and activity make the common nonlinear rank r .

Corollary 9.5 (Forest precision). *In the preceding forest setting, take $C_{je} = 4a_{je}$ in the feature allocation. Equation (73) holds with the exact volume V_F , and rational oracle input admits*

$$p_{\text{out}} \leq p_{\text{conv}} + A_r^{\text{tr}} + r - \log_2 V_F + 1/(2 \log 2) \leq p_{\text{conv}} + 6r + 1.$$

A path or star may have one original-coordinate Hessian block of arbitrarily large rank. Thus the forest conclusion is distinct from the fixed-block-rank guarantee. Cyclic edge systems need not supply independent diagonal square coordinates, so the forest proof does not extend by simply choosing a subset of their edges.

9.5 Why the actual domain must be preserved

Example 9.6 (A thin domain defeats independent-block transfer). Let $L \geq 1$, $\varepsilon = 4^{-L}$, and $\delta = \varepsilon/16$. On the original unit square consider $f_1(x) = x_1^2$ and $f_2(x) = (x_1 + \delta x_2)^2$ with separate tolerance ε . The coordinates $z_1 = x_1$, $z_2 = (x_1 + \delta x_2)/(1 + \delta)$ map the square to a parallelogram $\Omega \subset [0, 1]^2$ of area $\delta/(1 + \delta)$. The outputs become the independent positive squares $g_1(z) = z_1^2$, $g_2(z) = (1 + \delta)^2 z_2^2$.

On the actual domain, L binaries suffice: a shared square-prefix approximation v of x_1^2 has error at most $\varepsilon/4$. Introduce continuous $0 \leq d \leq 2\delta + \delta^2$ and set $w_1 = v$, $w_2 = v + d$. Every exact graph point is retained by taking $d = 2\delta x_1 x_2 + \delta^2 x_2^2$, and the second output error is at most $\varepsilon/4 + 2\delta + \delta^2 < \varepsilon$. Conversely, on the entire product square, each parity contact has coordinate widths at most $2\sqrt{\varepsilon}$ and $2\sqrt{\varepsilon}/(1 + \delta)$. Its area is at most $4\varepsilon/(1 + \delta)$, so

$$p_{\text{conv}}(g, [0, 1]^2) \geq \log_2 \frac{1 + \delta}{4\varepsilon} > 2L - 2.$$

Replacing the actual domain by its enclosing product therefore creates an unbounded count gap in fixed dimension. Original disjoint blocks, the product-zonotope argument, and the domain-volume terms above are substantive hypotheses. Copy variables do not supply independent domains. For $\delta = 1/M$, the primitive integer features are $(1, 0)$ and $(M, 1)$, whose row-length term $\log_2(M + 1)$ is consistent with this obstruction.

10 Limits on approximating the minimum integer count

The preceding algorithms construct a formulation with a near-minimal integer count. This is separate from optimizing a nonlinear objective or solving the resulting MILP. Even approximating the minimum count itself has a computational barrier.

For an unweighted graph $G = (V, E)$, let

$$f_G(x) = \sum_{\{i,j\} \in E} (x_i - x_j)^2, \quad x \in [0, 1]^n,$$

and let $M(G)$ be its maximum cut size. Successively maximizing over one cube coordinate moves it to an endpoint without decreasing a convex function, so $\max f_G = M(G)$. Also $f_G(1 - x) = f_G(x)$ and $f_G(\frac{1}{2}\mathbf{1}) = 0$.

Lemma 10.1 (Zero dimension). *At positive scalar tolerance ε ,*

$$p_{\text{conv}}(f_G) = 0 \iff p_{\text{bin}}(f_G) = 0 \iff M(G) \leq \varepsilon.$$

Proof. If $M(G) \leq \varepsilon$, the LP $x \in [0, 1]^n$, $0 \leq w \leq \varepsilon$ contains the graph and has error at most ε . Conversely, a zero-integer convex lift contains the midpoint of witnesses for the maximum-cut vertices v and $1 - v$. Its visible point is $(\frac{1}{2}\mathbf{1}, M(G))$, with error $M(G)$. $\square$

Taking $\varepsilon = k - 1/2$ for an integer cut threshold $k \geq 1$ makes zero dimension equivalent to $M(G) < k$. Thus zero recognition is coNP-complete on this restricted graph-quadratic family, using the established unweighted Max-Cut completeness theorem [18]. This is not a coNP-membership assertion for arbitrary convex-lift minimization. Any polynomial-time valid-formulation algorithm with a finite multiplicative approximation guarantee on all instances would distinguish zero from positive optimum, and hence would imply $P = NP$.

Theorem 10.2 (Positive-optimum approximation hardness). *For every fixed $\delta > 0$, unless $P = NP$, no polynomial-time algorithm always constructs a valid graph relaxation with count at most $p_{\min} + O(N^{1-\delta})$, where N is the input dimension and $p_{\min}$ is either p_{conv} or p_{bin} . Even under the promise $p_{\min} \geq 1$, neither that additive guarantee nor a multiplicative factor $O(N^{1-\delta})$ is possible. These conclusions hold for rational convex quadratic outputs on disjoint input blocks, with every component tolerance fixed at one.*

Proof. Replicate the graph quadratic on t independent input blocks and give each output tolerance $k - 1/2$. If $M(G) < k$, the product of the zero-integer LPs in lemma 10.1 has optimum zero. If $M(G) \geq k$, choose a maximum-cut vertex v and independently select v or $1 - v$ in each block. This gives 2^t exact graph points, any two of which have an inadmissible graph midpoint in some output. Any integer lift therefore needs distinct parities for these points, forcing $p \geq t$. The argument uses no boundedness of integer ranges or continuous lift size. Dividing each graph output by $k - 1/2$ changes its tolerance to one and preserves the two cases, with polynomial rational encoding.

To obtain a positive optimum, append $z \in [0, 1]$ and the output $8z^2$ with unit tolerance. This scalar graph has minimum exactly one: its endpoint midpoint has error two, while one binary β suffices by

$$z = (\beta + r)/2, \quad 0 \leq r \leq 1, \quad v = \beta r, \quad s \geq 0, \quad s \geq 2r - 1, \quad s \leq r, \quad w = 2(\beta + 2v + s).$$

Represent $v = \beta r$ by its exact linear binary-product inequalities. The residual square triangle has error at most $1/4$, so this output has error at most $1/2$ and contains the exact graph. In the low-Max-Cut case the full optimum is one. In the high case the extra scalar endpoints double the incompatible packing, forcing $p \geq t + 1$.

It suffices to consider $0 < \delta \leq 1$. If an additive guarantee has constant C , choose a fixed integer $q > 1/\delta$ and $t = n^q$. For all sufficiently large n , $C(tn + 1)^{1-\delta} < t$. Thus in the low case an additive algorithm returns fewer than $t + 1$ integers, and so does a multiplicative algorithm with that factor.

In the high case every valid result uses at least $t + 1$. This decides Max-Cut in polynomial time, since replication is a fixed polynomial in n . Finitely many smaller dimensions may be handled by enumeration or isolated-vertex padding. Larger δ would imply an already excluded guarantee. The same reasoning without the scalar appendage proves the unrestricted additive assertion. $\square$

Since the common nonlinear input rank is at most N , replacing N by that rank in any of these power-sublinear guarantees would also contradict theorem 10.2. This does not exclude, for example, an additive $N/\log N$ algorithm and does not prove a sharp linear barrier. Together with theorem 8.4, it locates the attainable dimension exponent up to subpolynomial factors. The separate geometric example 6.4 explains the logarithm lost by the particular Frobenius benchmark, not a universal impossibility for every other construction or invariant.

11 Scalar convex geometry at finite accuracy

For one convex output on an interval, chord geometry gives a uniform comparison with arbitrary convex integer lifts: two extra binaries suffice at every positive accuracy. We first prove this geometric statement, then develop the certified computations needed to obtain a short rational formulation. The latter requirement is substantive: a partition with a small logarithm of its cell count may still be too long to enumerate.

For a continuous convex function $f : [0, 1] \rightarrow \mathbb{R}$, let $N_\eta(f)$ be the smallest number of intervals in a partition of $[0, 1]$ on each of which the chord of f has error at most $\eta > 0$. This number is finite by uniform continuity. Chord errors decrease when an interval is restricted: the chord on a subinterval lies below the original chord and above f . Affine terms change neither these errors nor the formulation dimensions.

Lemma 11.1 (Chord refinement and scalar packing). *For every continuous convex f and $\eta > 0$,*

$$N_\eta(f) \leq 3N_{2\eta}(f), \quad N_{2\eta}(f) \leq 2^{p_{\text{conv}}(f;\eta)}, \quad p_{\text{bin}}(f;\eta) \leq \lceil \log_2 N_\eta(f) \rceil. \quad (77)$$

Consequently

$$p_{\text{conv}}(f;\eta) \leq p_{\text{bin}}(f;\eta) \leq p_{\text{conv}}(f;\eta) + 2. \quad (78)$$

There is a finite maximal set of $P \geq 1$ points pairwise satisfying $J_f(a, b) > \eta$. Every such set obeys

$$N_{2\eta}(f) \leq 2P, \quad N_\eta(f) \leq 6P. \quad (79)$$

These are existence statements with real coefficients and unrestricted continuous size.

Proof. On an interval of chord error at most 2η , write g for the chord minus f . It is continuous, concave, nonnegative, and zero at both endpoints. If $\max g > \eta$, its superlevel set $\{g \geq \eta\}$ is an interior interval $[u, v]$, with $g(u) = g(v) = \eta$. Split at u, v . On the outside pieces the new error, g minus its own chord, is at most $g \leq \eta$; on the middle piece it equals $g - \eta \leq \eta$. If $\max g \leq \eta$ no split is needed. This proves the first inequality.

A parity contact from lemma 2.2 has compact span $[a, b]$. The concave chord gap on this span has maximum at most twice its midpoint value: concavity between a maximizer and the farther endpoint proves this bound. Thus its chord error is at most 2η . The at most 2^p spans cover $[0, 1]$. A finite interval cover can be trimmed to a partition with no more intervals: starting at zero, take a covering interval reaching farthest to the right and continue; each resulting piece is contained in a covering member. Hence $N_{2\eta} \leq 2^p$. Each chord band $s - \eta \leq w \leq s$ contains the graph and has

error at most η . These bounded polyhedra have the binary encoding of lemma 2.3, proving the third inequality and (78).

Uniform continuity bounds the cardinality of every incompatible set, since sufficiently close points have gap at most η . Add points until a finite maximal set is reached. For fixed x , its compatible set $\{y : J_f(x, y) \leq \eta\}$ is a closed interval containing x . Indeed $J_f(a, b)$ increases under enlargement of $[a, b]$, by monotone secant slopes; equivalently, smooth approximation reduces this to $\partial_a J_f \leq 0$ and $\partial_b J_f \geq 0$. Maximality says these P intervals cover $[0, 1]$. Splitting each at its selected point gives at most $2P$ intervals of endpoint midpoint gap at most η and chord error at most 2η . Trim the cover and apply three-piece refinement to obtain (79). $\square$

The factor-two chord/midpoint comparison is also explicit in Simchowicz et al. [35, Lemma 2.1]. Their accuracy-dependent local approximation modulus and endpoint corrections concern sampling complexity; the dimension comparison here uses convex integer lifts. The comparison uses the established parity and disjunction mechanisms [25, 39]. Efficiently computing an exponentially long partition is a separate requirement; the finite two-bit bound alone supplies no polynomial-size rational formulation.

11.1 An accuracy-dependent curvature measure

Suppose $f \in C^2([0, 1])$ and $g = f'' \geq 0$ is nondecreasing. Define

$$A_\eta(x) = \sqrt{g(x)/\eta}, \quad \rho_\eta(x) = \min\{A_\eta(x), (1-x)A_\eta(x)^2\}, \quad M_\eta = \int_0^1 \rho_\eta(x) dx. \quad (80)$$

The linear branch suppresses curvature contributions too small to require separate intervals at the specified accuracy.

Theorem 11.2 (Truncated curvature mass). *Under these assumptions,*

$$N_\eta \leq 2M_\eta + 1, \quad M_\eta \leq 24N_\eta, \quad (81)$$

and

$$\log_2(1 + M_\eta) - \log_2 49 \leq p_{\text{conv}}(f; \eta) \leq p_{\text{bin}}(f; \eta) \leq \log_2(1 + M_\eta) + 2. \quad (82)$$

For every $[a, b] \subset [0, 1]$, with $m = \int_a^b \rho_\eta$ and $E = \int_a^b (b-t)A_\eta(t)^2 dt$, one has

$$E \leq m^2 + \frac{3}{2}m. \quad (83)$$

Proof. Suppress the subscript on A . On the set $(b-t)A(t) < 1$, the integrand of E is at most $\rho_\eta(t)$. On its complement, put $z = (b-t)A(t) \geq 1$. Monotonicity gives

$$m \geq \int_t^b \min\{A(s), (b-s)A(s)^2\} ds = z - \frac{1}{2}.$$

There $\rho_\eta(t) = A(t)$, so the contribution to E is at most $(m + 1/2)m$. This proves (83). If $m \leq 1/2$, then $E \leq 1$. Convexity bounds the chord error by the left-end Taylor remainder ηE . Dividing total mass into pieces of mass at most $1/2$, and absorbing zero-density intervals, proves the first bound in (81). Zero mass means f is affine.

For the reverse estimate, take an admissible interval, put $\ell = b - a$, $c = (a + b)/2$, and write $J = J_f(a, b) \leq \eta$. The right half of the tent-kernel representation gives $J \geq g(c)\ell^2/16$. The left half

of the Taylor remainder is at most $3g(c)\ell^2/8 \leq 6J$, and its right half is at most $2J$. Thus $E \leq 8$. Introduce the nonnegative potential $\chi(x) = (1-x)A(x)$. We claim for every interval that

$$m \leq \chi(b) - \chi(a) + 2E + 2\sqrt{2E}. \quad (84)$$

Put $d = 1 - b$. If $\ell \leq d$, then $m \leq \ell A(b)$ and

$$m - \chi(b) + \chi(a) \leq 2\ell A(a) - (d - \ell)(A(b) - A(a)) \leq 2\ell A(a) \leq 2\sqrt{2E}.$$

If $\ell > d$, split at $b - d$. On the left part, $\rho_\eta(t) \leq (1-t)A(t)^2 \leq 2(b-t)A(t)^2$, whose integral is at most $2E$. On the right its integral is at most $dA(b) = \chi(b)$. Hence $m - \chi(b) + \chi(a) \leq 2E + \chi(a) \leq 2E + 2\ell A(a) \leq 2E + 2\sqrt{2E}$. This includes $b = 1$. On an admissible interval (84) is at most the potential difference plus 24. Summing over a partition telescopes, with $\chi(1) = 0 \leq \chi(0)$, and proves $M_\eta \leq 24N_\eta$.

Pointwise $\rho_\eta \leq 2\rho_{2\eta}$. By lemma 11.1, $M_\eta \leq 48 \cdot 2^{p_{\text{conv}}(f;\eta)}$. Since $p \geq 0$, this yields the lower dimension bound. The upper follows from $N_\eta \leq 2(1 + M_\eta)$ and binary chord-band encoding. $\square$

Example 11.3 (Curvature and allocation obstructions). Let $f_M(x) = M^{-1} \sum_{j=1}^M x^{2^j}$, of degree $D = 2^M$. At fixed $\eta = 1/16$,

$$\int_0^1 \sqrt{f_M''(x)} dx \geq \frac{\sqrt{M}}{4\sqrt{2}}, \quad p_{\text{conv}} \leq p_{\text{bin}} \leq 4. \quad (85)$$

Indeed, on the disjoint layer $[1 - 1/k, 1 - 1/(2k)]$, $k = 2^j$, the j th monomial contributes curvature at least $k^2/(8M)$. Its length is $1/(2k)$; summing the resulting integrals proves the lower bound. The curvature estimate uses $k(k-1) \geq k^2/2$ and $(1 - 1/k)^{k-2} \geq 1/4$. The function increases from zero to one; inverse images of the sixteen equal output intervals yield sixteen chords with error at most $1/16$. Thus raw curvature arclength cannot characterize finite-accuracy count uniformly over functions. In contrast, $M_\eta \leq \eta^{-1}[f_M(1) - f_M(0) - f'_M(0)] = \eta^{-1}$.

At the different tolerance $\eta = 1/(512M)$, the same family satisfies

$$\lceil \log_2 M \rceil \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \lceil \log_2(512M) \rceil. \quad (86)$$

For the lower bound, choose $x_j = 1 - 2^{-j}$. If $j < \ell$ and $k = 2^j$, then $x_\ell - x_j \geq 1/(2k)$ and the selected monomial has curvature at least $k^2/8$ throughout that interval. Its midpoint gap is at least $1/256$, so $J_{f_M}(x_j, x_\ell) \geq 1/(256M) > \eta$. Parity proves the claim. The upper again partitions the range, now into $512M$ pieces. The coefficient-sum allocation defined in section 13 has $\Phi = \frac{1}{2} \log_2(512M)$ here. Consequently $p_{\text{conv}} - \Phi = \frac{1}{2} \log_2 \log_2 D + O(1)$, with matching upper and lower orders. This limits that allocation benchmark; it does not force a degree penalty relative to the true minimum.

12 Certified scalar compilation

A short indexed computation can describe exponentially many intervals. The construction has three parts: compute one cell from its index, interpolate its endpoints with continuous variables, and add a certified vertical band containing the graph. Only the cell index is declared integral. The geometric lower bounds then compare this count with all convex integer lifts, including those with unrestricted continuous size. The following established circuit principle specifies exactly which variables count as integers. Circuit-to-polyhedron compilation is explicit in Avis et al. [3, Section 3, Lemma 1]; products of discrete function values with continuous weights also have direct base-two formulations [1, Section 2].

Lemma 12.1 (Indexed endpoint compiler). *Fix rational instance data and an L -bit index, possibly restricted to $0 \leq k < K \leq 2^L$. Suppose a deterministic algorithm computes the two endpoint vectors of cell k in time polynomial in the instance length and L , with polynomial output length and a known common denominator of polynomial bit length for each output coordinate. Their line segment, with one common interpolation parameter, has a polynomial-size rational linear formulation using only the L index binaries. All internal wires and interpolation products are continuous variables. The construction is uniform in the stated running-time bound.*

Proof. Unroll the bounded computation into a polynomial-size Boolean circuit, with the instance as constants. Fix its worst-case running time, padding early exits with flags. Use continuous wires in $[0, 1]$, NOT equations $v = 1 - u$, and AND inequalities

$$0 \leq v, \quad v \leq u_1, \quad v \leq u_2, \quad v \geq u_1 + u_2 - 1.$$

Induction along the acyclic circuit forces every wire to its exact Boolean value for integral input bits. A comparison circuit excludes invalid indices. For $0 \leq \theta \leq 1$ and every endpoint output bit v , impose $0 \leq t \leq \theta$, $t \leq v$, $t \geq \theta + v - 1$. Then $t = \theta v$ exactly. Decode the fixed-denominator endpoint numerators linearly and form $(1 - \theta)a_k + \theta b_k$ using those products. Signed outputs are represented by a fixed rational offset and a nonnegative numerator. A fixed-width rational denominator can be incorporated in the linear decoding coefficients. No gate requires an additional integrality declaration. $\square$

12.1 Certified integration of monotone polynomial curvature

The next proof gives bit bounds, including the case of signed coefficients. Its root-isolation import is Sagraloff and Mehlhorn [33, Theorem 36 in the linked preprint]: integer univariate roots can be isolated and refined in time polynomial in degree, coefficient length, and requested precision bits. Rational denominator clearing and square-free preprocessing have polynomial cost. Exact sign testing then follows by evaluating at rational endpoints and between isolated roots; values at roots themselves are zero. Gaussian positivity and polynomial exactness are classical [28, Section 3.5(v)].

Lemma 12.2 (Cumulative curvature and inverse quantiles). *Let f be a densely encoded rational polynomial with $f'' \geq 0$ nondecreasing on $[0, 1]$, and let $\eta > 0$ be rational. For $F(x) = \int_0^x \rho_\eta(t) dt$, rational prefix queries x and requested absolute errors $\zeta > 0$ admit certified rational estimates in polynomial bit time. If f is nonaffine, the normalized quantile $F^{-1}(\theta F(1))$, for rational $\theta \in [0, 1]$, admits a dyadic approximation within any rational input tolerance $\delta > 0$ in polynomial bit time. The dependence on reciprocal tolerances is through their bit lengths; the dependence on degree is numerical, as appropriate for dense input. Endpoint quantiles are exact. All quantile outputs at a fixed requested accuracy can use a common dyadic precision.*

Proof. Put $H = f''/\eta$. Recognize $H = 0$ exactly and return zero integrals; the quantile assertion excludes this case. Otherwise set $d = \deg H$, $d_0 = \max(1, d)$, and $U = \max(1, H(1))$. Monotonicity and nonnegativity imply $H(x) > 0$ for $0 < x \leq 1$, since a zero at a positive point would force a polynomial to vanish on an interval. Both ρ_η and $\sqrt{H}$ are bounded by U . We may replace $\zeta > 1$ by one. Choose $\lambda = 2^{-J}$, $J \geq 1$, with $\lambda \leq \zeta/(16U)$ and omit $[0, \lambda]$, losing mass at most $\zeta/16$. The square-root branch is selected precisely where

$$P(x) = (1 - x)^2 H(x) - 1 \geq 0.$$

This is a nonzero polynomial because $P(1) = -1$. Isolate its distinct roots in disjoint rational neighborhoods of total length at most $\zeta/(16U)$. Omit those neighborhoods and determine the

branch on each remaining interval by a rational sign test. Tangencies and multiple roots are handled by square-free isolation. The branch $(1-x)H(x)$ has an exact rational antiderivative.

We first describe the simpler analytic panels available when H has nonnegative coefficients. Partition each dyadic interval $[2^{-j-1}, 2^{-j}]$, $0 \leq j < J$, into $16d_0$ equal intervals. For any subinterval of one panel, with center c and length ℓ , $\ell/c \leq 1/(16d_0)$. On $|z-c| \leq \ell$, every monomial $(z/c)^k$ has argument at most $1/8$ in absolute value and modulus less than two. Positive coefficients put $H(z)$ in an open right-half-plane sector, and $|H(z)| \leq 2H(c)$. Thus its principal square root is holomorphic there and bounded by $2U$. There are polynomially many panels. This argument does not extend directly to signed coefficients.

For general coefficients, start with $[\lambda, 1]$ and bisect a panel $I = [a, b]$, center c , length ℓ , until its exact rational Taylor coefficients $H(c+t) = \sum_{k=0}^d h_k(c)t^k$ satisfy

$$\sum_{k=1}^d |h_k(c)|\ell^k \leq H(c)/2. \quad (87)$$

Accepted panels have $\Re H(z) \geq H(c)/2 > 0$ and $|\sqrt{H(z)}| \leq 2U$ on the same disk. To bound the number of tests, let $\alpha_1, \dots, \alpha_d$ be the complex roots, with multiplicity; they are used only in the proof. If every $|c - \alpha_j| \geq 8d\ell$, factorization gives

$$\frac{\sum_{k \geq 1} |h_k(c)|\ell^k}{H(c)} \leq \prod_j \left(1 + \frac{\ell}{|c - \alpha_j|}\right) - 1 \leq e^{1/8} - 1 < \frac{1}{2}.$$

Thus every failed center lies within $8d\ell$ of some root's real part. Centers at one subdivision depth have spacing ℓ , so at most $d(16d+2)$ panels fail at that depth.

There are also only polynomially many depths. Clear denominators to an integer polynomial H_{int} and set $P_0 = H_{\text{int}}x(x-1)$, of degree $n = d+2$, with coefficients bounded by 2^T , $T \geq 1$. Every root has modulus at most 2^{T+1} . The primitive square-free part has leading coefficient dividing that of the primitive P_0 , and all coefficients bounded by 2^{B_0} , where $B_0 = (n+1)T + 2n$ suffices by expansion in its roots. Its nonzero integer discriminant gives the loose separation bound

$$|\alpha - \beta| \geq \sigma_* := 2^{-(B_0+2)n^2}$$

for distinct roots: bound all other differences by 2^{T+2} and the leading-coefficient factor by its height in the discriminant product. A real root of H is outside $(0, 1]$; its distance to $[\lambda, 1]$ is at least $\min(\lambda, \sigma_*)$. A nonreal root is separated from its conjugate, so its imaginary part has modulus at least $\sigma_*/2$. Consequently every root has distance at least $\rho_* := \min(\lambda, \sigma_*/2)$ from this interval. At depth K with $2^{-K} \leq \rho_*/(8d_0)$ every panel passes. Here K is polynomial in the input and precision lengths. The bound on failed panels per depth proves $O(d_0^2 K + 1)$ total tree nodes. For constant $H > 0$ the first panel passes. Exact Taylor tests at these rational centers have polynomial coefficient length.

Intersect the accepted panels with the branch intervals and $[0, x]$. If a subinterval has center c' and length ℓ' , then $|c' - c| + \ell' \leq (\ell + \ell')/2 \leq \ell$; its radius- ℓ' disk remains in the certified disk. On such an interval an order- q Gauss-Legendre rule has error at most $8U\ell'4^{-q}$. Indeed the degree- $(2q-1)$ Taylor polynomial has uniform error at most $4U4^{-q}$, and positivity, exactness, and weight sum ℓ' bound the integral and quadrature errors separately. Taking $q = \lceil \frac{1}{2} \log_2(128U/\zeta) \rceil$ makes their total at most $\zeta/16$, since panel lengths sum to at most one.

We spell out rational conditioning of the rule. Let P_q be the Legendre polynomial on $[-1, 1]$, with nodes r_i , and $H_q = (q+1)(2q)!$. Its coefficient absolute sum is at most H_q . The weights

mapped to an interval of length one satisfy

$$w_i = \frac{1}{(1 - r_i^2)P'_q(r_i)^2}, \quad (qH_q)^{-2} \leq w_i \leq 1. \quad (88)$$

Polynomial exactness follows by dividing by P_q and integrating Lagrange cardinal polynomials; applying exactness to their squares proves positivity, and the constant polynomial gives weight sum one. The derivative bound $|P'_q| \leq qH_q$ proves the lower weight bound. The denominator in (88) is at least one. Polynomial derivative bounds and root refinement therefore give positive rational weight estimates of relative error $\zeta/(64U)$ using polynomially many bits, without constructing a joint algebraic number field.

Put $V = \max(1, \sum_{k \geq 1} k|H_k|)$, a rational derivative bound on $[0, 1]$. Rational evaluations of the monotone H at the endpoints of an isolated node interval enclose its value; their difference is at most V times the width. The inequality $\sqrt{v} - \sqrt{u} \leq \sqrt{v - u}$ for $0 \leq u \leq v$ and rational square-root bisection give absolute function-value error at most $\zeta/64$ with polynomial precision. The positive weight sum bounds all numerical quadrature errors by $\zeta/16$. Exact polynomial integrals, rational sums, and denominator products retain polynomial bit length. Together the cutoff, branch neighborhoods, analytic quadrature, and rational evaluation cost less than $\zeta/4$, leaving slack for a certified error- ζ answer.

Finally let $0 < \delta \leq 1/2$; larger requests can be reduced to this range. Set

$$h_0 = H(\delta/4) > 0, \quad \kappa = (\delta/4) \min(1, h_0), \quad \mu = \delta\kappa/64. \quad (89)$$

On $[\delta/4, 1 - \delta/4]$ both branches of ρ_η are at least κ . Every input interval of length δ contains a subinterval of length $\delta/2$ in this range, so carries mass at least $\delta\kappa/2$. The exact rational h_0 has polynomial encoding even with cancellation. In the positive-coefficient case, one can instead use $h_0 = a(\delta/4)^k$ from any positive monomial ax^k of H , giving the same conclusion without cancellation. Compute both $F(s)$ and $F(1)$ within μ in a dyadic bisection for $F(s) - \theta F(1) = 0$. The difference error is at most 2μ . A certified nonzero sign updates the bracket. If the estimated magnitude is at most 2μ , the true magnitude is at most $4\mu < \delta\kappa/2$, which implies input distance less than δ from the unique quantile. Otherwise $\lceil \log_2(2/\delta) \rceil$ steps suffice. Endpoint quantiles are handled exactly; padding early outputs gives common precision. Since $\log(1/\mu)$ is polynomially bounded, so is the inverse computation. $\square$

12.2 Mass-accurate knots and a seven-bit comparison

For the formulation, accuracy in cumulative curvature mass suffices; the knots need not approximate exact quantiles in input distance. This permits a fixed absolute integration accuracy even where the density is very small. A continuous polygonal path from zero to one still covers the input interval if rounding reverses adjacent knots.

Theorem 12.3 (Compiled monotone curvature). *For a densely encoded rational polynomial with nonnegative monotone curvature on a nondegenerate rational interval and a positive rational tolerance η , a polynomial-time algorithm constructs a polynomial-size rational MILP graph relaxation of absolute error η using at most $p_{\text{conv}}(f; \eta) + 7$ binaries. For nondecreasing curvature on $[0, 1]$, the actual count L satisfies*

$$2^L \leq 5M_\eta + 2 \leq 120N_\eta + 2. \quad (90)$$

Its error is at most $15\eta/16$. Affine functions use no binaries.

Proof. Normalize the rational interval and reflect its coordinate if curvature is nonincreasing. Assume henceforth nonaffine nondecreasing curvature on $[0, 1]$. Put $\zeta = 1/512$. Compute $\widehat{M}$ within ζ using lemma 12.2 and set

$$U = \max(\zeta, \widehat{M} + \zeta), \quad L = \max\{0, \lceil \log_2((5/2)U) \rceil\}, \quad N = 2^L.$$

Then $M_\eta \leq U \leq M_\eta + 1/256$ and $U/N \leq 2/5$. The rational number $B = 1 + \sum_k k(k-1)|c_k|/\eta$ bounds the density. For interior index q , bisect for mass target $t = qU/N$, evaluating F to error ζ . Update the left bracket endpoint when $\widehat{F} + \zeta < t$, and the right endpoint when $\widehat{F} - \zeta > t$; stop otherwise, with mass error at most 2ζ . After $\lceil \log_2(B/\zeta) \rceil + 1$ strict steps stop at the bracket midpoint. When $t \leq M_\eta$ the bracket contains a preimage and has mass width at most ζ . When $t > M_\eta$, every strict step updates the left endpoint and its excess over total mass is at most 2ζ , so the mass error is at most 3ζ . With forced $R_0 = 0, R_N = 1$, all outputs therefore satisfy the common safe certificate

$$|F(R_q) - qU/N| \leq 4\zeta. \quad (91)$$

Common dyadic precision is obtained by padding. No density lower bound is used here, including for targets above the total mass.

Consecutive returned knots, whether ordered, reversed, or equal, enclose mass at most $U/N + 8\zeta \leq 133/320$. By (83), their chord error is at most

$$\eta \left[(133/320)^2 + \frac{3}{2}(133/320) \right] = \frac{81529}{102400} \eta < \frac{13}{16} \eta.$$

Compute each rational endpoint value of f downward to error at most $\eta/8$. Dense rational evaluation and downward dyadic rounding have polynomial bit cost. Add a fixed offset for negative values in the compiler. If y is their interpolant at the same continuous weight as the input, then $-\eta/8 \leq y - f(x) \leq 13\eta/16$. Impose

$$y - 13\eta/16 \leq w \leq y + \eta/8. \quad (92)$$

Every true graph point is admitted: the polygonal input path begins at zero and ends at one, hence covers the interval. Every admitted point has error at most $15\eta/16$. Determinism gives identical computed data for a shared endpoint. Lemma 12.1 uses exactly L binaries, without enumerating the cells.

Since $M_\eta \leq 48 \cdot 2^{p_{\text{conv}}}$, $(5/2)U \leq 120 \cdot 2^{p_{\text{conv}}} + 5/512 < 128 \cdot 2^{p_{\text{conv}}}$, proving the seven-bit count. Also $2^L \leq \max(1, 5U) \leq 5M_\eta + 2$, which proves (90). $\square$

For positive polynomials, endpoint values can alternatively be computed monomial by monomial using the rounded binary-power procedure in lemma 13.3. If $C = \sum_{k \geq 2} c_k > 0$, individual downward error $\min(1, \eta/(8C))$ suffices; restore the affine part exactly at the actual input. This implements the same band and will be useful when numerical degrees are too large for exact rational evaluation.

12.3 Arbitrary convex dense polynomials

Theorem 12.4 (Hybrid convex-polynomial compiler). *Let a densely encoded rational polynomial f be convex on $[0, 1]$. For rational $\eta > 0$, a deterministic polynomial-time algorithm constructs a polynomial-size rational MILP with error at most η and at most $p_{\text{conv}}(f; \eta) + 11$ binaries. Coefficients may have either sign and curvature need not be monotone. Convexity can be a promise or checked by exact univariate sign testing. Affine functions are exact with zero binaries.*

Proof. Assume degree $D \geq 2$. Set $V = \max(1, \sum_{k=1}^D k|c_k|) \geq \max|f'|$ and choose $h = 2^{-B}$, $B \geq 2$, with $h \leq \eta/(32V)$. The implicit grid has $2^B + 1$ points, but its indices have polynomial length. If a chord interval $[a, b]$ has error at most ξ , every subinterval of $[\max(0, a - h), \min(1, b + h)]$ has error at most $\xi + 2Vh$. Indeed the old and expanded chords differ by at most $2Vh$ at a, b and between them: the function is V -Lipschitz, endpoint value changes are at most Vh , and every chord slope has absolute value at most V . On each added strip the chord error is at most $2Vh$, and restriction does not increase it. Round an optimal $\eta/4$ partition down to the grid, keep zero and one, and discard repeated knots. Each resulting interval lies within one of the original intervals expanded by h : use the last original knot with its rounded left value. Thus there is a grid partition with at most $N_{\eta/4} \leq 9N_\eta$ cells of error at most $5\eta/16 < \eta/2$.

Greedily extend each grid interval to the farthest feasible endpoint at error $\eta/2$. Binary search uses $O(B)$ exact decisions of the polynomial predicate $s_{a,b} - f - \eta/2 \leq 0$ on $[a, b]$. Root isolation and sign testing decide it in polynomial bit time, including equality. Feasibility is monotone in the endpoint and a one-step interval is always feasible. The greedy partition minimizes count on this grid: by induction its k th endpoint is no earlier than that of any feasible partition, since the remaining part of a comparison interval is feasible by restriction. Its eventual count G satisfies

$$G \leq N_{\eta/4} \leq 9N_\eta. \quad (93)$$

Run only until completion or $9D$ cells. If it finishes, enumerate these polynomially many rational bands, using endpoint rounding and (92). Here $G \leq 27 \cdot 2^{p_{\text{conv}}}$, so at most $p_{\text{conv}} + 5$ binaries suffice.

If it has not finished, (93) proves $N_\eta > D$. Isolate the distinct roots of f''' in $(0, 1)$ in disjoint rational brackets of width at most h , with nonroot rational endpoints. Endpoint roots need no bracket; if $f''' = 0$ none are needed. Together the brackets and their complementary intervals form $b \leq 3D$ nondegenerate pieces. Every bracket has chord error at most $2Vh \leq \eta/16$ and uses one cell. On each complement f'' is monotone and nonnegative, so theorem 12.3 applies after rational affine normalization and, if needed, reflection. All endpoints, sign decisions, and substitutions have polynomial bit cost. The logarithm of root separation bounds the required precision; no uniform grid at inverse root separation is enumerated.

Let n_j be the optimal η chord count on piece j . Splitting a global optimal partition at its boundaries gives $\sum_j n_j \leq N_\eta + b - 1$. Local cell counts are at most $120n_j + 2$, including the one-cell brackets. Therefore

$$K \leq 120N_\eta + 122b \leq 120N_\eta + 366D < 486N_\eta \leq 1458 \cdot 2^{p_{\text{conv}}} < 2048 \cdot 2^{p_{\text{conv}}}. \quad (94)$$

Compute local counts and cumulative sums in binary. A single global index $0 \leq k < K$ selects a piece and its local index by comparisons and subtraction; unused binary codes are excluded. There are only polynomially many pieces, so running each indexed local routine and selecting its output is polynomial time. Reversal maps local knots back to their original rational interval. A common input denominator is the product of piece-endpoint denominators times the largest local dyadic denominator, of polynomial length. Endpoint values of f are rounded downward within $\eta/8$ using a fixed dyadic precision and offset. Apply lemma 12.1 and (92). Each local path covers its piece; the union covers $[0, 1]$ with error at most $15\eta/16$. The global index uses $\lceil \log_2 K \rceil \leq p_{\text{conv}} + 11$ binaries. $\square$

Greedy maximal segmentation is established in Codsí et al. [13, Proposition 1 and Corollary 1]. Their convex-corridor algorithm already yields a continuous PWL approximation (Remark 3), and computes each maximal segment with logarithmically many function and derivative oracle calls at its requested endpoint precision (Section 4.1). Their Lemma 4 bounds the extra segments caused

by splitting at changes of convexity by the number of split points. Here the split instead makes the curvature monotone. The hybrid's additional content is the lower-bound test that absorbs this splitting cost, together with certified rational random access to the remaining cells. It gives the eleven-bit comparison with arbitrary convex lifts in polynomial bit time for dense input; continuity, greedy segmentation, and logarithmic binary encoding are established ingredients.

12.4 Separable sums and independent outputs

Lemma 12.5 (Jensen superadditivity). *For every continuous convex ϕ and ordered points $a = x_0 < \dots < x_h = b$,*

$$J_\phi(a, b) \geq \sum_{j=0}^{h-1} J_\phi(x_j, x_{j+1}). \quad (95)$$

For $\phi(x) = \sum_{k \geq 2} c_k x^k$, $c_k \geq 0$, its feature curve $\psi(x) = (\sqrt{c_k} x^{k/2})_{k:c_k > 0}$ also satisfies

$$\frac{1}{4} \|\psi(a) - \psi(b)\|_2^2 \leq J_\phi(a, b) \leq \frac{1}{2} \|\psi(a) - \psi(b)\|_2^2, \quad (96)$$

and its squared secant lengths are superadditive along ordered points.

Proof. For C^2 convex ϕ , $J_\phi(a, b) = \frac{1}{2} \int_a^b \min(t - a, b - t) \phi''(t) dt$. The long tent dominates each shorter tent, whose interiors are disjoint. The same proof applies to a convex piecewise-linear function expressed as an affine term plus nonnegative hinges. Uniform convex piecewise-linear interpolation then proves (95) for every continuous convex function. For (96), convexity of $x^{k/2}$ gives the lower bound termwise, and $((a + b)/2)^k \geq (ab)^{k/2}$ gives the upper. Feature increments have nonnegative coordinates; expanding a squared sum of increments proves the last statement. Irrational feature coefficients occur only in the geometric proof. $\square$

Theorem 12.6 (Separable comparisons). *On the product box $[0, 1]^r$, consider either $f(x) = \ell(x) + \sum_i \phi_i(x_i)$ with scalar absolute tolerance η , or independent outputs $f_i(x) = \ell_i(x) + \phi_i(x_i)$ with separate tolerances η_i . Each ℓ_i is affine. For continuous convex summands and unrestricted real-coefficient size,*

$$p_{\text{bin}} \leq p_{\text{conv}} + 7r \quad \text{for sums}, \quad p_{\text{bin}} \leq p_{\text{conv}} + 4r \quad \text{for independent outputs.} \quad (97)$$

For the computational statements, assume rational polynomial data, including the affine terms, and positive rational tolerances. For dense positive polynomials the polynomial rational construction gives respectively $p_{\text{out}} \leq p_{\text{conv}} + 12r$ and $p_{\text{out}} \leq p_{\text{conv}} + 9r$. For arbitrary rational dense convex polynomials, it gives respectively $p_{\text{out}} \leq p_{\text{conv}} + 16r$ and $p_{\text{out}} \leq p_{\text{conv}} + 13r$. Affine-only coordinates are retained exactly; if $r = 0$, no integers are needed.

Proof. At local tolerance τ_i , take a maximal scalar incompatible set of size P_i as in lemma 11.1. For independent outputs use $\tau_i = \eta_i$. Any distinct product points violate some output's midpoint tolerance, so $2^{p_{\text{conv}}} \geq \prod_i P_i$. For sums take $\tau_i = \eta/r$. Adjacent selected points have gap strictly greater than τ_i ; lemma 12.5 gives gap greater than $\tau_i h$ for index distance $h > 0$. Select product index vectors greedily, deleting the lattice ball of ℓ_1 radius r after each selection. Such a ball has at most

$$\#\{z \in \mathbb{Z}^r : \|z\|_1 \leq r\} \leq 2^r \sum_{z \in \mathbb{Z}^r} 2^{-\|z\|_1} = 6^r$$

points. The retained collection has at least $\prod_i P_i / 6^r$ points with pairwise scalar gaps exceeding η . Thus

$$2^{p_{\text{conv}}} \geq \prod_i P_i / 6^r \quad \text{for sums.} \quad (98)$$

These packings prove lower bounds and are not enumerated by the construction.

Write L_i for the actual local number of binaries. Finite chord encoding has $2^{L_i} \leq 2N_{\tau_i} \leq 12P_i$. For the positive curvature compiler, $M_{\tau_i} \leq 2M_{2\tau_i} \leq 48N_{2\tau_i} \leq 96P_i$; its definition gives $2^{L_i} \leq \max(1, 5U_i) \leq 481P_i$. For the hybrid, either branch has at most $486N_{\tau_i}$ cells, so $2^{L_i} \leq 972N_{\tau_i} \leq 5832P_i$. Sum the logarithms. Independent outputs incur respectively $\log_2 12 < 4$, $\log_2 481 < 9$, and $\log_2 5832 < 13$ per coordinate. For sums (98) adds $\log_2 6$, yielding bounds below 7, 12, and 16. Compose the local graph relaxations and restore the affine outputs. Local errors sum to at most η in the sum model and remain within their separate tolerances in the independent model. All dense polynomial constructions have polynomial rational size and time; replacing η by η/r adds only $O(\log r)$ precision bits. $\square$

13 Positive polynomial systems and error allocation

Consider the multi-output system

$$f_j(x) = \ell_j(x) + \sum_{i=1}^r \sum_{k=2}^{D_i} c_{jik} x_i^k, \quad c_{jik} \geq 0, \quad x \in [0, 1]^r, \quad (99)$$

where every coordinate is nonlinear in some output and ℓ_j is affine. Inactive coordinates can be retained continuously. Let K be an unconditional error body, meaning it is invariant under every coordinate sign change. Convexity then implies

$$u \in K, \quad |v_j| \leq |u_j| \quad (1 \leq j \leq m) \implies v \in K : \quad (100)$$

the box generated by sign changes of u lies in K . Set $C_{ji} = \sum_k c_{jik}$ and

$$D_K = \max \left\{ \prod_i p_i : 0 \leq p_i \leq 1, \quad C p \in K \right\}, \quad \Phi_K = -\frac{1}{2} \log_2 D_K. \quad (101)$$

There is a strictly positive feasible allocation because zero is interior to K . An optimizer is positive. The case $r = 0$ is affine and uses no integers; all subsequent statements in this section assume $r \geq 1$. For a tolerance box the constraint is $\sum_i C_{ji} p_i \leq \eta_j$. For computational assertions data are rational, and K has a polynomial rational strong separation oracle with known positive rational inner and outer radii. Its scalar specialization in lemma 9.1 returns an exactly feasible positive rational p with

$$\prod_i p_i \geq e^{-1} D_K. \quad (102)$$

That proof includes rational logarithmic tangents, an explicit inner ball, and the central-ball repair (67); it requires neither nonnegative C nor unconditional K . The stronger hypotheses here are needed for graph-error geometry, not for allocation optimization. Final formulations use explicit rational output bands, not an exact linear representation of the potentially curved body K .

13.1 A direct transformed-covariance and Taylor construction

Proposition 13.1 (Baseline polynomial allocation). *Define*

$$A_{r,D} = \log_2[\omega_r(r+2)^{r/2}] + \sum_i \log_2 D_i - r/2, \quad B_{r,D} = \sum_i \log_2 D_i + r.$$

Then (99) satisfies

$$\max(0, \Phi_K - A_{r,D}) \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \Phi_K + B_{r,D}. \quad (103)$$

For rational dense input, the constructive count is at most $p_{\text{conv}} + 2 \sum_i \log_2 D_i + 4r + 1$, with polynomial rational size and time. Its direct prefix construction is retained below because it uses exact powers and an elementary Taylor rectangle.

Proof. For $2 \leq k \leq D$ and $a, b \in [0, 1]$, convexity of $x^{k/2}$ and the D/k -Lipschitz property of $u^{D/k}$ give

$$J_{x^k}(a, b) \geq \frac{1}{4}(a^{k/2} - b^{k/2})^2 \geq D^{-2}(a^{D/2} - b^{D/2})^2.$$

Map each compact parity support by the coordinate homeomorphism $t_i = x_i^{D_i/2}$. These transformed supports still cover the unit cube. On a positive-volume support, let Σ be the covariance of its uniform distribution and average over independent points. The original Jensen vector belongs to K and is nonnegative, and its expectation componentwise dominates $2 \sum_i C_{ji} \Sigma_{ii} / D_i^2$. By convexity and (100), $p_i = 2 \Sigma_{ii} / D_i^2$ is feasible; its caps follow from $\Sigma_{ii} \leq 1/4$. The covariance-volume inequality (24) and Hadamard's inequality give

$$\text{vol}(S) \leq \omega_r(r+2)^{r/2} \sqrt{\det \Sigma} \leq \omega_r(r+2)^{r/2} \prod_i (D_i / \sqrt{2}) \sqrt{D_K}.$$

A cover by at most 2^p such sets proves the lower bound. All volumes are in the transformed cube; the homeomorphism is not asserted to preserve volume. Zero-volume supports do not affect the argument.

For a positive allocation p , use original-axis depths $L_i = \lceil \log_2(D_i / \sqrt{p_i}) \rceil$ and widths $h_i = 2^{-L_i}$. Write $x_i = a_i + r_i$, where a_i is the binary prefix and $0 \leq r_i \leq h_i$. Every output has the exact Taylor expression

$$T_j = \ell_j(x) + \sum_{i,k} c_{jik}(a_i^k + k a_i^{k-1} r_i).$$

The coordinate second derivative is at most $D_i^2 C_{ji}$, so $0 \leq f_j - T_j \leq (Cp)_j/2$. The rational rectangle $T_j \leq w_j \leq T_j + (Cp)_j/2$ contains the graph and has $|w - f(x)| \leq Cp/2$, which implies error in K .

The prefix powers require no extra integers. With $a = \sum_{\ell=1}^L 2^{-\ell} z_\ell$, generate $v_0 = 1, t_0 = r$ and

$$v_k = \sum_{\ell} 2^{-\ell} (z_\ell v_{k-1}), \quad t_k = \sum_{\ell} 2^{-\ell} (z_\ell t_{k-1}). \quad (104)$$

Exact binary-times-bounded-continuous hulls, using $0 \leq v_k \leq 1$ and $0 \leq t_k \leq h$, force $v_k = a^k, t_k = a^k r$. Generate through D and $D-1$, respectively, and share these variables across outputs. This uses $O(D_i L_i)$ rows and continuous variables per coordinate. The count is at most $-\frac{1}{2} \log_2 \prod_i p_i + B_{r,D}$. An optimal allocation proves the finite upper. The rational allocation (102) adds only $1/(2 \ln 2)$. Since $A_{r,D} < \sum_i \log_2 D_i + 3r$ by the Gaussian volume bound, the stated constructive comparison follows. All exact rational lengths and recurrence counts are polynomial for dense input. $\square$

Radix disaggregation and polynomial MILP approximation already appear in Teles et al. [37]; the reused binary-product mechanism is established. The substantive comparison here includes a lower bound against every convex integer lift, not just a particular polynomial discretization. For diagonal PSD quadratics the specialized sharper constants remain: with $f_j = \frac{1}{2} \sum_i h_{ji} x_i^2 + \ell_j$, use $C_{ji} = h_{ji}$ in (101). The expected Jensen vector is $C \text{diag}(\Sigma)/4$, and the shared square upper has $|w - f| \leq Cp/8$. Thus

$$\max(0, \Phi_K - A'_r) \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \Phi_K + r, \quad A'_r = r + \log_2[\omega_r(r+2)^{r/2}] < 4r, \quad p_{\text{out}} \leq p_{\text{conv}} + 5r + 1. \quad (105)$$

This agrees with the structured quadratic analysis and extends its coordinatewise bands to the whole unconditional body.

13.2 Supporting scalarization removes degree from the lower bound

Theorem 13.2 (Double-logarithmic degree overhead). *Put $J_i = \lceil \log_2 D_i \rceil$, $S_i = \lceil \log_2(J_i + 1) \rceil$, and $A_r = r/2 + \log_2[\omega_r(r+2)^{r/2}] < 7r/2$. Then*

$$\max(0, \Phi_K - A_r) \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \Phi_K + r + \sum_i S_i. \quad (106)$$

For rational dense input, a polynomial-time rational construction has

$$p_{\text{out}} \leq p_{\text{conv}} + \frac{9}{2}r + \sum_i S_i + 1. \quad (107)$$

The same constructive bound holds for sparse nonnegative polynomial input with exponents in binary, with size and time polynomial in the sparse encoding, by the endpoint implementation below.

Proof. Let p^* maximize $\sum_i \log p_i$. Strict feasibility of a small positive common allocation and the convex normal-cone sum rule give an outward normal $\lambda \in N_K(Cp^*)$ and $\mu_i \geq 0$, where $N_K(z) = \{v : v^T(y - z) \leq 0 \text{ for all } y \in K\}$, such that

$$1/p_i^* = (C^T \lambda)_i + \mu_i, \quad \mu_i(p_i^* - 1) = 0. \quad (108)$$

We may choose $\lambda \geq 0$. At a positive coordinate of Cp^* , downward closure permits a small decrease, forcing the corresponding normal component nonnegative. A zero coordinate is a zero row of C , since $C \geq 0$ and $p^* > 0$. Set its normal component to zero: the support function of an unconditional body is unchanged by sign changes and monotone in absolute coordinates, so the normal inequality remains valid while its value at Cp^* is unchanged. Set $a_i = (C^T \lambda)_i \geq 0$ and $b = \lambda^T Cp^* = h_K(\lambda)$. The scalar allocation $0 \leq p \leq 1$, $\sum_i a_i p_i \leq b$ has the same optimum D_K . Indeed, concavity and (108) give

$$\sum_i \log(p_i/p_i^*) \leq \sum_i (p_i - p_i^*)/p_i^* = a^T(p - p^*) + \sum_i \mu_i(p_i - p_i^*) \leq 0.$$

Points with zero product cannot improve the optimum. If $\lambda = 0$, all $p_i^* = 1$ and the lower bound is trivial. An index with $a_i = 0$ also has $p_i^* = 1$.

For $a_i > 0$, normalize $\phi_i(x) = a_i^{-1} \sum_k (\sum_j \lambda_j c_{jik}) x^k$ and let $g_i = \sqrt{\phi_i}$. This is an increasing homeomorphism of $[0, 1]$ onto itself and is convex: it is the Euclidean norm of nonnegative convex coordinates $\sqrt{d_{ik}} x^{k/2}$ with $\sum_k d_{ik} = 1$. For $a_i = 0$, use $g_i(x) = x$. Since a nonnegative convex g satisfies $J_{g^2}(x, y) \geq (g(x) - g(y))^2/4$, each parity support transformed by $t_i = g_i(x_i)$ obeys $\frac{1}{4} \sum_i a_i (t_i - s_i)^2 \leq b$. Taking independent uniform points gives $\frac{1}{2} \sum_i a_i \Sigma_{ii} \leq b$. Thus $p_i = \Sigma_{ii}/2$ is feasible for the scalar allocation and $\prod_i \Sigma_{ii} \leq 2^r D_K$. The covariance-volume bound now gives $\text{vol}(S) \leq 2^{r/2} \omega_r(r+2)^{r/2} \sqrt{D_K}$. Coverage of the transformed cube proves the degree-independent lower bound. Multipliers and transformed coordinates are proof devices; the construction need not compute or rationalize them.

For the upper, partition each coordinate into $J_i + 1$ layers

$$[1 - 2^{-j}, 1 - 2^{-j-1}] \quad (0 \leq j < J_i), \quad [1 - 2^{-J_i}, 1]. \quad (109)$$

If a layer has left endpoint ℓ and width s , then for every $2 \leq k \leq D_i$ its normalized monomial curvature is

$$k(k-1)s^2 x^{k-2} \leq 2. \quad (110)$$

On the final layer $s \leq 1/D_i$ gives a bound of one. Else $s \leq 1/2$ and $x \leq 1 - s$. For $k = 2$ the bound is at most $1/2$; for $k \geq 3$ put $v = (k-2)s$, giving $k(k-1)s^2(1-s)^{k-2} \leq (v+1)^2 e^{-v} \leq 4/e < 2$.

Encode each layer using S_i binaries and continuous selectors $\theta_j \geq 0$ summing to one, each bounded by the matching bit literals. At integral bits the matching selector is one and all others zero; unused codes are infeasible. Such implied-binary selectors can be used in exact binary-product hulls without additional declared integers. For allocation p_i , set $L_i = \lceil \frac{1}{2} \log_2(1/p_i) \rceil$, $h_i = 2^{-L_i}$. Write the normalized layer input as $a_0 + u$, with L_i prefix bits and $0 \leq u \leq h_i$. In original coordinates impose

$$a = \sum_j \theta_j(\ell_j + s_j a_0), \quad v = \sum_j s_j \theta_j u, \quad x = a + v.$$

For any bounded variable q , the product aq is represented as $\sum_j [\ell_j(\theta_j q) + s_j(\theta_j(a_0 q))]$; $a_0 q$ uses the prefix bits and selector products use their forced binary values. Generate exact $a^k, a^k v$ by (104) with this implementation. Bounds $0 \leq a^k \leq 1$, $0 \leq a^k v \leq h_i$ suffice. At $L_i = 0$, the prefix is zero and those products disappear. The size per coordinate is $O(D_i(L_i + J_i + 1) + J_i S_i)$, polynomial for dense input.

The resulting exact Taylor expression T_j satisfies $0 \leq f_j - T_j \leq \sum_i C_{ji} h_i^2 \leq (Cp)_j$ by (110). Use $T_j \leq w_j \leq T_j + (Cp)_j$. The graph is contained and every error belongs to K by (100). Counting the $L_i + S_i$ bits proves the finite upper in (106). The allocation (102) adds at most $1/(2 \ln 2)$ to the count. Combining with $A_r < 7r/2$ proves (107). The sparse implementation is proved next and uses exactly the same layer and cell bits. $\square$

Lemma 13.3 (Rounded binary powers). *For dyadic $x \in [0, 1]$, integer $k \geq 1$ in binary, and rational $\xi > 0$, downward-rounded repeated squaring returns $A \in [0, 1]$ with $0 \leq x^k - A \leq \xi$ in polynomial time in the input length, $\log k$, and precision encoding. It suffices to use P fractional bits at least the input precision and $\lceil \log_2(k/\xi) \rceil$.*

Proof. All rounded factors stay below their exact values in $[0, 1]$. If factors representing exponents $a, b \geq 1$ have errors at most $(a-1)2^{-P}$ and $(b-1)2^{-P}$, multiplication and one downward rounding have error at most $(a+b-1)2^{-P}$. The exact base has zero error. Induction over $O(\log k)$ products proves error at most $(k-1)2^{-P}$. An initial exact accumulator one requires no rounding when multiplied by an already representable factor. All arithmetic uses polynomially many bits. $\square$

Sparse construction for theorem 13.2. Let E_i list the exponents, in binary. Obtain the same rational allocation, layer count J_i , and depth L_i . The only integer inputs are the S_i -bit layer index $j \leq J_i$ and L_i -bit cell index q . Compute the exact dyadic endpoints $a = \ell_j + s_j q h_i$, $b = \ell_j + s_j (q+1) h_i$ at common precision $J_i + L_i + 1$. For each listed k , compute downward endpoint powers A_{ik}, B_{ik} with error at most $\xi_i = p_i/4$ using lemma 13.3; a common precision $P_i \geq \max(J_i + L_i + 1, \lceil \log_2(D_i/\xi_i) \rceil)$ suffices. Only listed exponents are evaluated. Compile all endpoint values and interpolate with one common weight per input, giving $x_i = (1 - \theta_i)a + \theta_i b$ and $z_{ik} = (1 - \theta_i)A_{ik} + \theta_i B_{ik}$. The exact input grid covers the coordinate interval. The curvature bound (110) and the chord bound yield

$$-\xi_i \leq z_{ik} - x_i^k \leq h_i^2/4.$$

Set $T_j = \ell_j(x) + \sum_{i,k} c_{jik} z_{ik}$ and impose

$$T_j - \sum_i C_{ji} h_i^2/4 \leq w_j \leq T_j + \sum_i C_{ji} \xi_i. \quad (111)$$

These bands contain the graph and have $|w - f| \leq Cp/2$, hence error in K . All circuit computations are polynomial in sparse exponent lengths and allocation length; no exact numerator of a numerical-degree power is formed. By lemma 12.1 the count remains $\sum_i (L_i + S_i)$. $\square$

14 Pure powers and inverse interpolation

For one power per coordinate shared across all outputs,

$$f_j(x) = \ell_j(x) + \sum_{i=1}^r c_{ji} x_i^{\alpha_i}, \quad c_{ji} \geq 0, \quad (112)$$

the degree penalty disappears. We retain both an explicit rational reciprocal construction for dense integer degrees and an indexed construction polynomial in binary rational exponent lengths. The continuous descriptions differ even where their integer counts agree. Throughout, K is unconditional and each active coordinate has some positive coefficient; $r = 0$ is affine and exact.

Lemma 14.1 (Power-coordinate chords). *For $\alpha \geq 2$, a chord of x^α on $[a, b] \subset [0, 1]$ has error at most $4(b^{\alpha/2} - a^{\alpha/2})^2$. Moreover $J_{x^\alpha}(a, b) \geq (a^{\alpha/2} - b^{\alpha/2})^2/4$. For all $\alpha > 1$, set $s = \min(1, \alpha - 1)$. Then*

$$J_{x^\alpha}(a, b) \geq \frac{s}{8}(b^{\alpha/2} - a^{\alpha/2})^2. \quad (113)$$

On a uniform power-coordinate cell $[t, t+h]$, $t = kh$, interpolating $(t^{2/\alpha}, t^2)$ and $((t+h)^{2/\alpha}, (t+h)^2)$ gives

$$0 \leq y - x_0^\alpha \leq 4sh^2. \quad (114)$$

Proof. The stronger Jensen bound for $\alpha \geq 2$ follows from convexity of $x^{\alpha/2}$. Write $A = a^{\alpha/2}$, $B = b^{\alpha/2}$. If $A \leq B/2$, the chord is at most $B^2 \leq 4(B - A)^2$. Otherwise the usual curvature chord bound and the mean-value estimate give error at most

$$\frac{\alpha(\alpha-1)b^{\alpha-2}}{8}(b-a)^2 \leq \frac{\alpha-1}{2\alpha}(b/a)^{\alpha-2}(B-A)^2 \leq 2(B-A)^2,$$

since $(b/a)^{\alpha/2} < 2$. This proves the first assertion and (114) for $\alpha \geq 2$.

If $1 < \alpha < 2$ and $b > 0$, the curvature on (a, b) is at least $\alpha(\alpha-1)b^{\alpha-2}$. Subtract its quadratic and apply convexity, extending to $a = 0$ by continuity. Since $u^{\alpha/2} \geq u$ on $[0, 1]$, $b^{\alpha/2} - a^{\alpha/2} \leq b^{\alpha/2-1}(b-a)$. This proves (113); the zero interval is immediate. For the first power-coordinate cell, $y - x_0^\alpha = h^2(\theta - \theta^\alpha) \leq sh^2\theta(-\log \theta) \leq sh^2/e$. For another cell put $u = t + h \leq 2t$ and $\beta = 2/\alpha \in (1, 2)$. Its input length is at most $\beta u^{\beta-1}h$ and its maximum curvature is $\alpha st^{2-2\beta}$. The chord error is at most $[s/(2\alpha)](u/t)^{4/\alpha-2}h^2 \leq 2sh^2$, proving the safe bound (114). $\square$

Proposition 14.2 (Finite pure-power dimension). *For real exponents $\alpha_i \geq 2$, with $C = (c_{ji})$ and the allocation (101),*

$$\max(0, \Phi_K - A_r) \leq p_{\text{conv}} \leq p_{\text{bin}} \leq \Phi_K + 2r, \quad A_r = r/2 + \log_2[\omega_r(r+2)^{r/2}] < 7r/2. \quad (115)$$

Thus $p_{\text{bin}} \leq p_{\text{conv}} + 11r/2$ with unrestricted real coefficients and size. For all $\alpha_i > 1$, set $s_i = \min(1, \alpha_i - 1)$, $C_{ji}^{\text{eff}} = c_{ji}s_i$, and define Φ_{eff} by (101) with C^{eff} in place of C . Then

$$p_{\text{conv}} \geq \Phi_{\text{eff}} - A'_r, \quad A'_r = r + \log_2[\omega_r(r+2)^{r/2}] < 4r. \quad (116)$$

Proof. Map parity supports by $t_i = x_i^{\alpha_i/2}$, an increasing homeomorphism for every positive exponent. For $\alpha_i \geq 2$, the nonnegative Jensen vector dominates $C[(t-u)^2]/4$. Independent uniform points in a transformed support of covariance Σ give $C \text{diag}(\Sigma)/2 \in K$. Thus $p_i = \Sigma_{ii}/2$ is feasible, and the covariance-volume proof gives the first lower bound. The scaled Jensen inequality instead gives $C^{\text{eff}} \text{diag}(\Sigma)/4 \in K$; using $p_i = \Sigma_{ii}/4$ proves (116). The caps follow from $\Sigma_{ii} \leq 1/4$ in both cases.

For the finite upper, use a positive allocation and power-coordinate widths $h_i = 2^{-L_i}$ with $L_i = \lceil \frac{1}{2} \log_2(1/p_i) \rceil + 1$. The knots $(kh_i)^{2/\alpha_i}$ give chord error at most $4h_i^2 \leq p_i$ by lemma 14.1. Encode each union of chord bands with L_i binaries and set the outputs to the positive combination of these coordinate approximations. Then $|w - f| \leq Cp \in K$ in the domination sense, and the exact graph is contained. Sum the counts at an optimal allocation. The generally irrational knots and potentially exponential row count do not supply a compact rational construction. $\square$

For rational dense integer degrees D_i , the direct original-axis prefix construction also retains the bound $p_{\text{out}} \leq \Phi_K + \sum_i \log_2 D_i + r + 1/(2 \ln 2)$, hence $p_{\text{out}} \leq p_{\text{conv}} + \sum_i \log_2 D_i + 9r/2 + 1$ by the stronger pure-power lower bound. The inverse constructions below remove this degree contribution while changing the continuous representation.

14.1 Positive Stieltjes approximation with rational coefficients

Positive resolvent approximations of fractional powers are established. For example, substituting $\lambda = 1/t$ in Bonito and Pasciak [8, Section 3.3, equation (37) and Lemma 3.4] gives positive sums of $t/(t+b)$ approximating t^γ . The next proof supplies explicit rational coefficient lengths, exact endpoint normalization, and uniform parameter bounds for the present construction.

Lemma 14.3 (Certified rational inverse power). *Given an integer $D \geq 3$ and rational $0 < \delta < 1/2$, one can construct positive rational a_k, b_k such that*

$$R(t) = \sum_{k=1}^M a_k \frac{t}{t + b_k}, \quad R(0) = 0, \quad R(1) = 1, \quad \sup_{0 \leq t \leq 1} |R(t) - t^{2/D}| \leq \delta.$$

The term count is $M = O(D(1 + \log(1/\delta) + \log D)^2)$; time and total coefficient length are polynomial in D and the precision encoding. For $D = 2$ use $R(t) = t$ exactly.

Proof. Put $\gamma = 2/D \in (0, 1)$ and use the Stieltjes integral

$$I_\gamma(t) = \int_0^\infty \frac{t}{t+s} s^{\gamma-1} ds, \quad I_\gamma(0) = 0.$$

Substitution gives $I_\gamma(t) = t^\gamma I_\gamma(1)$, and splitting at one gives $2 \leq I_\gamma(1) \leq D/2 + 3$. Let $e = \min(\delta, 1/4)$, let p be the least integer with $2^{-p} \leq e$, and put $d_* = \lceil \log_2 D \rceil$,

$$L = \lceil (D/2)(p + d_* + 8) \rceil, \quad U = 3(p + 8), \quad N = L + U.$$

The omitted tails below 2^{-L} and above 2^U are bounded, uniformly for $0 \leq t \leq 1$, by $2^{-L\gamma}/\gamma \leq e/512$ and $2^{-U(1-\gamma)}/(1-\gamma) \leq 3e/256$, respectively. Their sum is less than $e/64$.

On each dyadic panel substitute $s = 2^j v$, $1 \leq v \leq 2$, $-L \leq j < U$. The integrand is $2^{j\gamma} v^{\gamma-1} t/(t + 2^j v)$. On $|v - 3/2| \leq 1$ it is holomorphic and has modulus at most four: $\Re v \geq 1/2$, $|v^{\gamma-1}| \leq 2$; for $j \leq 0$ the other factors have product modulus at most one, and for $j \geq 0$ use $|t/(t + 2^j v)| \leq 2^{1-j}$. The $t = 0$ integrand is zero. The degree- $(2q - 1)$ Taylor remainder on $[1, 2]$ is at most $8 \cdot 4^{-q}$. Gaussian positivity and exactness give panel error at most $16 \cdot 4^{-q}$. Taking $q = \lceil [p + \lceil \log_2(256N) \rceil]/2 \rceil$ bounds the sum of these errors by $e/16$. Writing the common nodes and weights as $v_i \in (1, 2)$, $w_i > 0$, $\sum_i w_i = 1$, yields the positive sum

$$Q(t) = \sum_{j=-L}^{U-1} \sum_{i=1}^q A_{ji} \frac{t}{t + B_{ji}}, \quad A_{ji} = 2^{j\gamma} w_i v_i^{\gamma-1}, \quad B_{ji} = 2^j v_i.$$

It satisfies $\|Q - I_\gamma\|_\infty < 5e/64$ and $0 \leq Q(t) \leq Q(1) \leq D + 4$.

Choose $\tau = e/[128(D + 4)]$. Approximate every A_{ji}, B_{ji} by positive rationals with relative error at most τ . A positive term changes relatively by at most 4τ , because its perturbed ratio lies between $(1-\tau)/(1+\tau)$ and its reciprocal. Consequently $\|\hat{Q} - Q\|_\infty \leq e/32$ and $E := \|\hat{Q} - I_\gamma\|_\infty < 7e/64 < e/8$.

We verify polynomial precision, including small weights. With $H_q = (q + 1)(2q)!$, (88) gives

$$2^{-L} \leq B_{ji} \leq 2^U, \quad \frac{2^{-L}}{2(qH_q)^2} \leq A_{ji} \leq 2^U.$$

All logarithmic magnitude bounds are polynomial in L, U, q . Root isolation and rational interval evaluation of the Legendre weight formula therefore give the required relative precision in polynomial bit time. For the remaining powers use $A_{ji} = (w_i/v_i)(2^j v_i)^{2/D}$: certified bisection of a positive D th root solves $z^D = (2^j v_i)^2$. All arguments have explicit positive upper and lower bounds of polynomial bit length. Derivatives of this fixed-depth expression and the root map on those bounds have logarithmic magnitudes polynomial in D, L, U, q ; refining its input intervals and squared comparisons thus costs polynomially many bits. Raising rational test values to degree D has polynomial time and length in numerical D , as asserted. No unbounded-precision real-arithmetic operation is assumed.

Normalize exactly by the positive rational $\hat{Q}(1)$: $a_{ji} = \hat{A}_{ji}/\hat{Q}(1)$, $b_{ji} = \hat{B}_{ji}$. The endpoints of $R = \hat{Q}/\hat{Q}(1)$ are exactly zero and one, and

$$|R(t) - t^\gamma| \leq \frac{2E}{2 - E} \leq e \leq \delta.$$

Positive coefficients make R strictly increasing. Rational sums and normalization retain polynomial total encoding, since denominator products add their bit lengths. Finally $N = O(D(p + \log D))$ and $q = O(p + \log D)$, giving the stated term count. This degree dependence is compatible with dense input, not with a claim polynomial in $\log D$. $\square$

Lemma 14.4 (Rational endpoint products with existing bits). *Let $a = \sum_{\ell=1}^L 2^{-\ell} z_\ell$, $h = 2^{-L}$, and $R = P/Q$ for supplied rational polynomials with a supplied certificate $Q(t) \geq q_{\min} > 0$ on $[0, 1]$. For either $t = a$ or $t = a + h$ and $0 \leq \theta \leq 1$, the value $\theta R(t)$ has an exact linear representation using only the L existing binaries and $O((d + 1)(L + 1))$ rows and continuous variables, where $d = \max(\deg P, \deg Q)$, with the zero numerator treated directly. The coefficient length is polynomial in the supplied data, including $q_{\min}$. Signs of polynomial coefficients are unrestricted.*

Proof. For $t = a + sh$, $s \in \{0, 1\}$ fixed, impose

$$0 \leq v_k \leq 1/q_{\min}, \quad v_k = \sum_{\ell} 2^{-\ell} (z_\ell v_{k-1}) + sh v_{k-1} \quad (1 \leq k \leq d), \quad \sum_{k=0}^d Q_k v_k = \theta.$$

Products use exact bounded binary-product hulls. Recurrences force $v_k = t^k v_0$, so positivity of $Q(t)$ forces $v_0 = \theta/Q(t)$. These values satisfy the bounds; therefore $\sum_k P_k v_k = \theta R(t)$ exactly. The same proof covers $L = 0$, where $a = 0, h = 1$ and the prefix products are absent. Using weights $1 - \theta, \theta$ represents the interpolation of $R(a), R(a + h)$. If $R(0) = 0, R(1) = 1$, its polygonal path covers $[0, 1]$ even when R is nonmonotone; retain $0 \leq x \leq 1$ to clip any excursions. A uniform error $\|R - g\|_\infty \leq \delta$ implies error at most δ between interpolated endpoint values. The lemma supplies neither a denominator-certification algorithm nor a short rational approximation for an arbitrary g . $\square$

Theorem 14.5 (Dense reciprocal construction). *For (112) with integer degrees $D_i \geq 2$, rational dense input and the allocation oracle assumptions, a rational MILP of polynomial size is constructible in polynomial time with $p_{\text{out}} \leq p_{\text{conv}} + 13r/2 + 1$.*

Proof. For a positive feasible rational allocation, set $L = \lceil \frac{1}{2} \log_2(1/p) \rceil + 2$, $h = 2^{-L}$, $\delta = p/(16D)$ in each coordinate. Take the rational approximation of lemma 14.3. Each term has reciprocal variables

$$(a + b_k)v_k^- = 1 - \theta, \quad (a + h + b_k)v_k^+ = \theta, \quad 0 \leq v_k^\pm \leq 1/b_k.$$

Expanding a in its bits makes these exact linear constraints via bounded binary products. They impose

$$x = \sum_k a_k [1 - b_k(v_k^- + v_k^+)] = (1 - \theta)R(a) + \theta R(a + h).$$

For $D = 2$ use $x = a + h\theta$ instead. The auxiliary $y = a^2 + 2ha\theta + h^2\theta$ is linearized with the same bits, since $a^2 = \sum_\ell 2^{-\ell} z_\ell a$ and similarly for $a\theta$. The size is $O((M+1)(L+1))$, and reciprocal bounds have polynomial coefficient length. No new integers occur.

Let $x_0 = (1 - \theta)a^{2/D} + \theta(a + h)^{2/D}$. Then $|x - x_0| \leq \delta$ and lemma 14.1 gives $0 \leq y - x_0^D \leq 4h^2$. Since x^D is D -Lipschitz, $-D\delta \leq y - x^D \leq 4h^2 + D\delta$. Impose the rational band

$$y - (4h^2 + D\delta) \leq q \leq y + D\delta. \quad (117)$$

The increasing endpoint-normalized R covers every input, so the whole graph is admitted, while $|q - x^D| \leq 4h^2 + 2D\delta \leq 3p/8$. Set $w_j = \ell_j(x) + \sum_i c_{ji}q_i$; unconditionality controls all errors. Using (102), the count is at most $\Phi_K + 3r + 1/(2 \ln 2)$. Combine with proposition 14.2. Polynomial numerical-degree complexity follows from the positive rational approximation lemma. $\square$

14.2 Rational powers with binary exponent encoding

Lemma 14.6 (Random access to inverse powers). *For an L -bit dyadic grid and a rational exponent $\alpha > 1$ in binary, one can compute dyadic knots $R_k \in [0, 1]$ within rational error $0 < \delta < 1/2$ of $(k/2^L)^{2/\alpha}$ in time polynomial in L , exponent encoding, and precision encoding. The endpoints are exact, shared indices give identical outputs, and all outputs may use one fixed precision. For integer exponents there is also a direct rounded-power bisection algorithm.*

Proof. The case $L = 0$ consists only of the two exact endpoints. For the direct integer procedure let $D \geq 2$, $0 < k < 2^L$, $u = (k/2^L)^2 \geq 2^{-2L}$, and $\tau = \delta 2^{-2L}/16$. Bisection for $s^D = u$ uses the enclosure $[A, A + \tau]$ from lemma 13.3, at precision $P \geq \max(R + 1, \lceil \log_2(D/\tau) \rceil)$, where $R = \lceil \log_2(2/\delta) \rceil$. A separated enclosure gives a strict sign. If it contains u , then $|s^D - u| \leq \tau$ and both power values are at least $u/2$. On the intervening segment, $Dz^{D-1} \geq z^D \geq u/2$, so the mean-value theorem gives $|s - u^{1/D}| \leq 2\tau/u \leq \delta/8$. Otherwise R steps make the bracket sufficiently short. Padding outputs to $R + 1$ fractional bits works. This requires $O(\log(1/\delta))$ bisections, $O(\log D)$ rounded products per step, and $P = O(\log D + L + \log(1/\delta))$ bits, even for enormous D .

For rational α , put $\beta = 2/\alpha \in (0, 2)$ and normalize the interior dyadic t as $v = 2^e t \in [1, 2)$, $1 \leq e \leq L$. For $w \in [1, 2]$, define

$$H_N(w) = 2 \sum_{j=0}^{N-1} \frac{((w-1)/(w+1))^{2j+1}}{2j+1}.$$

Integrating the geometric series gives $0 \leq \log w - H_N(w) \leq 3 \cdot 9^{-N}$. Choose N with $6(L+1)9^{-N} \leq \delta/8$ and compute the rational $A = \beta(H_N(v) - eH_N(2))$. Then $-2L \leq A \leq 0$ and $|A - \beta \log t| \leq \delta/8$.

Let M be a power of two with $4L \leq M < 8L$, so $q = A/M \in [-1/2, 0]$. For an odd J large enough that $2^{-(J+1)} \leq \delta/(16M)$, the exact rational Taylor sum $P_J(q) = \sum_{j=0}^J q^j/j!$ obeys

$$\frac{1}{2} \leq P_J(q) \leq e^q \leq 1, \quad 0 \leq e^q - P_J(q) \leq \delta/(16M).$$

The first lower bound follows by pairing the terms after $1 + q$; the other bounds are the alternating-series estimate. The M th-power map is M -Lipschitz on $[0, 1]$, and the exponential is one-Lipschitz on the nonpositive axis. Compute $P_J(q)^M$ exactly and round downward with error at most $\delta/4$. Its total error from t^β is at most $\delta/16 + \delta/8 + \delta/4 < \delta$. All series have $O(\log(L+1) + \log(1/\delta))$ terms. Exact rational powering enlarges their bit lengths by at most $M = O(L)$, so the algorithm has polynomial bit cost. For $\alpha \geq 2$, one may use the smaller constants $3(L+1)$, $-L \leq A \leq 0$, and $2L \leq M < 4L$. Endpoints and common output precision are handled separately as above. $\square$

Theorem 14.7 (Binary rational-exponent construction). *For rational (112) with binary-encoded $\alpha_i > 1$ and the unconditional allocation oracle assumptions, a polynomial-time construction gives a rational MILP of polynomial total encoding length with*

$$p_{\text{out}} \leq p_{\text{conv}} + 7r + 1. \quad (118)$$

If all $\alpha_i \geq 2$, the stronger count $p_{\text{out}} \leq p_{\text{conv}} + 13r/2 + 1$ holds. Size depends polynomially on the exponent bit lengths, not their numerical magnitudes.

Proof. For general $\alpha_i > 1$ allocate with $C_{ji}^{\text{eff}} = c_{ji}s_i$, $s_i = \min(1, \alpha_i - 1)$. Set $L_i = \lceil \frac{1}{2} \log_2(1/p_i) \rceil + 2$, $h_i = 2^{-L_i}$, and $\delta_i = s_i p_i / (16\alpha_i)$. The indexed algorithm computes both inverse knots and the exact power-coordinate squares $(kh_i)^2, ((k+1)h_i)^2$. Apply lemma 12.1 to impose their interpolants (x_i, y_i) with one weight. Relative to the exact inverse interpolant x_{0i} , $|x_i - x_{0i}| \leq \delta_i$. By (114) and the α_i -Lipschitz bound,

$$-\alpha_i \delta_i \leq y_i - x_i^{\alpha_i} \leq 4s_i h_i^2 + \alpha_i \delta_i.$$

The band $y_i - (4s_i h_i^2 + \alpha_i \delta_i) \leq q_i \leq y_i + \alpha_i \delta_i$ therefore has error at most $3s_i p_i / 8$. Its graph containment follows from the continuous polygonal path joining exact endpoint knots zero and one; reversed or duplicate interior segments cause no problem. Assemble outputs as positive combinations of the q_i . Every error is dominated by $3C^{\text{eff}} p / 8$, and so lies in K . The count is at most $\Phi_{\text{eff}} + 3r + 1/(2 \ln 2)$; use (116). If all exponents are at least two, take $s_i = 1$ and the stronger lower bound in proposition 14.2. All precisions, including $\log(1/s_i)$ when α_i approaches one, have polynomial rational encoding. Only the external cell bits are integral. $\square$

15 Relative error and rational encoding boundaries

The preceding counts concern positive absolute error. Two distinct limitations arise if the error vanishes at a domain endpoint or if short rational coefficients are required for a sharply varying root.

Proposition 15.1 (Relative accuracy near zero). *For $q > 1$ and any finite $\eta \geq 0$, no convex lift with finitely many integer coordinates can contain all (x, x^q) , $0 < x \leq 1$, and require $w \leq (1 + \eta)x^q$ at every admitted point. Thus finite integer dimension is impossible for two-sided pure relative error. For $0 < q < 1$, two-sided relative error is impossible with finite integer dimension if $0 \leq \eta < 1$, whereas $p_{\text{conv}} = 0$ if $\eta \geq 1$. Including $x = 0$ does not change these conclusions. For fixed $q > 1$, $\eta > 0$ on the truncated domain $[a, 1]$,*

$$p_{\text{conv}}(a), p_{\text{bin}}(a) = \Theta_{q,\eta}(\log \log(1/a)) \quad (a \downarrow 0). \quad (119)$$

Proof. For $q > 1$ choose an integer $M \geq 2$ with $M^{q-1} > 2^q(1 + \eta)$. In a purported p -integer lift choose exact witnesses for $M^p + 1$ points of the sequence M^{-k} . Two witness indices have the same residue modulo M . Label their inputs $0 < x < y$, so $x/y \leq 1/M$. The convex combination of weight $1/M$ on the witness at y has an integer index and visible coordinates

$$u = y/M + (1 - 1/M)x \leq 2y/M, \quad v = y^q/M + (1 - 1/M)x^q \geq y^q/M.$$

Thus $v/u^q \geq M^{q-1}/2^q > 1 + \eta$, a contradiction. The proof uses only finite convex combinations and applies to nonclosed lifts and unbounded integer witnesses.

For $0 < q < 1$, $\eta < 1$, choose M with $M^{q-1} + M^{-q} < 1 - \eta$ and use the sequence M^{-2k} . The same residue argument gives $x/y \leq M^{-2}$ and $u \geq y/M$, so $v/u^q \leq M^{q-1} + M^{-q} < 1 - \eta$, again a contradiction. If $\eta \geq 1$, the convex set $\{(x, w) : 0 < x \leq 1, 0 \leq w \leq x^q\}$ contains the graph and has relative error at most one. Adding $(0, 0)$ handles the closed domain.

On $[a, 1]$, the $N + 1$ inputs $1, M^{-1}, \dots, M^{-N}$, $N = \lfloor \log_M(1/a) \rfloor$, must have distinct residues. Therefore $p_{\text{conv}}(a) \geq \log_M(N + 1)$. For the upper let $\rho = (1 + \eta)^{1/q} > 1$ and divide $[a, 1]$ into at most $K = \max(1, \lceil \log(1/a)/\log \rho \rceil)$ geometric intervals $[\ell, u]$ with $u/\ell \leq \rho$. Each rectangle $\ell \leq x \leq u$, $\ell^q \leq w \leq u^q$ contains the graph and has $1/(1 + \eta) \leq w/x^q \leq 1 + \eta$, which implies the required two-sided error. Encode the finite union with $\lceil \log_2 K \rceil$ binaries. These bounds prove (119); no matching leading constants are asserted. The real-coefficient upper uses $O(\log(1/a))$ cells. At zero error on any nondegenerate interval, strict convexity makes arbitrarily many distinct graph points pairwise midpoint-incompatible, so finite integer dimension is again impossible. $\square$

The residue proof is an elementary extension of the midpoint method [25]. Affine slices transfer its convex-power obstruction to nonnegative bilinear and multilinear products by setting all inputs equal, and to quadratic-over-linear graphs on a fixed positive-denominator slice approaching zero numerator. A positive absolute error floor or truncation away from zero removes this specific obstruction.

15.1 A fixed-error root graph requires long rational MILPs

Theorem 15.2 (Tight rational encoding barrier). *For integer $D \geq 2$, let $f_D(x) = x^{1/D}$ on $[0, 1]$. Every rational MILP whose projection contains its graph and has absolute vertical error at most $1/16$ has ordinary explicit total binary encoding length $\Omega(D)$. This includes arbitrary numbers of continuous auxiliaries and unbounded general integer variables. Conversely a rational MILP with four binaries and total encoding $O(D)$ achieves this error. Thus minimum rational MILP encoding is $\Theta(D)$, although four binaries suffice for every D .*

Proof. Let s be total explicit encoding, counting rows, variables, and rational coefficients. Freeze any integer witness z_* for the exact graph point $(2^{-D}, 1/2)$. Add $w \geq 1/4$ and minimize x over the resulting rational polyhedron. This LP is feasible, has finite attained optimum v , and

$$0 < v \leq 2^{-D}. \tag{120}$$

Indeed the tube bounds inputs below by zero, while an attained point with $x = 0, w \geq 1/4$ would violate its error bound at zero.

Clear denominators separately in each original row and the new row. If ℓ_i is the sum of coefficient numerator and denominator bit lengths in row i , including integer columns and right-hand side, and t_i its nonzero count, then $\sum_i \ell_i = O(s)$ and $\sum_i t_i = O(s)$. Continuous coefficients after clearing are integers of magnitude at most 2^{ℓ_i} . Fixing z_* changes only the integer right-hand side, whose numerator can be arbitrarily large. Split free continuous variables into nonnegative differences and

add slacks. The resulting integer row norms are at most $\sqrt{2t_i + 1} 2^{\ell_i}$. After dependent equations are removed, the feasible standard-form LP has an optimal basic solution; this argument does not require a vertex in the original free-variable polyhedron. Cramer's rule makes its positive objective value have a denominator dividing the nonzero integer basis determinant. Hadamard bounds every such determinant by

$$\log_2 |\det B| \leq \sum_i \ell_i + \frac{1}{2} \sum_i \log_2 (2t_i + 1) \leq C_0 s$$

for a universal encoding constant C_0 . Thus $v \geq 2^{-C_0 s}$, independently of witness magnitude. With (120), this proves $s \geq D/C_0$. The row-wise total-length accounting is essential to this linear bound in D .

For the upper choose rational knots $a_j = (j/16)^D$, $0 \leq j \leq 16$. On segment j impose

$$x = (1 - \theta)a_j + \theta a_{j+1}, \quad t = (j + \theta)/16, \quad t \leq w \leq t + 1/16, \quad 0 \leq \theta \leq 1.$$

Concavity and monotonicity give $t \leq f_D(x) \leq t + 1/16$. The segments cover $[0, 1]$. Their finite union uses four binaries; explicitly, continuous selectors λ_j bounded by their four matching literals sum to one, and $0 \leq v_j \leq \lambda_j$ gives

$$x = \sum_j [a_j \lambda_j + (a_{j+1} - a_j) v_j], \quad t = \sum_j (j \lambda_j + v_j)/16.$$

The row and variable counts are constant, and each rational knot has $O(D)$ bits. This proves the upper encoding bound. $\square$

This is a denominator bound for rational linear systems, not a lower bound on real-coefficient formulation size. The binary exponent $1/D$ has $O(\log D)$ bits, whereas the point of height one half is at input 2^{-D} . A small integer count alone cannot ensure a short rational graph formulation.

15.2 The same four bits with a short conic formulation

A mixed-integer second-order cone formulation (MISOCP) uses rational linear equations and second-order cone constraints with integer variables. Its count is compared below with rational MILP encoding; it is not included in the definition of p_{bin} .

Lemma 15.3 (A fixed conic optimum as a coefficient). *Suppose the primal-dual conic pair*

$$\min\{c^T z : Az = b, z \in \mathcal{K}\}, \quad \max\{b^T u : A^T u + s = c, s \in \mathcal{K}^*\}$$

has attained equal finite optimum v . Then, for $\theta \geq 0$, the conic linear constraints

$$Az = \theta b, z \in \mathcal{K}, \quad A^T u + s = \theta c, s \in \mathcal{K}^*, \quad c^T z = b^T u = t \quad (121)$$

project exactly to $t = \theta v$. They introduce no bilinear constraints.

Proof. For $\theta > 0$, divide the variables by θ and use weak duality; equality of the objective values forces $t = \theta v$. Scale attained optimizers for the converse. At $\theta = 0$, use any original feasible $z_0, (u_0, s_0)$ to get $c^T z = s_0^T z \geq 0$ and $b^T u = -z_0^T s \leq 0$. Their equality forces $t = 0$, which is feasible using zero variables. Thus no nonzero recession value enters at zero weight. $\square$

Theorem 15.4 (Rational MILP–MISOCP encoding separation). *For $D = 2^B$, $B \geq 1$, the fixed-error graph relaxation in theorem 15.2 has a rational MISOCP with exactly four binaries and $O(B)$ constant-size numerical coefficients in a structured cone-list description. Ordinary sparse matrix encoding has $O(B \log(B + 2))$ total bits. Every rational MILP for the same graph error has $\Omega(2^B)$ total encoding.*

Proof. For fixed rational $0 < a < 1$, the SOCP

$$\min z_B, \quad z_0 = a, \quad z_i \geq z_{i-1}^2 \quad (1 \leq i \leq B)$$

has optimum $v_B = a^{2^B}$, attained by $v_i = a^{2^i}$. Each squaring constraint is $(z_i, 1/2, z_{i-1}) \in \mathcal{Q}_r$, where $\mathcal{Q}_r = \{(u, v, w) : u, v \geq 0, 2uv \geq w^2\}$ is the self-dual rotated second-order cone. The rational point $z_i = a + i(1-a)/(B+1)$ is strictly feasible. There is also an explicit dual certificate: put $\lambda_B = 1$ and $\lambda_i = 2v_i\lambda_{i+1}$ for $1 \leq i < B$. The cone vectors $\lambda_i(1, 2v_{i-1}^2, -2v_{i-1})$ lie in $\mathcal{Q}_r$. Their pairings with $(z_i, 1/2, z_{i-1})$ sum to $z_B - v_B$; coefficients of intermediate z_i cancel, and the constants telescope using $\lambda_i v_i = 2\lambda_{i+1} v_{i+1}$. They are nonnegative and vanish at the primal optimizer, certifying attained equal dual value. These possibly long certificate coordinates are witnesses, not coefficients in the SOCP. Conversion to conic standard form has linear size. Apply lemma 15.3 to encode $t = \theta a^{2^B}$ with $O(B)$ short rational conic data. Values $a = 0, 1$ use direct equations.

Use the sixteen-segment construction of theorem 15.2, with four index bits and selectors λ_j . Write $0 \leq v_j \leq \lambda_j$, $r_j = \lambda_j - v_j$, $s_j = v_j$. Two copies of the conic-value gadget impose $X_j = (j/16)^D r_j$ and $Y_j = ((j+1)/16)^D s_j$. Set

$$x = \sum_j (X_j + Y_j), \quad t = \sum_j (j\lambda_j + v_j)/16, \quad t \leq w \leq t + 1/16.$$

Only the selected segment contributes; its graph containment and error were proved above. There are constantly many $O(B)$ gadgets, and all numerical data have constant length. Sparse variable and row indices contribute at most $O(B \log(B+2))$ bits. The MILP lower bound is theorem 15.2 with $D = 2^B$. $\square$

Conic strong duality, repeated-squaring lifts, and short conic systems with long solution encodings are established phenomena [9, 30, 41]. The separation above concerns fixed vertical graph accuracy with the same four declared binaries. It uses exact cone feasibility and exact primal-dual value equality; it asserts no tolerance-stable numerical solution or polynomial-time exact-output algorithm. In particular, a polynomial-size rational MILP outer approximation of these conic constraints that still contains the whole exact graph must violate the required vertical error somewhere, by theorem 15.2.

16 Shared-input convex vector graphs

Several outputs sharing an input must be approximated at the same input point. Separate scalar formulations need not achieve a good joint integer count. We first construct common partitions, then replace output count by the dimension of the component curvatures. All finite statements allow real coefficients and unrestricted finite continuous size. All computational statements in this section assume dense rational polynomial data and positive rational tolerances. They concern polynomial construction time and total rational encoding, including the supplied error-body data.

For a vector function on an interval, write $g_F = T_F - F$ for its vector chord gap. Componentwise convexity means $g_F \geq 0$ coordinatewise. For continuous functions, curvature rank means linear dimension modulo affine functions; no derivative is required. The first comparison pays for the number of outputs. Selecting a basis from the original convex outputs replaces that dependence by curvature rank. For general unconditional bodies, positive polar directions control the error and an explicit inner polytope supplies the final linear band. A shared basis then extends the construction to separable inputs. The separation examples distinguish which hypotheses each comparison needs.

16.1 Level refinement and an implicit common partition

Lemma 16.1 (Refinement at an arbitrary factor). *If a continuous convex function has chord error at most $H\tau$ on an interval, where $H \geq 1$ is an integer and $\tau > 0$, it has a partition into at most $2H - 1$ intervals with chord error at most τ . An interval cover by N intervals having a given scalar or componentwise convex chord-error bound can be replaced by a partition with at most N intervals and the same bound.*

Proof. Let g be the original nonnegative concave chord gap. For each level $j\tau$, $1 \leq j < H$, strictly below its maximum, cut at both endpoints of $\{g \geq j\tau\}$. Along the two sides of the maximum, consecutive cuts enclose a range of g of width at most τ . The middle interval has equal endpoint values at the highest retained level and maximum at most one further level. Its range also has width at most τ . If no level is retained the original maximum is at most τ . On a new interval the local chord gap equals g minus its endpoint interpolant, hence is nonnegative and at most this range width. There are at most $2H - 2$ cuts. Plateaus, repeated cuts and singleton intervals cause no difficulty.

For the cover statement, begin at the left endpoint and select a covering interval containing that point and extending farthest right. At each new endpoint repeat this choice among covering intervals containing it. Unless the right domain endpoint has been reached, the finite covering property ensures strict progress. No selected interval is needed twice. Restrict each selected interval to the new partition cell. Convex chord errors decrease under restriction, component by component. $\square$

Proposition 16.2 (Finite output-count comparison). *For continuous componentwise convex $F : [0, 1] \rightarrow \mathbb{R}^m$ and positive box tolerances,*

$$p_{\text{bin}} \leq p_{\text{conv}} + \lceil \log_2(2m + 1) \rceil.$$

For $K = \{e : |Ae| \leq b\}$ with $A \geq 0$, $b > 0$, and q rows, compactness of K gives the finite bound $p_{\text{bin}} \leq p_{\text{conv}} + \lceil \log_2(8q + 1) \rceil$.

Proof. On the interval hull of a parity contact from lemma 2.2, each component has midpoint gap at most ε_j , and hence full chord gap at most $2\varepsilon_j$. The three-piece proof in lemma 11.1 uses at most two cuts for each component. Overlaying these cuts yields at most $2m + 1$ cells, all with $0 \leq g_j \leq \varepsilon_j$. The bands $T_j - \varepsilon_j \leq w_j \leq T_j$ contain the exact vector graph and admit only the prescribed errors. At most $(2m + 1)2^p$ bands result from a p -integer lift; lemma 2.3 proves the first bound. For a facet row use the convex function $(AF)_k/b_k$. Its gap on a parity hull is at most two. Two three-piece refinements give gap at most $1/2$ using at most eight cuts. Overlaying all rows gives at most $8q + 1$ cells with $g_F \in K/2$. The band $w - T_F \in K/2$ contains the graph and admits errors in $K/2 + K/2 = K$. It is linear using continuous absolute-value auxiliaries. Degenerate hulls can be kept as singleton cells throughout. $\square$

The half-body band matters for coupled budgets. A downward shift by an arbitrary nonnegative vector in K can produce an error outside K : $(1, 0)$ and $(0, 1)$ both belong to the unit ℓ_1 ball, but their difference has norm two.

Lemma 16.3 (Sorted random-access overlay). *Suppose polynomially many nondecreasing rational arrays $0 = a_{j,0} \leq \dots \leq a_{j,K_j} = 1$ have polynomial-time indexed evaluators. Assume K_j have polynomial bit length and all knots are integer multiples of $1/H$ for a known positive integer H of polynomial bit length. The sorted multiset of all right endpoints $a_{j,1}, \dots, a_{j,K_j}$, with a single zero prepended, has a polynomial-time indexed evaluator. Its $K = \sum_j K_j$ cells cover the interval, and every positive-length cell lies inside a source cell of every array. Duplicates are retained as zero-length cells.*

Proof. For an integer $v \in [0, H]$, binary search each source index to count its right endpoints at most v/H . Sum these counts. The numerator of the k th multiset endpoint is the least v whose count is at least k , found by binary search in $[0, H]$. This uses $O(\log(H+1) \sum_j \log(K_j+1))$ indexed evaluations and polynomial rational arithmetic. Multiplicity makes equality an exact integer comparison. If a source knot lay strictly inside a positive merged cell it would also appear in the merged multiset, a contradiction. For source-cell metadata at a left endpoint $a < 1$, take the last source knot at most a and its successor; the successor is at least the right merged endpoint. At a repeated merged endpoint any source cell containing that point suffices; at one use the final source cell. $\square$

The mass-knot procedure in theorem 12.3 can be made nondecreasing without increasing its precision. Use the same deterministic mass evaluation and accuracy at each bisection node for every target. At a node the targets going left, returning the midpoint, and going right are ordered ranges. All outputs of its left subtree are at most the midpoint, and all outputs of its right subtree are at least it. Induction on the fixed tree depth proves monotonicity, even if the approximate mass values at different nodes are not monotone. Forced endpoints preserve order. Reverse local indices when reflecting a piece. Greedy knots are already increasing; concatenating the hybrid's consecutive pieces therefore gives the arrays required in lemma 16.3. The common denominator is the product of the polynomially many rational piece-endpoint denominators times the largest local dyadic denominator. Its bit length is polynomial; individual short denominators alone would not establish this common-grid property.

Theorem 16.4 (Compiled convex vector overlay). *For dense rational componentwise convex polynomial outputs with positive rational box tolerances, a polynomial rational MILP has*

$$p_{\text{out}} \leq p_{\text{conv}} + 11 + \lceil \log_2 m \rceil.$$

An entirely affine graph has an exact zero-integer formulation.

Proof. Use the sorted scalar hybrid arrays at tolerance ε_j . Their actual counts obey $K_j \leq 1458 2^p$ by (94), because projecting any p -integer vector lift gives a scalar lift. Each source cell has chord gap at most $13\varepsilon_j/16$. An affine component can use one cell. Apply lemma 16.3 to their right endpoints. Evaluate each original polynomial at each selected merged endpoint and round downward with error at most $\varepsilon_j/8$, using fixed dyadic output denominators and offsets for signed values. Let y_j interpolate these rounded values at the common continuous weight. Since restriction preserves convex chord error, the band

$$y_j - 13\varepsilon_j/16 \leq w_j \leq y_j + \varepsilon_j/8 \tag{122}$$

contains the graph and admits absolute error at most $15\varepsilon_j/16$. This remains true at zero-length cells. All components use the same input and interpolation weight. Lemma 12.1 needs only $\lceil \log_2 K \rceil$ binary index variables: the endpoint arithmetic and internal Boolean wires remain continuous. Exclude codes at least K . Since $K \leq 1458m2^p < 2048m2^p$, the count follows. The algorithm computes actual counts and never uses the comparator lift. $\square$

Merging explicit breakpoints and sharing an SOS2 index for several outputs are established formulation tools; Lyu et al. [26, Section 3, Proposition 1] give that common-input construction. The point of the present overlay is its polynomial-bit random access when the underlying lists can be exponentially long, together with a comparison against every convex integer lift. No sparse huge-degree evaluation claim is made here.

16.2 Original-output curvature bases

Lemma 16.5 (Barycentric spanners of curvature rows). *Let $G_1, \dots, G_m$ be continuous convex functions whose classes modulo affine functions span dimension $r > 0$. There are r original functions G_{i_s} such that*

$$G_j = \alpha_j + \beta_j x + \sum_{s=1}^r c_{js} G_{i_s}, \quad |c_{js}| \leq 1. \quad (123)$$

For rational dense polynomials, a polynomial-time rational algorithm returns original functions with the weaker bound $|c_{js}| \leq 2$. If $\Psi = \sum_s G_{i_s}$ and the coefficient bound is c , then every interval satisfies $0 \leq g_{G_j} \leq cg_\Psi$.

Proof. Choose coordinates on the quotient and maximize the absolute determinant of r original rows. Replacing one row by any other multiplies the determinant by its representation coefficient, so Cramer's rule proves (123). Affine terms cancel in chord gaps. The selected gaps are nonnegative, hence signed coefficients of magnitude at most c still give $g_{G_j} \leq c \sum_s g_{G_{i_s}}$.

For rational polynomials form the nonlinear coefficient matrix and retain r independent columns. Start with a nonsingular row minor. Whenever a representation coefficient has magnitude greater than two, replace the corresponding basis row. Its determinant more than doubles. If the retained matrix has total encoding length L , the product of its entry denominators is at most 2^L . Nonzero row determinants are at least 2^{-rL} and all are at most $r!2^{rL}$. Thus at most $2rL + \log_2(r!) + 1$ exchanges occur. Every current basis is a submatrix of the original matrix; inverses and exact comparisons have polynomial bit length. Relations on independent columns extend to all columns, so the resulting coefficients represent the entire polynomial rows. $\square$

The maximum-determinant argument and determinant exchange are the barycentric-spanner constructions of Awerbuch and Kleinberg [4, Section 2.3, Propositions 2.2 and 2.4]. The requirement to select original convex outputs, rather than an arbitrary algebraic basis, preserves nonnegative chord gaps.

Theorem 16.6 (Box and facet curvature rank). *For box tolerances normalize $G_j = F_j/\varepsilon_j$, and let r be the dimension of their span modulo affine functions. For $r \geq 1$,*

$$p_{\text{bin}} \leq p_{\text{conv}} + \lceil \log_2(4r - 1) \rceil, \quad p_{\text{out}} \leq p_{\text{conv}} + \lceil \log_2 r \rceil + 12. \quad (124)$$

The first bound is finite for continuous convex components; the second is a polynomial rational construction for dense polynomial data. For a compact body $K = \{e : A|e| \leq b\}$, $A \geq 0$, $b > 0$, let r instead be the curvature rank of $H_k = (AF)_k/b_k$. Then

$$p_{\text{bin}} \leq p_{\text{conv}} + \lceil \log_2(8r - 1) \rceil, \quad p_{\text{out}} \leq p_{\text{conv}} + \lceil \log_2 r \rceil + 13. \quad (125)$$

The computational facet statement includes rational A, b in its input. In either model $r = 0$ forces an affine graph and needs no integers.

Proof. Choose an exact original-output basis and sum it to obtain Ψ . Each normalized basis function has midpoint gap at most one on a parity hull, so $g_\Psi \leq 2r$ there. For a box, refine to scalar gap one using at most $4r - 1$ cells. Lemma 16.5 bounds all normalized component gaps by one, and downward chord bands give the finite claim. For facets refine to $1/2$, using at most $8r - 1$ cells. Then $g_F \in K/2$; the symmetric band $w - T_F \in K/2$ gives containment and admissibility.

For a rational box use the factor-two basis, so $g_{G_j} \leq 2g_\Psi$. Refining parity hulls to tolerance $1/2$ gives $N_{1/2}(\Psi) \leq (8r - 1)2^p$ by lemma 16.1. The hybrid has at most $486N_{1/2}$ cells, each

with $g_\Psi \leq 13/32$. Evaluate all original normalized outputs at their common endpoints, round downward within $1/8$, and use (122) with unit normalized tolerances. The cell count is less than $3888r2^p < 2^{12}r2^p$.

For rational facets use the factor-two basis of the H_k and tolerance $1/4$. Now $N_{1/4}(\Psi) \leq (16r - 1)2^p$ and every hybrid cell has $g_\Psi \leq 13/64$, hence $g_F \in (13/32)K$. Put

$$M_A = \max_k \sum_j A_{kj}/b_k > 0, \quad \delta = \min\{1, (16M_A)^{-1}\}.$$

Round each original endpoint coordinate downward within δ . Its rounding error belongs to $K/16$, as does the interpolated error. The resulting center y satisfies $y - F \in (15/32)K$. The band $w - y \in K/2$ contains the graph and admits errors in $(31/32)K$. Its explicit linear form is

$$s_j \geq w_j - y_j, \quad s_j \geq y_j - w_j, \quad As \leq b/2, \quad s \text{ continuous.} \quad (126)$$

The count is less than $486 \cdot 16r2^p < 2^{13}r2^p$. All additional component evaluations and bands add only continuous variables to the shared index compiler.

Finally, rank zero in the box model means each output is affine. For facets, compactness and nonnegativity imply that each column of A has a positive entry. Vanishing nonnegative weighted chord gaps therefore force every component gap to vanish. This also proves the rank-zero assertion for continuous functions. $\square$

For weighted ℓ_1 error, $K = \{e : \sum_j a_j |e_j| \leq b\}$ with $a_j > 0$, there is one facet scalarization. Unless the graph is affine, its rank is one; (125) has finite overhead three and compiled overhead thirteen, independent of output count and degree. These are upper comparisons, not assertions that the logarithmic rank loss is necessary.

16.3 Direct allocation for unconditional bodies

Proposition 16.7 (Maximum-product box transfer). *For any unconditional error body $K \subset \mathbb{R}^m$ and continuous componentwise convex curve,*

$$p_{\text{bin}} \leq p_{\text{conv}} + \lceil \log_2(4m - 1) \rceil.$$

For dense rational polynomial components and a polynomial rational strong separation oracle for K , with known positive rational radii $\rho B_2^m \subset K \subset RB_\infty^m$, a polynomial rational construction has $p_{\text{out}} \leq p_{\text{conv}} + 14 + \lceil \log_2 m \rceil$. The final formulation uses explicit linear bands.

Proof. Maximize $\prod_j b_j$ over nonnegative $b \in K$. A maximizer is strictly positive, because K is compact and contains a positive point. Sign changes and convexity give $[-b, b] \subset K$. Differentiating the log product along a feasible segment from b to $v \in K \cap \mathbb{R}_+^m$ yields

$$\sum_j v_j/b_j \leq m. \quad (127)$$

Thus $\Psi = \sum_j F_j/b_j$ has full gap at most $2m$ on every parity hull. Refine it to gap one with at most $4m - 1$ cells. Every normalized component gap is at most one, so downward bands in $[-b, b]$ prove the finite statement.

The scalar specialization of lemma 9.1, applied to RI_m and coordinate cap one, returns rational $b > 0$ in K with product at least e^{-1} times optimal. The cap loses no positive feasible point, since all coordinates of K are at most R . This approximate maximizer obeys

$$\sum_j v_j/b_j \leq 7m \quad (v \in K \cap \mathbb{R}_+^m). \quad (128)$$

Indeed for $m \geq 2$ set $s = 1 - 1/m$, $a_j = v_j/b_j$, $S = \sum_j a_j$. The feasible point $sb + v/m$ has product ratio at most e , whereas expansion with nonnegative terms gives ratio at least $s^{m-1}(s + S/m)$. Since $s^{m-1} \geq e^{-1}$, $S \leq m(e^2 - 1) + 1 < 7m$. For $m = 1$ the ratio bound directly gives $S \leq e < 7$.

For this rational Ψ the parity-hull gap is at most $14m$; therefore $N_1(\Psi) \leq (28m - 1)2^p$. Its hybrid cells have scalar gap at most $13/16$. Each component F_j/b_j has no larger gap. Evaluate and round these normalized values downward within $1/8$ and impose the normalized version of (122). The error belongs to $(15/16)[-b, b] \subset K$ and the graph is contained. The count is below $486 \cdot 28m2^p < 2^{14}m2^p$. The rational oracle bounds b , $1/b_j$, and all subsequent precision lengths polynomially. An affine graph is retained exactly. $\square$

Maximum-product allocation and the first-order inequality (127) are classical; in network allocation they are the proportional-fairness relation of Kelly et al. [24, Section 2]. This direct construction avoids any facet enumeration. The next result uses the effective nonlinear image to replace its output-count penalty.

17 Curvature rank with a strong error-body oracle

Let K be unconditional and put

$$S = \text{span}\{F(x) - (1 - x)F(0) - xF(1) : 0 \leq x \leq 1\}, \quad r = \dim S. \quad (129)$$

Choose a full-column-rank $m \times r$ matrix V spanning S and write $F(x) = a + bx + Vq(x)$, with $q(0) = q(1) = 0$. For rational dense polynomials, select independent nonlinear coefficient columns for V and use a rational left inverse J to obtain q . Elimination has polynomial bit cost. The effective body $C = \{z : Vz \in K\}$ is symmetric, compact, and full dimensional in $\mathbb{R}^r$, although it need not be unconditional there. If $\rho B_2^m \subset K \subset RB_\infty^m$, known radii are

$$\frac{\rho}{1 + \sum |V_{ij}|} B_2^r \subset C \subset mR(1 + \sum |J_{ij}|) B_2^r. \quad (130)$$

A separator for K at Vz pulls back by V^T . At an exterior query its pulled-back normal is nonzero, because zero belongs to C .

17.1 Exactly feasible rational spanners

We give the oracle interface in detail, because approximate optimization alone neither proves exact feasibility nor controls denominator growth through repeated basis exchanges. The imported weak-optimization theorem is Grötschel et al. [20, Definition (5), Theorem (3.1)]: given a rational weak separator and explicit inner and outer balls for a convex body P , it returns a rational z with

$$\text{dist}(z, P) \leq \eta, \quad d^T z \geq \max_{x \in P} d^T x - \eta. \quad (131)$$

The objective comparison is with all of P . In dimension one apply the theorem to $P \times [-1, 1]$ and objective $(d, 0)$, then project; inner radius $\min(\sigma, 1)$ and outer radius $R_0 + 1$ suffice.

Lemma 17.1 (Fixed-grid feasible spanner). *Let $B_2(c, \sigma) \subset P \subset B_2(c, R_0)$ be a compact convex body with a polynomial rational weak separator and explicit rational c, σ, R_0 , $0 < \sigma \leq R_0$. Let $W \in \mathbb{Q}^{d \times r}$, $r \geq 1$, and suppose rational seeds $\lambda_1, \dots, \lambda_r \in P$ have independent image rows $\lambda_i^T W$. A polynomial-time rational algorithm returns feasible such rows which represent every $\lambda^T W$, $\lambda \in P$, with coefficients of magnitude at most $9/4 < 3$. Complexity is polynomial in the entire input, balls, and seeds.*

Proof. Let B_0 be the seed row matrix and $\Delta_0 = |\det B_0| > 0$. Choose the rational bounds

$$Z = 1 + \sum_i |c_i| + R_0, \quad M = 1 + \sum_{ij} |W_{ij}|, \quad A = 1 + dMZ, \quad U = 1 + r!A^{r-1}/\Delta_0, \quad L = 1 + dMU.$$

Every feasible image coordinate has magnitude at most A . As long as determinants never decrease, cofactors bound every inverse basis entry by U . Each representation-coordinate objective $d_i = WB^{-1}e_i$ and its negative then has ℓ_1 norm at most L . These bounds have polynomial bit length, including when r varies.

Choose one positive dyadic δ before starting exchanges, such that

$$2d\delta \leq 1, \quad d\delta[1 + L + 2L(R_0 + 1)/\sigma] \leq 1/4. \quad (132)$$

For any objective d' of ℓ_1 norm at most L , apply (131) with $\eta = d\delta$. Round its output coordinatewise to multiples of δ , obtaining q with $\|q - z\|_2 \leq d\delta$, and set

$$\epsilon = 2d\delta, \quad \lambda = \frac{\sigma q + \epsilon c}{\sigma + \epsilon}. \quad (133)$$

Since $\text{dist}(q, P) \leq \epsilon$, choose $x \in P$ within that distance. The repaired point is the convex combination, with weights $\sigma/(\sigma + \epsilon)$ and $\epsilon/(\sigma + \epsilon)$, of x and $c + (\sigma/\epsilon)(q - x) \in B_2(c, \sigma)$. Thus $\lambda \in P$ exactly. Also $\|c - q\|_2 \leq R_0 + 1$, so

$$\max_{x \in P} (d')^T x - (d')^T \lambda \leq d\delta + L[d\delta + 2d\delta(R_0 + 1)/\sigma] \leq 1/4.$$

Every returned point uses the same grid and the same repair weights. Its coordinates have a fixed common rational denominator of polynomial bit length and uniformly bounded numerators. Including seed denominators preserves this statement for every possible basis.

At each basis position optimize both signs of d_i . If any returned coefficient has magnitude greater than two, exchange that row. The determinant more than doubles. Otherwise every exact feasible coefficient has magnitude at most $2 + 1/4$, proving the spanner guarantee. Since all determinants are at most $r!A^r$, there are at most $1 + \lceil \log_2(r!A^r/\Delta_0) \rceil$ exchanges. Fixed denominators ensure that every inverse, later objective, oracle input, and comparison retains polynomial bit length. This proves uniform polynomial complexity, not merely a per-call bound in a potentially growing current input. $\square$

To apply the lemma to positive polar directions, a weak separator must itself be constructed. Replace the coordinate outer radius by the rational Euclidean radius $R_E = mR$, so $\rho B_2^m \subset K \subset R_E B_2^m$. For rational a and $\tau > 0$, weak optimization over K followed by central repair returns rational $x \in K$ with

$$a^T x \leq h_K(a) \leq a^T x + \tau. \quad (134)$$

Indeed take $\eta \leq 1$ satisfying $\eta[1 + \|a\|_1(R_E + 1)/\rho] \leq \tau$ in (131) and put $x = \rho z/(\rho + \eta)$. The inner-ball convex-combination argument proves exact feasibility; the support loss is at most the displayed quantity. The zero objective can return zero.

Define the full-dimensional positive polar

$$P_+ = \{\lambda \geq 0 : h_K(\lambda) \leq 1\}.$$

Choose a positive dyadic $a_0 \leq \min\{1, (8R_E m)^{-1}\}$. An inner ball has center $c = a_0 \mathbf{1}$ and radius $a_0/2$: its coordinates are positive and its Euclidean norm is less than $1/R_E$. An outer ball centered at c has radius $1/\rho + ma_0$, since $P_+ \subset \rho^{-1}B_2^m$. For a rational query λ and distance accuracy $\eta > 0$,

first separate any coordinate outside $[0, 1/\rho]$ by the appropriate coordinate inequality. Otherwise set $\tau = \eta/[2(1 + m/\rho)]$ and obtain x from (134). If $\lambda^T x > 1$, the inequality $x^T \mu \leq 1$ separates the query from all of P_+ . Its rational nonzero normal can be rescaled by its infinity norm to meet the norm convention in the weak separator. If $\lambda^T x \leq 1$, then $h_K(\lambda) \leq 1 + \tau$ and $\lambda/(1 + \tau) \in P_+$ lies within $\tau\|\lambda\|_2 \leq \eta/2$ of the query. This is a valid weak-membership answer, not an exact membership test. All accuracies and returned coefficients have polynomial bit length.

Choose independent rows $j_1, \dots, j_r$ of V . The feasible seeds $2a_0 e_{j_s} \in P_+$ have independent image rows $2a_0 e_{j_s}^T V$. Lemma 17.1 therefore constructs feasible nonnegative λ_s whose images span and 3-approximately represent all $\lambda^T V$, $\lambda \in P_+$. For C in (130), use its strong separator, center zero, image matrix I_r , and small coordinate-axis seeds from its inner ball. The same lemma returns a feasible factor-three basis of C . This derives both oracle calls from the original strong oracle; it assumes neither a polyhedral body nor a strong separator for its polar. Positive-polar access is a quantitative implementation of the classical polar and nonnegative anti-blocker oracle equivalences [20, Corollaries (3.4)–(3.5)]. Approximate optimization combined with barycentric-spanner exchange also predates this construction; Plevrakis and Hazan [31, Section 3.3] use that combination in online control. The explicit fixed denominator and exact-feasibility repair here supply the rational interface needed by the graph compiler, without asserting a new general spanner or polar-oracle principle.

17.2 A linear band in the effective image

The two spanners serve different purposes. The positive polar spanner reduces all output budgets to one convex scalar function. The spanner of the effective body turns that scalar gap certificate into an explicit linear band. Summing the polar directions contributes one rank factor to the refinement count, and the inner polytope contributes another. Together they yield an overhead proportional to twice the logarithm of the rank. The band’s auxiliary coordinates remain continuous, so the approximation index is still the only source of binaries.

Theorem 17.2 (Oracle-body curvature rank). *For a continuous componentwise convex curve and unconditional error body, with r as in (129) and $r \geq 1$,*

$$p_{\text{bin}} \leq p_{\text{conv}} + \lceil \log_2(8r^2 - 1) \rceil. \quad (135)$$

Under the dense rational and strong-oracle hypotheses above, a polynomial rational MILP has

$$p_{\text{out}} \leq p_{\text{conv}} + 17 + 2\lceil \log_2 r \rceil. \quad (136)$$

No oracle constraint remains in that MILP. For $r = 0$ the graph is affine.

Proof. For nonnegative e , unconditionality and polar duality give

$$\|e\|_K = \sup_{\lambda \in P_+} \lambda^T e.$$

Indeed the polar is unconditional, and replacing a polar vector by its coordinatewise absolute value can only increase its pairing with $e \geq 0$. Select feasible polar directions λ_s whose images form a spanner with coefficient bound c , and put $\Psi = \sum_s \lambda_s^T F$. All these scalar functions are convex. The same signed-coefficient argument as in lemma 16.5 gives, on every interval,

$$\|g_F\|_K \leq c g_\Psi. \quad (137)$$

The selected sum is nonaffine when $r > 0$: if its nonnegative summand gaps vanished, each selected scalarization would be affine, contradicting independence in the nonlinear image.

Let B have independent feasible columns in C giving coefficient bound d . Symmetry and convexity imply

$$B\{u : \|u\|_1 \leq 1\} \subset C \subset B[-d, d]^r.$$

Put $P_0 = r^{-1}B[-1, 1]^r$ and $P = VP_0$. Then

$$P \subset K \cap S \subset drP, \quad w - y \in P/2 \iff w - y = VBu/(2r), \quad -1 \leq u_i \leq 1. \quad (138)$$

The right-hand band is linear with continuous u .

For the finite theorem take maximum-absolute-determinant bases in the compact spanning positive-polar image and in C . Cramer's rule gives $c = d = 1$. On every parity hull, each selected polar function has midpoint gap at most one, so $g_\Psi \leq 2r$. Refine to scalar tolerance $1/(2r)$ with at most $8r^2 - 1$ cells. Equations (137) and (138) put the chord gap in $P/2$. The exact chord center with band $P/2$ contains the graph and admits errors in $P \subset K$. Finite disjunction proves (135). Only the midpoint Jensen vectors, which automatically lie in S , are used from the comparator lift; its other admitted errors may lie outside S .

For the rational theorem use the two factor-three spanners just constructed, so $c = d = 3$. Set $\tau = 1/(36r)$. Refinement gives $N_\tau(\Psi) \leq (144r^2 - 1)2^p$. A hybrid cell has $g_\Psi \leq 13/(576r)$, hence its vector gap belongs to

$$(13/(192r))K \cap S \subset (13/64)P.$$

Evaluate the effective coordinate polynomials q at the common knots and round each coordinate within $\delta = (16rL_B)^{-1}$, where $L_B = 1 + \sum_{ij} |(B^{-1})_{ij}|$. Each rounding vector belongs to $P_0/16$, because its image under B^{-1} has infinity norm at most $1/(16r)$. Interpolation preserves this property. Restore the affine part exactly to get $y = a + bx + Vy_q$. Then $y - F \in (17/64)P$. The explicit band in (138) contains the graph and admits errors in $(49/64)P \subset K$. Rounding in effective coordinates keeps every error exactly in S . The count is less than $486 \cdot 144r^2 2^p = 69984r^2 2^p < 2^{17} r^2 2^p$. The fixed-grid spanner bounds, rational coordinate evaluations, and lemma 12.1 prove the encoding and time claims. $\square$

18 Separable coupled vector outputs

Let

$$F_j(x) = \ell_j(x) + \sum_{i=1}^n \phi_{ji}(x_i), \quad x \in [0, 1]^n, \quad (139)$$

where every ϕ_{ji} is continuous and convex, and every ℓ_j is affine. Remove coordinates on which all outputs are affine; n below counts the remaining coordinates. In the computational assertions the displayed separated representation is supplied, with rational affine terms and dense rational polynomial summands. Mixed-coordinate nonlinear terms are not covered by this representation.

For box tolerances, form one row for each normalized output by concatenating its classes ϕ_{ji}/ε_j modulo affine functions over all input coordinates. Let r be the rank of these rows. For explicit facets use the concatenated rows of $(AF)_k/b_k$ instead. For a general unconditional body, subtract the endpoint chord from each vector summand and let S be the span of all resulting vectors over all coordinates; set $r = \dim S$. For polynomial data these are ordinary rational coefficient-matrix ranks. In particular, the basis is shared across coordinates: selecting unrelated output bases for different inputs would not justify the comparison below.

Lemma 18.1 (Product packing for a shared scalarization). *Suppose $\Psi = \text{affine} + \sum_i \psi_i(x_i)$ is a sum of r original normalized outputs or feasible positive-polar scalarizations. For a local tolerance*

$\tau > 0$, let P_i be the size of a maximal scalar packing with pairwise $J_{\psi_i} > \tau$. If $q = r/(n\tau)$ is a positive integer, then

$$2^{p_{\text{conv}}} \geq \frac{\prod_i P_i}{[3(2q+1)]^n}. \quad (140)$$

A finite chord construction has local binary capacities at most $12P_i$; a dense hybrid construction has capacities at most $5832P_i$.

Proof. Order each scalar packing. By lemma 12.5, an index difference $h > 0$ gives midpoint gap greater than $h\tau$. Two distinct product points with index vectors u, v therefore have $J_\Psi > \tau\|u - v\|_1$. If this exceeds r , one selected normalized output or polar function has midpoint gap greater than one, making the original vector graph midpoint inadmissible. Greedily retain product indices and delete the integer ℓ_1 ball of radius qn after each choice. With $t = q/(q+1)$, its size satisfies

$$\#\{z \in \mathbb{Z}^n : \|z\|_1 \leq qn\} \leq t^{-qn} \left(\sum_{k \in \mathbb{Z}} t^{|k|} \right)^n = (1 + 1/q)^{qn} (2q+1)^n < [3(2q+1)]^n.$$

The retained set has at least the product cardinality divided by this bound, and pairwise inadmissible midpoints. Parity proves (140); neither the packings nor this code are computational inputs. Finally lemma 11.1 gives $N_\tau(\psi_i) \leq 6P_i$. Finite binary encoding has capacity $2^{\lceil \log_2 N_\tau \rceil} \leq 2N_\tau \leq 12P_i$. The hybrid has at most $486N_\tau$ cells and capacity at most $972N_\tau \leq 5832P_i$, including one-cell cases. $\square$

Theorem 18.2 (Separable vector rank comparisons). *For $n, r \geq 1$ in (139), the finite and polynomial rational bounds are as follows:*

Error description	Finite upper for $p_{\text{bin}} - p_{\text{conv}}$	Compiled upper for $p_{\text{out}} - p_{\text{conv}}$
Box	$n\lceil \log_2 r \rceil + 7n$	$n\lceil \log_2 r \rceil + 17n$
Nonnegative facets	$n\lceil \log_2 r \rceil + 8n$	$n\lceil \log_2 r \rceil + 18n$
Unconditional oracle	$2n\lceil \log_2 r \rceil + 8n$	$2n\lceil \log_2 r \rceil + 21n$

The finite oracle-body statement requires only the body itself; the compiled statement uses the strong rational oracle and known rational inner/outer radii of theorem 17.2. Rank zero or no active input coordinates means an exact affine graph with no integers.

Proof. For a box select one exact or factor-two barycentric spanner of the concatenated original normalized output rows. Its representation coefficients apply in every coordinate. Summing the selected functions gives $\Psi = \text{affine} + \sum_i \psi_i$, with $0 \leq g_{ji} \leq cg_{\psi_i}$, where $c = 1$ in the finite case and $c = 2$ in the compiled case. If some ψ_i is affine, every corresponding original summand is affine, so that coordinate was removable.

For the finite box construction use $\tau = 1/n$ and hence $q = r$ in lemma 18.1. The packing denominator is at most $(9r)^n$. Choose finite coordinate partitions of error τ and interpolate every original summand at the same coordinate knots and weight. Each normalized output's total gap is between zero and one. The band $T_j - 1 \leq w_j \leq T_j$ contains the graph and admits unit normalized error. One finite disjunction per coordinate encodes that coordinate's input and all its component chords. Adding the output bands and restoring affine terms adds only continuous equations. The count is at most $p_{\text{conv}} + n \log_2(12 \cdot 9r)$, which is less than $p_{\text{conv}} + n \log_2 r + 7n$.

For the compiled box construction set $\tau = 1/(2n)$, so $q = 2r$ and the packing denominator is at most $(15r)^n$. Each hybrid coordinate cell has $g_{\psi_i} \leq 13/(32n)$, implying $0 \leq \sum_i g_{ji} \leq 13/16$. Round each normalized original summand endpoint downward within $1/(8n)$ and sum interpolants to obtain y_j , restoring the affine part exactly. The total downward rounding is at most $1/8$. The

band $y_j - 13/16 \leq w_j \leq y_j + 1/8$ contains the graph and admits error at most $15/16$. The count factor per coordinate is $5832 \cdot 15r = 87480r < 2^{17}r$.

For explicit facets apply the same common-row construction to the normalized facet functions. In the finite case choose $\tau = 1/(2n)$, so the total gap of each facet is at most $1/2$. Thus $g_F \in K/2$, and the band $w - T \in K/2$ works. The packing denominator is $(15r)^n$ and the count factor is $12 \cdot 15r < 2^8r$. In the compiled case use $\tau = 1/(4n)$, so the packing denominator is at most $(27r)^n$. Summing factor-two gap bounds gives $g_F \in (13/32)K$. Round each original summand endpoint within $\min\{1, (16nM_A)^{-1}\}$, with M_A defined in the proof of theorem 16.6. The total rounding error belongs to $K/16$; therefore the center differs from F by an element of $(15/32)K$. Use (126). The count factor is $5832 \cdot 27r = 157464r < 2^{18}r$.

For an arbitrary unconditional body choose the same effective coordinates and two spanners as in theorem 17.2, now for the shared image of all coordinate blocks. Write $F = \ell + V \sum_i q_i(x_i)$ and $\Psi = \sum_s \lambda_s^T F = \text{affine} + \sum_i \psi_i$. Equations (137) and (138) hold for every coordinate gap, using the same c, d and P . In the finite case $c = d = 1$ and choose $\tau = 1/(2rn)$, so $q = 2r^2$. The total gap has K norm at most $1/(2r)$ and lies in S , hence belongs to $P/2$. The exact chord center with band $P/2$ works. The packing denominator is at most $(15r^2)^n$ and the count factor is $12 \cdot 15r^2 < 2^8r^2$.

For the compiled oracle case $c = d = 3$ and set $\tau = 1/(36rn)$, so $q = 36r^2$. The packing denominator is at most $(219r^2)^n$. The total scalar hybrid gap is at most $13/(576r)$, and the vector gap belongs to $(13/64)P$. Round every entry of each q_i at its knots within $(16nrL_B)^{-1}$. Its error belongs to $P_0/(16n)$; summing and restoring the affine part gives a center with $y - F \in (17/64)P$. The explicit band in (138) admits only $(49/64)P \subset K$. The count factor is $5832 \cdot 219r^2 = 1277208r^2 < 2^{21}r^2$. In every construction all outputs share each coordinate's index and continuous interpolation weight. All extra evaluations, basis maps, rounding data and bands have polynomial rational size in the specified computational models. The extra factors n, r in tolerances cost only logarithmic additional precision. Compact nonnegative-facet bodies and positive-polar domination give the asserted affine degeneracies. $\square$

For weighted ℓ_1 error the single facet has rank one whenever the graph is nonlinear. The separable finite and compiled overheads are therefore $8n$ and $18n$. General mixed-coordinate polynomials require additional arguments: replacing the sum of univariate chords by multilinear corner interpolation does not itself supply the necessary compact linear encoding.

19 Limits of scalarization and simultaneous refinement

The rank comparisons do not follow by taking the largest of the separate scalarized minimum dimensions. The following family makes every such scalar minimum small while the simultaneous integer dimension grows.

Proposition 19.1 (Separated positive-power layers). *For $M \geq 1$, put*

$$D_j = 128 \cdot 1024^{j-1}, \quad F_j(x) = \frac{7}{4}x^{D_j}, \quad x_j = 1 - 64/D_j, \quad K = [-1, 1]^M.$$

Every exact-graph midpoint has component error strictly below $7/8$. For every real λ with $\|\lambda\|_1 = 1$, the scalar graph $\lambda^T F$ at its induced tolerance one has a finite binary linear formulation with at most one binary. Nevertheless

$$\lceil \log_3 M \rceil \leq p_{\text{conv}} \leq p_{\text{bin}}, \quad \lceil \log_2 M \rceil \leq p_{\text{bin}} \leq \lceil \log_2(M+1) \rceil. \quad (141)$$

If $N_F(\eta)$ is the least number of interval chords with vector error at most η in the infinity norm, then $N_F(2) = 1$ and $N_F(1) \geq M$. The binary lower bound holds even for arbitrary convex binary lifts.

Proof. For convex differentiable f , $J_f(a, b)$ is nonincreasing in a and nondecreasing in b , as follows by differentiating the midpoint formula. Hence $0 \leq J_{F_j}(a, b) \leq J_{F_j}(0, 1) = 7/8 - (7/4)2^{-D_j} < 7/8$.

For $j < k$, Bernoulli's inequality gives $x_k^{D_j} \geq 1 - 64D_j/D_k \geq 15/16$. At $\xi = (x_j + 2x_k)/3 \leq 1 - 64/(3D_j)$, $\xi^{D_j} \leq e^{-64/3} \leq 3/67$. Therefore the thirds chord error in component j is at least

$$\frac{7}{4} \left(\frac{2}{3} \frac{15}{16} - \frac{3}{67} \right) = \frac{2177}{2144} > 1. \quad (142)$$

Selected exact graph witnesses at these M inputs must have distinct integer residues modulo three; otherwise their indicated weighted average has integer coordinates and violates the error body. Thus $3^p \geq M$. In a binary lift, equal assignments admit every convex weight, so the same witnesses require $2^p \geq M$. These arguments use neither bounded integer labels nor closedness.

Each component increases from zero to $7/4$. Its half-height point $a_j = 2^{-1/D_j}$ splits its range into intervals of length $7/8$. Overlay all half-height points to obtain $M + 1$ input intervals. On each, the rectangle formed from the input interval and all component ranges contains the vector graph and has component error at most $7/8$. Finite disjunction proves the upper binary count; its coefficients may be real algebraic. The full-domain chord error is at most $7/4 < 2$. An interval with unit vector chord error cannot contain two selected x_j, x_k : its component chord lies above the chord through those two contacts and would inherit (142). Thus $N_F(1) \geq M$.

Finally set $\Psi_\lambda = \sum_j |\lambda_j| F_j$. It increases continuously from zero to $7/4$, and $|\lambda^T F(b) - \lambda^T F(a)| \leq \Psi_\lambda(b) - \Psi_\lambda(a)$ for $a \leq b$. Split at $\Psi_\lambda = 7/8$ and use the scalar input-output range rectangle on each side. Each range has length at most $7/8$, giving one binary. Negative scalarization coefficients are allowed. The induced scalar tolerance is exactly $h_K(\lambda) = \|\lambda\|_1 = 1$. $\square$

The statement quantifies separately over scalarizations: each may choose its own partition and formulation. It does not question polar duality, which exactly describes the norm, or rule out algorithms combining several scalarizations. It also does not establish an unbounded difference $p_{\text{bin}} - p_{\text{conv}}$. The family has curvature rank M , so its behavior is consistent with the rank-dependent upper bounds.

Proposition 19.2 (Cap-set contact restriction). *For the family of proposition 19.1, let $\text{cap}(p)$ be the largest subset of $\mathbb{F}_3^p$ with no three distinct elements summing to zero. Every p -integer convex lift satisfies*

$$M \leq \text{cap}(p) \leq 3c_*^p, \quad c_* = (1 + t_* + t_*^2)t_*^{-2/3} < 3, \quad t_* = (\sqrt{33} - 1)/8. \quad (143)$$

In particular $p_{\text{conv}} \geq \max\{0, \lceil \log_{c_}(M/3) \rceil\}$, and for $M = 3$ the exact counts are $p_{\text{conv}} = p_{\text{bin}} = 2$.*

Proof. The selected witness residues are distinct by (142). If three distinct residues summed to zero, their uniform lifted average would be integral. Let j be the smallest selected input index. Its weight is $1/3$ and the other two inputs have component- j powers at least $15/16$. The average input is at most $1 - 64/(3D_j)$. Equation (142) again gives a forbidden error. Thus the residues form a cap set. Over $\mathbb{F}_3$, a zero-sum triple with two equal entries has all three equal, so this is precisely the absence of nontrivial three-term progressions.

Theorem 4 of Ellenberg and Gijswijt [17], specialized to $\mathbb{F}_3$, bounds its size by $3\#\{\alpha \in \{0, 1, 2\}^p : \sum_i \alpha_i \leq 2p/3\}$. For $0 < t < 1$, this monomial count is at most $t^{-2p/3}(1 + t + t^2)^p$, by bounding the indicator with $t^{\sum \alpha_i - 2p/3}$ and summing. Differentiating its base gives $4t^2 + t - 2 = 0$ at the unique minimum, proving (143). For a simpler rational base, $t = 1/2$ gives $(7/4)2^{2/3} < 14/5$, so $M \leq 3(14/5)^p$. A cap set in $\mathbb{F}_3$ has at most two elements. Thus $M = 3$ needs at least two general integers, while (141) gives two binaries. $\square$

There is also a direct one-integer obstruction which illustrates the role of repeated convexification. For any selected $x_j < x_k$, every chord weight $t \in [2/3, 4/5]$ on the larger input has component- j gap at least

$$\frac{7}{4} \left(\frac{2}{3} \frac{15}{16} - \frac{25}{2048} \right) = \frac{8785}{8192} > 1,$$

because the averaged input is at most $1 - 64/(5D_j)$ and $e^{-64/5} \leq 25/2048$, using $e^v \geq v^2/2$. If two integer labels differ in magnitude by $q \geq 3$, some k/q lies in this weight interval: choose $k = \lceil 2q/3 \rceil$; $q = 3, 4$ are direct and $q \geq 5$ gives $k \leq (2q + 2)/3 \leq 4q/5$. The corresponding chord has an integer label and is forbidden. Equal labels are forbidden as well, so three selected labels would have to be three consecutive integers. The midpoint of the extreme labeled witnesses belongs to the middle integer section. Mix it with the third witness so that each original witness has weight $1/3$. This stays in that section and violates (142). Checking only integer points on segments between initially assigned labels would miss this second convexification. The cap-set theorem supplies the higher-dimensional version; its exponential base greater than two and the absence of a matching upper construction leave the binary/general-integer gap unresolved.

20 Exact separations and the role of error geometry

20.1 Fixed-degree convex graphs with exact product counts

Theorem 20.1 (Exact convex box counts). *For $n \geq 1$, let*

$$F_n(x) = \left(\frac{7}{4}(1 - x_i)^{32}, \frac{7}{4}x_i^{32} \right)_{i=1}^n, \quad x \in [0, 1]^n,$$

with unit error in each of the $2n$ components. Then

$$p_{\text{conv}}(F_n) = n, \quad p_{\text{bin}}(F_n) = \lceil n \log_2 3 \rceil. \quad (144)$$

The lower binary count is valid for arbitrary convex binary lifts. Requiring closed convex lifts leaves both minima unchanged. Both upper bounds have rational MILP realizations; the general-integer one uses $3n$ continuous auxiliaries and $13n$ inequalities, with rational coefficients from a fixed finite set independent of n . The binary construction asserted here may have exponential continuous size. The linear size claim concerns variable and row counts, without a claim of linear serialized bit length or a solver running-time bound.

Proof. For one input put $A = 7/4$, $c = 1/48$, $L = A(1-c)^{32}$, $d = Ac^{32}$, and $Q(x) = (A(1-x)^{32}, Ax^{32})$. We have $A - 1 < L < 1$, $d < 1/4$, and $(A + d)/2 < 1$. Indeed Bernoulli gives $(47/48)^{16} \geq 2/3$ and hence $L \geq 7/9 > 3/4$. The binomial expansion gives $(48/47)^{32} \geq 1 + 32/47 + 496/47^2 > 7/4$, so $L < 1$. The remaining inequalities follow from $d < A/48 < 1/4$. Consider the rational boxes

$$\begin{aligned} B_0 &= [0, c] \times [L, A] \times [0, d], \\ B_1 &= [c, 1 - c] \times [0, L]^2, \\ B_2 &= [1 - c, 1] \times [0, d] \times [L, A]. \end{aligned}$$

Each contains the graph on its input interval and has unit graph error: the outer component range widths are $A - L < 1$ and $d < 1$, and both true and admitted middle outputs lie in $[0, L]$. Append labels 0, 1, 2 and take their convex hull $\mathcal{C}$. At integer label zero or two only the corresponding box remains. At label one, consolidating within each box gives weights $t, 1 - 2t, t$, $0 \leq t \leq 1/2$. Its input is at least $(1 - 2t)c + t(1 - c) = c + t(1 - 3c) \geq c$ and, by symmetry, at most $1 - c$. Every output is

in $[0, (1 - 2t)L + t(A + d)] \subset [0, \max\{L, (A + d)/2\}] \subset [0, 1)$. Every true output there also lies in $[0, L]$. Thus the entire middle integer section, including all mixtures, is valid.

Write $B_j = [\ell_j, u_j]$ coordinatewise in $v = (x, w_1, w_2)$. An explicit rational linear lift of $\mathcal{C}$ is

$$\begin{aligned} \lambda_j &\geq 0 \quad (j = 0, 1, 2), \quad \sum_{j=0}^2 \lambda_j = 1, \quad z = \lambda_1 + 2\lambda_2, \\ \sum_{j=0}^2 \lambda_j \ell_{j,k} &\leq v_k \leq \sum_{j=0}^2 \lambda_j u_{j,k} \quad (k = 1, 2, 3). \end{aligned}$$

For fixed weights, the weighted sum of boxes is exactly the box with these weighted endpoints, since its coordinate intervals vary independently. Hence projection gives precisely the labeled hull. Only z is integer. There are three continuous auxiliaries, nine inequalities and two equalities, or thirteen inequalities after splitting the equalities. Products prove $p_{\text{conv}}(F_n) \leq n$ with the stated counts and fixed rational data.

The three contacts at $0, 1/2, 1$ are pairwise incompatible under an oriented thirds combination. For $0, 1/2$, give weight $2/3$ to zero; the first-component gap at $1/6$ is

$$\frac{2}{3}A + \frac{1}{3}A2^{-32} - A(5/6)^{32} > 1.$$

Since $4 \cdot 5^8 < 6^8$, $(5/6)^{32} < 1/256$, and the gap exceeds $7/6 - 7/1024 > 1$. The pair $1/2, 1$ is symmetric. For $0, 1$, weight $2/3$ on zero gives gap $\frac{2}{3}A - A(2/3)^{32} > 1$ by the same bound.

Choose witnesses at all 3^n product inputs in $\{0, 1/2, 1\}^n$. Two integer codes equal modulo three would permit both thirds combinations. They differ in some input coordinate, where one orientation is forbidden by the preceding calculation. Thus all codes have distinct residues and $3^p \geq 3^n$, proving $p \geq n$. Equal binary codes would permit every convex weight, so these same contacts need distinct binary assignments: $2^p \geq 3^n$. For equality, take the 3^n products of the three valid boxes and apply lemma 2.3 with $\lceil \log_2(3^n) \rceil$ bits. This is a finite rational polyhedral construction. Both rational upper constructions use closed lifted sets, while the lower bounds allow all convex lifts, proving the closedness assertion as well. $\square$

Adding $56x_i$ to the first component of each pair makes both components nondecreasing: its derivative becomes $56[1 - (1 - x_i)^{31}] \geq 0$. This affine shear preserves errors, both minima, and the closed-lift minima. Substitution in the displayed formulation preserves its $3n$ auxiliary and $13n$ inequality counts. It does not make all monomial coefficients nonnegative: the first component's cubic coefficient is -8680 . Equation (144) shows that an additive loss proportional to input dimension can be necessary even with fixed degree, tolerance and numerical data. It does not amplify the gap with only one input.

A piecewise-linear precursor explains the three-section geometry. Take $a = 1/8$, $A = 3/2$ and $H(x) = (A \max\{1 - x/a, 0\}, A \max\{(x - 1 + a)/a, 0\})$. Its three exact affine graph segments, labeled $0, 1, 2$, have a valid one-integer convex hull at unit box error. In the middle integer section, the outer weights are t, t and their averaged input lies in $[(1 - a)/2, (1 + a)/2] \subset [a, 1 - a]$. The true graph there is zero; the admitted outputs are between zero and $tA \leq 3/4$. The contacts $0, 1/2, 1$ cannot share a binary section pairwise: the first pair's chord at a has first output $A(1 - 2a) = 9/8$, the last pair is symmetric, and the outer-pair chord gives $A(1 - a) = 21/16$. Thus this hinge curve also has exact counts one and two.

This strict separation survives a convexity-preserving polynomial approximation. For a scalar L -Lipschitz function h , its Bernstein polynomial $B_N h(x) = \sum_k h(k/N) \binom{N}{k} x^k (1 - x)^{N-k}$ satisfies

$$|B_N h(x) - h(x)| \leq L \mathbb{E}|X/N - x| \leq L \sqrt{x(1 - x)/N} \leq L/(2\sqrt{N}), \quad X \sim \text{Bin}(N, x). \quad (145)$$

For convex h , its sampled second differences are nonnegative, and

$$(B_N h)'' = N(N-1) \sum_{k=0}^{N-2} [h((k+2)/N) - 2h((k+1)/N) + h(k/N)] \binom{N-2}{k} x^k (1-x)^{N-2-k} \geq 0.$$

The hinge components are 12-Lipschitz. Taking $N = 2^{18}$ gives uniform error at most $3/256 < \delta = 1/64$. Thicken the labeled polytope in outputs by $[-\delta, \delta]^2$. It contains the exact Bernstein graph and its integer-section error is at most $3/4 + 2\delta < 1$. Each forbidden chord loses at most 2δ , so the binary lower remains two; the three separately thickened segments give two binaries. This is a distinct stability mechanism, although the degree-32 family in theorem 20.1 is simpler and sharper. The probabilistic Bernstein estimate and its convexity-preserving second-difference identity are classical approximation tools.

20.2 Nonconvex scalar polynomials and degree dependence

Theorem 20.2 (Two general integers and growing binary dimension). *For every integer $M \geq 2$ there is a dense rational polynomial q_M of degree at most $1024M^2$ such that, at fixed absolute error $1/4$,*

$$p_{\text{conv}}(q_M; 1/4) \leq 2, \quad \lceil \log_2 M \rceil \leq p_{\text{bin}}(q_M; 1/4) \leq \lceil \log_2 M \rceil + 1. \quad (146)$$

The lower bound holds even for arbitrary convex binary lifts. For a continuous scalar function with s convex or concave pieces,

$$p_{\text{bin}} \leq p_{\text{conv}} + \lceil \log_2(3s) \rceil. \quad (147)$$

Consequently a degree- D nonaffine polynomial has finite overhead $\lceil \log_2(3 \max\{1, D-1\}) \rceil$.

Proof. Let T_M be the triangular wave of M periods: for $t = Mx - j \in [0, 1]$, $j = 0, \dots, M-1$, set $T_M(x) = 2t$ for $t \leq 1/2$ and $T_M(x) = 2 - 2t$ otherwise. It is $2M$ -Lipschitz. Take $N = (32M)^2$ and $q_M = B_N T_M$. Equation (145) gives $|q_M - T_M| \leq 1/32$, and positivity of Bernstein weights gives $0 \leq q_M \leq 1$. Sampled values are rational of polynomial encoding; binomial expansion gives a dense monomial representation of polynomial bit length in N and $\log M$.

An integer period $z \in \{0, \dots, M-1\}$ and one binary orientation b encode the triangular-wave graph exactly by $x = (z+t)/M$ and the two bounded linear branches

$$b = 0 : 0 \leq t \leq 1/2, y = 2t; \quad b = 1 : 1/2 \leq t \leq 1, y = 2 - 2t.$$

A two-branch disjunction with finite rational big- M bounds realizes these equations. Period boundaries and $x = 1$ are included. Adding $|w - y| \leq 1/32$ contains the entire q_M graph and admits error at most $1/16 < 1/4$. The period and orientation together count as two general integers.

At the M peaks $(j+1/2)/M$, $q_M \geq 31/32$; at each intervening trough k/M , $q_M \leq 1/32$. If two peak contacts had witnesses in the same binary section, their full chord would be admitted. At a trough between them its value is at least $31/32$, yielding error at least $30/32 > 1/4$. Thus peaks need distinct binary codes. Conversely replace the period integer in the explicit upper construction by $\lceil \log_2 M \rceil$ digits and exclude unused codes, retaining the orientation bit. This proves (146).

For (147), restrict any p -integer lift to each shape interval and negate a concave output. The scalar chord lemma gives at most $3 \cdot 2^p$ error-valid chord bands per piece. Their finite union has at most $3s2^p$ members. A polynomial of degree $D \geq 2$ has at most $D-2$ distinct roots of its second derivative, and hence at most $D-1$ such intervals. An affine polynomial is exact with zero integers. Roots may be irrational in this finite real-coefficient argument. $\square$

The polynomial degrees in this family tend to infinity: alternating peaks and troughs force at least $2M$ distinct zeros of $q_M - 1/2$. Together with the upper degree bound, (146) shows that logarithmic degree overhead is the correct worst-case order. The general binary/general-integer distinction and binarization of bounded integer variables are established; the example imposes the additional requirement of one connected rational polynomial graph at fixed positive tolerance. It is a lower bound on optimum counts, not only on one approximation construction.

Theorem 20.3 (Compiled arbitrary polynomial vectors). *Let the nonaffine outputs of $F : [0, 1] \rightarrow \mathbb{R}^m$ be dense rational polynomials of degrees $D_j \geq 2$, with positive rational box tolerances. A polynomial-time rational MILP construction has*

$$p_{\text{out}} \leq p_{\text{conv}} + 12 + \left\lceil \log_2 \sum_{j:D_j \geq 2} D_j \right\rceil. \quad (148)$$

Affine outputs are represented exactly, and an entirely affine graph uses no integers. For one output this gives overhead $12 + \lceil \log_2 D \rceil$.

Proof. For output $f_j = \sum_k c_{jk} x^k$ put $M_j = \max\{1, \sum_{k \geq 1} k |c_{jk}| \} \geq \max |f'_j|$. Isolate each distinct interior zero of f'_j in a disjoint rational bracket of width at most $\min\{1, \varepsilon_j/(32M_j)\}$, with nonroot endpoints. Squarefree preprocessing, root isolation and refinement have polynomial bit cost by the univariate root routine used in theorem 12.4; endpoint roots need no bracket. The brackets and complementary intervals give at most $2D_j$ pieces. On a complement the curvature has constant sign; normalize its rational interval and apply the convex hybrid to f_j or $-f_j$. Restriction and output negation preserve the comparator's integer count. Each such piece has fewer than 2^{p+11} cells, where p is the integer dimension of any comparator vector lift. A bracket has one cell: a derivative bound gives absolute chord error at most $2M_j h \leq \varepsilon_j/16$, including on every subinterval. Thus the scalar list has

$$K_j \leq 4096 D_j 2^p. \quad (149)$$

Use ordered local quantiles, reverse reflected indices, and concatenate in original input order. The indexed evaluator records source-cell type: convex, concave, or bracket. All knots have a polynomial-bit common denominator as in lemma 16.3. Merge their right endpoints with multiplicity. On every positive merged cell recover each output's containing source cell and type using the search in that lemma; at a repeated endpoint any containing source cell is valid.

Evaluate each original polynomial at the selected merged endpoints. Round downward within $\varepsilon_j/16$ for convex cells and brackets, and upward within $\varepsilon_j/16$ for concave cells. If y_j is the interpolant, use

$$\begin{cases} y_j - 13\varepsilon_j/16 \leq w_j \leq y_j + \varepsilon_j/8, & \text{convex or bracket,} \\ y_j - \varepsilon_j/8 \leq w_j \leq y_j + 13\varepsilon_j/16, & \text{concave.} \end{cases} \quad (150)$$

For convex cells $y_j - f_j \in [-\varepsilon_j/16, 13\varepsilon_j/16]$; for concave cells it lies in $[-13\varepsilon_j/16, \varepsilon_j/16]$. For brackets it lies in $[-\varepsilon_j/8, \varepsilon_j/16]$. Each band contains the graph and has absolute admitted error at most $15\varepsilon_j/16$. The same checks hold on singleton cells. The type flags are computed Boolean wires and choose the offsets linearly; they require no new integer declarations. A common interpolation weight and input cell guarantee simultaneous vector graph containment.

The total merged count is $K = \sum_j K_j$, so (149) proves (148). All numerator searches, source-cell lookups, directed evaluations, flags and invalid-code exclusions have polynomial bit complexity. Lemma 12.1 declares only the $\lceil \log_2 K \rceil$ input index bits integer. The algorithm uses actual computed counts, not p or the comparator lift. $\square$

20.3 Tilted error bodies and conditioning

Proposition 20.4 (A two-output tilted-body gap). *There are rational componentwise convex polynomial curves with two outputs and rational symmetric parallelogram error bodies for which $p_{\text{conv}} \leq 2$ and $p_{\text{bin}} \geq \lceil \log_2 M \rceil \rightarrow \infty$. The Euclidean outer-to-inner radius ratio of these bodies is at least $8 + 60M^2$; this is not a uniformly conditioned family.*

Proof. Use q_M from theorem 20.2, with dense coefficients c_k , and put

$$C_M = 1 + \sum_{k \geq 2} k(k-1)|c_k|, \quad F_M = (q_M + C_M x^2, C_M x^2), \quad K_M = \{e : |e_1 - e_2| \leq 1/4, |e_2| \leq C_M\}.$$

Since $|q_M''| \leq C_M - 1$, both components are convex. The body is compact, symmetric and has interior, but is not unconditional: it contains (C_M, C_M) and not its first-coordinate sign flip. Take the two-integer scalar lift in theorem 20.2, with output s of error at most $1/16$. Add $0 \leq w_2 \leq C_M$ and $w_1 = s + w_2$. Exact vector graph points lift with $s = q_M(x)$, $w_2 = C_M x^2$; every admitted error obeys $|e_1 - e_2| \leq 1/16$ and $|e_2| \leq C_M$. This gives the general-integer upper bound with rational linear constraints. Any binary convex lift projects by $s = w_1 - w_2$ to a scalar lift of q_M with tolerance $1/4$. Its binary count is at least $\lceil \log_2 M \rceil$; replacing the period integer also gives an upper count of $\lceil \log_2 M \rceil + 1$.

The body has circumradius at least $\sqrt{2}C_M$ and inradius at most $1/(4\sqrt{2})$ because of its narrow difference strip. At a peak a and adjacent troughs $a \pm h$, $h = 1/(2M)$, the uniform Bernstein estimate gives $q_M(a - h) - 2q_M(a) + q_M(a + h) \leq -15/8$. The identity

$$q_M(a - h) - 2q_M(a) + q_M(a + h) = \int_{-h}^h (h - |t|) q_M''(a + t) dt$$

implies $\max |q_M''| \geq (15/2)M^2$. Thus $C_M \geq 1 + (15/2)M^2$, proving the claimed ratio. $\square$

Adding a sufficiently large quadratic to make a smooth function convex is the elementary difference-of-convex construction; the general framework predates this application [21]. The large permitted error in the common output direction absorbs that quadratic. The narrow difference direction retains the nonconvex scalar obstruction. The error geometry, as well as componentwise convexity, is therefore material.

Proposition 20.5 (Finite conditioning comparisons). *Let $F : [0, 1] \rightarrow \mathbb{R}^m$ be continuous and componentwise convex. If $aB_\infty^m \subset K \subset bB_\infty^m$, $0 < a \leq b$, then*

$$p_{\text{bin}} \leq p_{\text{conv}} + \lceil \log_2 (2 \lceil 2mb/a \rceil - 1) \rceil. \quad (151)$$

The same bound holds with Euclidean inner and outer balls. In that Euclidean case there is also the bound

$$p_{\text{bin}} \leq p_{\text{conv}} + \left\lceil \log_2 \left(2 \left\lceil 2\sqrt{2m} b/a \right\rceil - 1 \right) \right\rceil. \quad (152)$$

These are finite real-coefficient statements. They hold even for a nonsymmetric compact convex permitted-error set containing the given inner ball, using the same graph-sandwich definition.

Proof. On a parity hull the nonnegative midpoint vector belongs to K . For $h = \sum_j F_j$, its midpoint gap is at most mb under infinity containment and at most $\sqrt{m}b$ under Euclidean containment. The full gap is at most twice that. In the infinity case refine to tolerance a , so lemma 16.1 uses at most $2\lceil 2mb/a \rceil - 1$ cells per parity hull. Each component gap is between zero and a , and downward component bands have error in $aB_\infty^m \subset K$. In the Euclidean case refine instead to $a/\sqrt{m}$. Downward

bands then have error in aB_2^m , with the same refinement factor. Finite disjunction proves (151) in both cases.

For the sharper Euclidean variant, if the total scalar gap is at most s , use the simplex band

$$w = T - v, \quad v \geq 0, \quad \sum_j v_j \leq s. \quad (153)$$

It contains the graph by $v = g_F$. Every admitted error is $g_F - v$, where both nonnegative vectors have ℓ_1 norm at most s . Thus $\|g_F - v\|_2^2 \leq \|g_F\|_2^2 + \|v\|_2^2 \leq 2s^2$. Choose $s = a/\sqrt{2}$ and refine from $2\sqrt{m}b$ to s . This proves (152). Only the outer bound on midpoint vectors and the inner bound for final errors were used; symmetry of the intervening permitted-error set is unnecessary. $\square$

A related normalization explains why unconditionality avoids arbitrary conditioning losses. Let $a_j = \max_{e \in K} |e_j| > 0$ and scale each output by $1/a_j$. The normalized body K' satisfies

$$B_1^m \subset K' \subset B_\infty^m.$$

For the first inclusion, take a point attaining a coordinate maximum and average its sign flips in all other coordinates. The resulting positive and negative unit coordinate vectors lie in K' ; their convex hull is B_1^m . Consequently $m^{-1}B_\infty^m \subset K'$ and (151) gives finite overhead $\lceil \log_2(4m^2 - 1) \rceil$. The simplex band improves this elementary normalization bound: now the parity-hull gap of $\sum_j F_j/a_j$ is at most $2m$; refine to $s = 1/2$. Since $\|g_F - v\|_1 \leq 2s = 1$, the band lies in $B_1^m \subset K'$ and has overhead $\lceil \log_2(8m - 1) \rceil$. The direct maximum-product bound in proposition 16.7 is sharper, but the normalization and simplex arguments give additional geometric comparisons without computing an optimal allocation.

20.4 The remaining one-input box question

For one compact input interval and continuous componentwise convex outputs with box error, it remains unresolved here whether a universal constant C satisfies $p_{\text{bin}} \leq p_{\text{conv}} + C$, independently of output count or curvature rank. Theorem 20.1 rules out equality of the two minima, but its unbounded product gap increases input dimension. The positive-power family proves failure of scalar-minimum and uniform refinement tests; neither its residue lower bound nor the cap-set strengthening supplies a growing difference between the two minima. The tilted-body construction changes the error geometry and loses uniform conditioning, so it does not settle the box question either.

One can express a concrete obstacle using lattice-free convex sets. For an admissible convex lift $\mathcal{C}$ and a convex component F_j , the strict upper-violation set

$$\{(x, w, z, a) \in \mathcal{C} : w_j > F_j(x) + \varepsilon_j\}$$

is convex. Its projection to the integer coordinates is convex and has no integer point; any such point would have an invalid feasible integer witness. The union of these sets over outputs need not be convex. Taking its convex hull is unjustified: mixing violations of different components can remove every violation. Thus applying lattice-free geometry to one output does not automatically control all outputs.

The mixed Helly identity $h(\mathbb{R}^q \times \mathbb{Z}^p) = (q + 1)2^p$ gives finite infeasibility certificates for convex sets [2, Theorem 1.1]; it does not supply an interval cover preserving the output error. Its dependence on the unrestricted continuous dimension also precludes a direct count bound. Similarly, an integral Radon partition would need a common integer index and compatible input supports producing a

violation in one output; parity alone does not ensure those properties. These observations delimit what the present contact arguments establish. None of the proved comparisons depends on resolving the constant-gap question.

21 Conclusion

The minimum integer count for quadratic graph approximation is governed by a joint algebraic invariant: half the noncommutative rank of the Hessian space is its exact logarithmic precision coefficient. This coefficient applies equally to arbitrary convex integer lifts and binary linear lifts, although their finite-accuracy counts can differ. At finite accuracy, contact covariance leads to a determinant benchmark and constructive bounds with explicit dimension losses. Positive structure, effective input coordinates and the actual error body can sharpen those losses, while correlated domains can prevent naive transfers.

For scalar convex graphs, finite chord geometry makes binary linear lifts competitive within two variables with arbitrary convex integer lifts. Certified integration, rational endpoint algorithms and a common index compiler give an eleven-bit polynomial rational comparison for dense convex polynomials, together with separate constructions for the specified sparse polynomial and power representations. Coupled outputs require simultaneous control: original-output or positive-polar curvature bases give such control, and a shared basis across separable inputs preserves the relevant packing lower bound. These results depend on convexity and the stated error geometry. The exact product counts, positive-power obstruction, nonconvex polynomial family and tilted-body example identify different ways those assumptions matter.

Several questions remain. The finite covariance benchmark has an order- $n \log n$ gap on an explicit family; a different efficiently computable invariant might give smaller losses, subject to the proved count-approximation hardness. Local and global smooth Hessian ranks can differ, and rank-changing scalar maps are not characterized by the constant-rank theorem. For one input and arbitrarily many convex outputs with box error, the present logarithmic-rank upper comparison and strict one-bit example leave open whether a universal constant additive gap exists. The cap-set restriction strengthens one lower-bound family but does not decide that question. Extending the separable vector construction to mixed-coordinate nonlinear terms also requires a new compact encoding argument.

The encoding separation supplies a further boundary: even four binary variables can require exponentially long rational linear data relative to a binary exponent, whereas second-order-cone constraints encode the same graph accuracy with short data. Integer count, continuous size, coefficient length, construction time and optimization time should therefore remain separate when these formulations are used or compared.

References

- [1] Warren P. Adams and Stephen M. Henry. Base-2 expansions for linearizing products of functions of discrete variables. *Operations Research*, 60(6):1477–1490, 2012. doi: 10.1287/opre.1120.1106. URL <https://www.osti.gov/servlets/purl/1648449>. Section 2 in the linked author manuscript, SAND2012-0505P.
- [2] Gennadiy Averkov and Robert Weismantel. Transversal numbers over subsets of linear spaces. *Advances in Geometry*, 12(1):19–28, 2012. doi: 10.1515/advgeom.2011.028. URL <https://arxiv.org/abs/1002.0948v2>. Theorem locator refers to arXiv:1002.0948v2.

- [3] David Avis, David Bremner, Hans Raj Tiwary, and Osamu Watanabe. Polynomial size linear programs for problems in P. *Discrete Applied Mathematics*, 265:22–39, 2019. doi: 10.1016/j.dam.2019.03.016. URL <https://arxiv.org/abs/1408.0807>.
- [4] Baruch Awerbuch and Robert D. Kleinberg. Adaptive routing with end-to-end feedback: Distributed learning and geometric approaches. In *Proceedings of the 36th Annual ACM Symposium on Theory of Computing*, pages 45–53, 2004. URL <https://www.cs.cornell.edu/~rdk/papers/OLSP.pdf>.
- [5] Benjamin Beach, Robert Hildebrand, and Joey Huchette. Compact mixed-integer programming formulations in quadratic optimization. *Journal of Global Optimization*, 84(4):869–912, 2022. doi: 10.1007/s10898-022-01184-6. URL <https://arxiv.org/abs/2011.08823>.
- [6] Benjamin Beach, Robert Burlacu, Andreas Bärmann, Lukas Hager, and Robert Hildebrand. Enhancements of discretization approaches for non-convex mixed-integer quadratically constrained quadratic programming: Part I. *Computational Optimization and Applications*, 87: 835–891, 2024. doi: 10.1007/s10589-023-00543-7. URL <https://arxiv.org/abs/2211.00876>.
- [7] Benjamin Beach, Robert Burlacu, Andreas Bärmann, Lukas Hager, and Robert Hildebrand. Enhancements of discretization approaches for non-convex mixed-integer quadratically constrained quadratic programming: Part II. *Computational Optimization and Applications*, 87: 893–934, 2024. doi: 10.1007/s10589-024-00554-y. URL <https://arxiv.org/abs/2302.01164>.
- [8] Andrea Bonito and Joseph E. Pasciak. Numerical approximation of fractional powers of elliptic operators. *Mathematics of Computation*, 84(295):2083–2110, 2015. doi: 10.1090/S0025-5718-2015-02937-8. URL <https://arxiv.org/abs/1307.0888>.
- [9] Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004. URL <https://web.stanford.edu/~boyd/cvxbook/>.
- [10] Jop Briët, Fernando Mário de Oliveira Filho, and Frank Vallentin. The positive semidefinite Grothendieck problem with rank constraint. In *Automata, Languages and Programming, Part I*, volume 6198 of *Lecture Notes in Computer Science*, pages 31–42, 2010. doi: 10.1007/978-3-642-14165-2_4. URL <https://arxiv.org/abs/0910.5765>.
- [11] Weiming Cao. An interpolation error estimate on anisotropic meshes in $\mathbb{R}^n$ and optimal metrics for mesh refinement. *SIAM Journal on Numerical Analysis*, 45(6):2368–2391, 2007. doi: 10.1137/060667992. URL <https://epubs.siam.org/doi/10.1137/060667992>.
- [12] Alfonso Cevallos, Stefan Weltge, and Rico Zenklusen. Lifting linear extension complexity bounds to the mixed-integer setting. In *Proceedings of the Twenty-Ninth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 788–807. SIAM, 2018. URL <https://arxiv.org/abs/1712.02176v1>.
- [13] Julien Codsì, Sandra Ulrich Ngueveu, and Bernard Gendron. LinA: A faster approach to piecewise linear approximations using corridors and its application to mixed-integer optimization. *Mathematical Programming Computation*, 17(2):265–306, 2025. doi: 10.1007/s12532-024-00274-8. URL <https://doi.org/10.1007/s12532-024-00274-8>.
- [14] Christopher Criscitiello and Nicolas Boumal. An accelerated first-order method for non-convex optimization on manifolds. *Foundations of Computational Mathematics*, 23:1433–1509, 2023. doi: 10.1007/s10208-022-09573-9. URL <https://arxiv.org/abs/2008.02252>. Appendix I is also available in arXiv:2008.02252v2.
- [15] Daniel Dadush, Chris Peikert, and Santosh Vempala. Enumerative lattice algorithms in any norm via M -ellipsoid coverings, 2011. URL <https://sites.cc.gatech.edu/fac/cpeikert/pubs/svp-anynorm.pdf>. Full manuscript dated April 13, 2011; Theorem B.5.

- [16] Alberto Del Pia. Rational Jacobi rotations and the complexity of approximating mixed integer quadratic programming. arXiv:2607.29386, 2026. URL <https://arxiv.org/abs/2607.29386>.
- [17] Jordan S. Ellenberg and Dion Gijswijt. On large subsets of $\mathbb{F}_q^n$ with no three-term arithmetic progression. *Annals of Mathematics*, 185(1):339–343, 2017. doi: 10.4007/annals.2017.185.1.8. URL <https://arxiv.org/abs/1605.09223v1>. Theorem locator refers to arXiv:1605.09223v1.
- [18] Michael R. Garey, David S. Johnson, and Larry Stockmeyer. Some simplified NP-complete graph problems. *Theoretical Computer Science*, 1(3):237–267, 1976. doi: 10.1016/0304-3975(76)90059-1. URL <https://www.math.ucdavis.edu/~deloera/MISC/LA-BIBLIO/trunk/JohnsonDavid2.pdf>.
- [19] Ankit Garg, Leonid Gurvits, Rafael Oliveira, and Avi Wigderson. Operator scaling: Theory and applications. *Foundations of Computational Mathematics*, 20:223–290, 2020. doi: 10.1007/s10208-019-09417-z. URL <https://www.math.ias.edu/~avi/PUBLICATIONS/GargGOW20.pdf>.
- [20] Martin Grötschel, László Lovász, and Alexander Schrijver. The ellipsoid method and its consequences in combinatorial optimization. *Combinatorica*, 1(2):169–197, 1981. URL <https://ir.cwi.nl/pub/10046/10046D.pdf>.
- [21] Philip Hartman. On functions representable as a difference of convex functions. *Pacific Journal of Mathematics*, 9(3):707–713, 1959. URL <https://msp.org/pjm/1959/9-3/pjm-v9-n3-p09-p.pdf>.
- [22] Lars Hörmander. Oscillatory integrals and multipliers on FL^p . *Arkiv för Matematik*, 11:1–11, 1973. doi: 10.1007/BF02388505. URL https://archive.ymsc.tsinghua.edu.cn/pacm_download/116/7179-11512_2006_Article_BF02388505.pdf.
- [23] Gábor Ivanyos, Youming Qiao, and K. V. Subrahmanyam. Constructive non-commutative rank computation is in deterministic polynomial time. *Computational Complexity*, 27(4):561–593, 2018. doi: 10.1007/s00037-018-0165-7. URL <https://arxiv.org/abs/1512.03531v6>. Theorem locators refer to arXiv:1512.03531v6.
- [24] F. P. Kelly, A. K. Maulloo, and D. K. H. Tan. Rate control for communication networks: Shadow prices, proportional fairness and stability. *Journal of the Operational Research Society*, 49(3):237–252, 1998. doi: 10.1057/palgrave.jors.2600523.
- [25] Miles Lubin, Juan Pablo Vielma, and Ilias Zadik. Mixed-integer convex representability. *Mathematics of Operations Research*, 47(1):720–749, 2022. doi: 10.1287/moor.2021.1146. URL <https://arxiv.org/abs/1706.05135>.
- [26] Bochuan Lyu, Illya V. Hicks, and Joey Huchette. Building formulations for piecewise linear relaxations of nonlinear functions. *Operations Research*, 74(1):484–499, 2026. doi: 10.1287/opre.2023.0187. URL <https://arxiv.org/abs/2304.14542v1>. Section and proposition locators refer to arXiv:2304.14542v1.
- [27] Garth P. McCormick. Computability of global solutions to factorable nonconvex programs: Part I—convex underestimating problems. *Mathematical Programming*, 10:147–175, 1976. doi: 10.1007/BF01580665.
- [28] National Institute of Standards and Technology. Digital Library of Mathematical Functions, section 3.5: Quadrature, 2026. URL <https://dlmf.nist.gov/3.5>. Accessed September 5, 2026; Section 3.5(v).
- [29] Fabio Nicola. Boundedness of fourier integral operators on fourier lebesgue spaces and affine fibrations. *Studia Mathematica*, 198(3):207–219, 2010. doi: 10.4064/sm198-3-1. URL <https://arxiv.org/abs/0805.4122>.

- [30] Ryan O'Donnell. SOS is not obviously automatizable, even approximately. In *8th Innovations in Theoretical Computer Science Conference*, volume 67 of *Leibniz International Proceedings in Informatics*, pages 59:1–59:10, 2017. doi: 10.4230/LIPIcs.ITCS.2017.59. URL <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ITCS.2017.59>.
- [31] Orestis Plevrakis and Elad Hazan. Geometric exploration for online control. In *Advances in Neural Information Processing Systems*, volume 33, pages 7637–7647. Curran Associates, Inc., 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/565e8a413d0562de9ee4378402d2b481-Abstract.html>.
- [32] Steffen Rebennack and Josef Kallrath. Continuous piecewise linear delta-approximations for univariate functions: Computing minimal breakpoint systems. *Journal of Optimization Theory and Applications*, 167(2):617–643, 2015. doi: 10.1007/s10957-014-0687-3. URL <https://josefkallrath.github.io/files/RebKal2014a.pdf>.
- [33] Michael Sagraloff and Kurt Mehlhorn. Computing real roots of real polynomials. *Journal of Symbolic Computation*, 73:46–86, 2016. doi: 10.1016/j.jsc.2015.03.004. URL <https://arxiv.org/abs/1308.4088v2>. Theorem 36 in the linked arXiv version 1308.4088v2.
- [34] Jamico Schade, Makrand Sinha, and Stefan Weltge. Lower bounds on the complexity of mixed-integer programs for stable set and knapsack. *Mathematical Programming*, 216:135–176, 2026. doi: 10.1007/s10107-025-02234-z. URL <https://link.springer.com/article/10.1007/s10107-025-02234-z>.
- [35] Max Simchowitz, Kevin Jamieson, Jordan W. Suchow, and Thomas L. Griffiths. Adaptive sampling for convex regression. arXiv:1808.04523v3, 2018. URL <https://arxiv.org/abs/1808.04523>.
- [36] Suvrit Sra and Reshad Hosseini. Geometric optimisation on positive definite matrices for elliptically contoured distributions. In *Advances in Neural Information Processing Systems*, volume 26, 2013. URL <https://proceedings.neurips.cc/paper/2013/hash/3948ead63a9f2944218de038d8934305-Abstract.html>.
- [37] João P. Teles, Pedro M. Castro, and Henrique A. Matos. Multi-parametric disaggregation technique for global optimization of polynomial programming problems. *Journal of Global Optimization*, 55:227–251, 2013. doi: 10.1007/s10898-011-9809-8.
- [38] Lieven Vandenbergh, Stephen Boyd, and S.-P. Wu. Determinant maximization with linear matrix inequality constraints. *SIAM Journal on Matrix Analysis and Applications*, 19(2):499–533, 1998. URL <https://web.stanford.edu/~boyd/papers/maxdet.html>.
- [39] Juan Pablo Vielma. Embedding formulations and complexity for unions of polyhedra. *Management Science*, 64(10):4721–4734, 2018. doi: 10.1287/mnsc.2017.2856. URL <https://arxiv.org/abs/1506.01417>.
- [40] Jurij Volčič. Hilbert’s 17th problem in free skew fields. *Forum of Mathematics, Sigma*, 9:e61, 2021. doi: 10.1017/fms.2021.54. URL <https://arxiv.org/abs/2101.02314>.
- [41] Jie Wang. Weighted geometric mean, minimum mediated set, and optimal simple second-order cone representation. *SIAM Journal on Optimization*, 34(2):1490–1514, 2024. doi: 10.1137/22M1531257. URL <https://arxiv.org/abs/2206.05924>.
- [42] Thomas H. Wolff. Lectures in harmonic analysis, 2002. URL https://personal.math.ubc.ca/~ilaba/wolff/notes_march2002.pdf. Revised lecture notes, March 2002.
- [43] Dmitry Yarotsky. Error bounds for approximations with deep ReLU networks. *Neural Networks*, 94:103–114, 2017. doi: 10.1016/j.neunet.2017.07.002. URL <https://arxiv.org/abs/1610.01145v3>. Proposition locator refers to arXiv:1610.01145v3.

- [44] Hongyi Zhang and Suvrit Sra. First-order methods for geodesically convex optimization. In *Proceedings of the 29th Annual Conference on Learning Theory*, volume 49 of *Proceedings of Machine Learning Research*, pages 1617–1638, 2016. URL <https://proceedings.mlr.press/v49/zhang16b.html>.